

GENERAL THETA FOUNDATIONS I: CAUSAL EXPERIMENTS, PREDICTIVE QUOTIENTS, AND RESOURCE-AWARE REDUCTION

QIAN QI

ABSTRACT. We study causal experiments in which one finite-state encoder must serve several decision tasks without access to the unknown parameter. For standard Borel observations with dominated, continuously parameterized loss effects, we prove compactness of the attainable private risk set and an exact shared-randomness minimax duality over a compact family of decision rows. All measurable encoder kernels are admitted. The proof constructs executable representatives of weak star policy classes and retains the common encoder throughout the compactness argument. Finite observation approximations converge uniformly over all memory widths. For finite marked instruments we characterize the least retained state preserving every restart prediction risk, including randomized statistics, and give exact causal reconstruction of the joint report–target law. Rational continuation spaces yield an output-sensitive synthesis algorithm. A three-task example separates taskwise, private and shared compression, and a delayed-query Gaussian experiment exhibits a temporal penalty of order $S^{-2/d}$. Positive hidden Markov models, nonlinear finite-state semigroups and latent-phase moment problems provide explicit consequences. Predictive-tree, measure-valued, Gaussian and count regimes are developed with their respective information and numerical resource specifications.

CONTENTS

1. Introduction	3
1.1. The main results	3
1.2. Related work and the distinction of scope	4
1.3. Consequences and organization	4
2. The marked decision spectrum	5
2.1. Experiments, tasks, and signatures	5
2.2. Typed simulations and invariance	6
2.3. The terminal decision-complete slice	7
3. Compact duality for dominated causal experiments	8
3.1. The experiment and its information pattern	8
3.2. The topology and an executable realization	9
3.3. Finite observation approximations, uniformly in memory	12
4. The minimal marked continuation quotient	12
4.1. An operational necessity theorem	13
4.2. Two exact marked causal channels	14
5. Synthesis from continuation spaces	15
6. A common-encoder obstruction with an interior dual witness	17
7. Controlled common encoders and the minimax risk body	18
7.1. Finite instruments and the information pattern	18
7.2. The variational theorem	19

Date: Tenth revision, September 23, 2026.

2020 Mathematics Subject Classification. Primary 62B05, 94A34; Secondary 60G35, 62C10, 68Q45, 37D35.

Key words and phrases. causal experiment, statistical deficiency, predictive quotient, finite-state realization, Wasserstein entropy, finite precision.

7.3. Controlled continuation partitions	20
7.4. Private randomization is a different finite problem	20
8. One variational principle: occupancies, Bayes envelopes and pressure	21
8.1. An exact support-function recursion	21
8.2. The common quadratic task family	22
8.3. Exponential criteria and hidden-phase control	24
9. Minimal posterior-and-report-predictive coordinates	24
10. Finite compilation and several charged resources	27
11. A Gaussian experiment with a genuine temporal penalty	29
12. Marked projections and finite numerical acquisition	31
12.1. A finite table for the delayed Gaussian problem	32
13. Positive observation models and history reduction	33
13.1. A proved forgetting estimate	34
14. Nonlinear finite-state semigroups	35
15. Hidden phases and risk-sensitive continuation	36
16. A prior-independent experiment behind predictive coordinates	38
16.1. The causal factorization	39
17. A necessity theorem for causal realization	40
18. Finite-prefix, finite-description realization	42
19. The measure-valued slice: sharp presentation entropy	44
20. A dynamic finite-experiment consequence	46
20.1. A theorem-level downstream arrow	47
Supporting theory and model verification	48
Supporting theory and model verification	48
21. Positive experiments and predictive quotients	48
21.1. Actual histories	48
21.2. The future-test quotient	48
21.3. Attainable Bayes geometry	49
21.4. Normalized response and observation modules	50
22. Causal morphisms and charged approximation	50
23. A structural theorem for predictive trees	51
23.1. Local hypotheses and the finite profile	51
23.2. Balance, dilation and exact closure	53
23.3. Static lower bound and the causal converse	54
24. Two model verifications	55
24.1. Markov-renewal orbit geometry	55
24.2. An infinite hidden shift with noisy continual reports	56
25. Finite presentations of measure-valued filters	57
25.1. A nonlinear physical realization	58
26. Sharp finite-experiment budgets	59
26.1. The cardinality obstruction	59
26.2. Positive second-order reconstruction	60
27. A sharp budget for the physical count-to-contact experiment	61
27.1. The A2 matrices and the scope of the consequence	63
References	63

1. INTRODUCTION

A statistic may preserve every parameter likelihood and yet fail to preserve the target of a decision. A compressor may be optimal for each task separately and yet fail when its update rule must be chosen before the task. Both distinctions become structural when information arrives sequentially and retained memory is bounded. We study them through the attainable risks of an executable causal experiment, with its target marks and information pattern specified together.

The principal object is the set of risk functions obtained by one common encoder and a family of task controllers. The encoder may use the current report, the current control and its retained label, but neither the task index nor the unknown parameter is an additional input. A task controller may depend on its task and retained label; it must work across all parameter rows of that task. Randomness is also part of the specification. Fresh private draws do not provide a free retained random variable, whereas an independent shared seed may correlate all stages of an execution. Expectations over that seed are taken before the worst decision row is selected.

1.1. The main results. Fix finite controls, decisions and a horizon T , but allow standard Borel report spaces and a compact parameter set. For a width vector $\mathbf{S} = (S_0, \dots, S_T)$ let $K_{\mathbf{S}}$ be the set of all private common-encoder designs and let $R(k) \in C(J)$ be their risk functions on a compact set J of task–parameter rows. Under the dominated loss-effect condition of Section 3, Theorem 3.3 gives

$$\min_{\lambda \in \mathcal{P}(K_{\mathbf{S}})} \max_{j \in J} \{\mathbb{E}_{\lambda} R_j - b_0(j)\} = \max_{w \in \mathcal{P}(J)} \left\{ \min_{k \in K_{\mathbf{S}}} \int R_j(k) w(dj) - \int b_0(j) w(dj) \right\}. \quad (1.1)$$

The private attainable risk set is compact, and its shared-seed barycenter set is compact and convex. Positive measures on J characterize the closed upper risk body. These conclusions concern all measurable finite-state updates, not just a compact family postulated in advance or encoders enumerated on a finite report tree.

The proof uses a product of weak star simplex spaces. Its main analytic step is a tensor-testing lemma: convergence of one bounded encoder factor in each report coordinate can be tested against an arbitrary L^1 path effect. Compact families of effects give uniform convergence of the risk functions. A separate representative lemma makes public mixtures of weak star classes jointly measurable and executable. Thus compactness is proved within the common-encoder information pattern. Corollary 3.5 treats countably many tasks by compactness and finite-subproblem duality; in that setting the dual is a supremum and no uniform finite seed is asserted.

The finite observation theorem, Theorem 3.7, compares the actual joint report–mark laws by causal cell conversion and within-cell reconstruction. Its error tends to zero uniformly in the memory width, task and controller. Finite experiments are therefore an approximating family for the dominated theory, not its sole domain. Corollary 3.6 gives the price of replacing a public mixture by a finite privately retained index.

Our second main result concerns the state carried by a marked experiment. For a finite instrument, two histories have the same continuation state when their conditional laws of the parameter, future reports and terminal mark agree under every future control word. Theorem 4.3 derives this state from decision risks: among all source-consuming randomized causal statistics, it has the least number of labels preserving every restart prediction risk. No prescribed posterior coordinates appear in that operational definition. The proof uses a strict conditional-variance identity for binary-event prediction. Theorem 4.5 then constructs parameter-independent causal report channels preserving the actual terminal mark. An explicit binary example shows why parameter sufficiency alone does not give this conclusion.

For rational finite instruments, Theorem 5.2 computes these classes by backward continuation spaces and forward exploration of distinct positive states. Its cost is polynomial in the input bit length and the number of output states, rather than in the entire expanded history tree. The

dimension of a linear representation is used to test equivalence; it is not confused with the number of positive retained states. A binary source with linear dimension two and $t + 1$ distinct states at time t illustrates the difference.

Two quantitative examples make the information pattern visible. Theorem 6.1 has three individually lossless two-label tasks, but common-encoder worst risk $1/2$ with private randomness and $2/9$ with shared randomness. All nine decision rows receive positive mass in an optimal dual witness. The delayed-query Gaussian theorem, Theorem 11.2, has pre-query error of order $S^{-2/d}$ while its query-aware checkpoint has order S^{-2} . Proposition 12.2 gives a finite acquisition and table implementation retaining the causal exponent.

1.2. Related work and the distinction of scope. Weak star policies and existence in stochastic teams are developed by Saldi [22]; strategic measures, public mixtures and failures of weak closure are treated by Yüksel and Saldi [23]. In particular, convexification by common randomness and compactness methods are not new principles here. Our result specifies a bounded retained register shared by several tasks, proves continuity for its coupled encoder factors, and obtains the risk-function body and its width-uniform marked approximations. Nonconvexity of the private set is consistent with the classical fact that randomization cannot improve a single fixed linear team objective: our strict example concerns a worst-row objective.

Causal source coding and common-information control provide the closest information-structure antecedents. See Witsenhausen [21], Yüksel [24], and Nayyar, Mahajan and Teneketzis [20]. An offline occupancy or belief recursion does not by itself constitute an online finite-memory implementation; we keep that distinction in the finite support recursion and the compiler. Comparison of filtered experiments, as in Norberg [15], is likewise part of the background of causal comparison, rather than a novelty claimed for the word “causal”.

Continuation equivalence is related to automata and predictive-state constructions [16, 13]. The linear span computation uses the weighted-automaton equivalence method presented by Kiefer [25]; neither rational elimination nor linear automaton minimization is asserted as new. The operational minimality here is for the declared marked continuation and for source-consuming statistics under all restart preparations. It is not minimality among independent simulators with unpriced retained memory. The target mark and this comparison category cannot be omitted from the statement.

The positive-model applications include established mechanisms: positive-matrix filter contraction, the logarithmic control-semigroup construction of Nisio [26], and the risk-sensitive information-state augmentation of Bäuerle and Rieder [27]. We prove the needed estimates here and specify how their actual marked effects enter the common-encoder theorem. No priority claim is made for these individual mechanisms.

1.3. Consequences and organization. The marked decision spectrum of Section 2 fixes the experiment, the task family, the information pattern and the resource budget before optimization. The compact theorem and the finite common-encoder theorem give exact variational descriptions in their stated classes. Quadratic targets give conditional variance; complete terminal loss families give statistical comparison; tree geometry gives weighted distortion estimates; and presenting an actual probability measure gives a Wasserstein law task. These are typed consequences of one risk construction, not an assertion that one scalar rate or one domination assumption describes every model. In particular, presenting a realized measure and sampling a single marginal probe are different tasks.

The following sections develop the compact theorem, marked quotient, compiler and common-task example, followed by the finite variational and saturation theory. The Gaussian temporal example and numerical acquisition result precede three model applications: positive hidden Markov observations and history reduction, nonlinear finite-state semigroups, and hidden-phase exponential moments. The remaining sections supply the predictive-tree, measure, terminal Gaussian and

count developments with full proofs. The separate complete-development edition also includes the earlier mathematical bodies and their historical introductions. None of their local hypotheses is removed when the results are used here.

2. THE MARKED DECISION SPECTRUM

We first fix the object whose finite presentations are to be compared. This requires specifying both the decision tasks and the information available to the decoder. These specifications cannot be recovered from a dimension or a cardinality alone.

2.1. Experiments, tasks, and signatures. An experiment is a standard Borel causal instrument, with an initial law and transition–observation kernels indexed by a parameter θ . Actions, when present, are chosen before the corresponding transition. The same observation-based policy is used at every value of θ . A *mark* d specifies a horizon, decision times, action spaces, a loss in $[0, 1]$, and a risk functional. The latter is either expectation under a declared prior or the supremum over a declared parameter set, taken after selecting one decoder. A minimax decoder does not know the unknown parameter. A mark may instead specify a fixed Bayesian prediction problem with an integrable squared loss; assertions involving total variation below apply only after a uniform loss bound is supplied. When a loss concerns a latent target, the marked experiment includes that target, and a simulation must preserve its joint law with the observations. Equality of report laws alone does not suffice for such a mark.

An encoder Q is an initialized causal transducer

$$I_t \sim q_t(\cdot \mid I_{t-1}, Y_t, A_{t-1}), \quad I_t \in \{1, \dots, S\}. \quad (2.1)$$

After an update, a decoder sees only (t, I_t) and any explicitly declared current side input. It does not see the discarded observation, an output transcript, old actions, or a data-dependent random seed. All persistent decoder state belongs to I_t . Fresh randomization is allowed. A public seed, when allowed, is independent of the experiment; a data-dependent position or modification of that seed is charged. A terminal reconstruction kernel may have a continuous output even when the physical statistic has a finite alphabet.

We use the resource signature

$$b = (S, A, L, P, p, W, \tau; R). \quad (2.2)$$

Here S counts persistent states; A counts physical terminal labels; L bounds raw symbols read per acquisition; P bounds fixed description bits; p is numerical precision; W is transient work space; τ is update time; and R records the permitted independent randomization. A coordinate ∞ means that no bound is asserted. For a precision constraint, the order is the order on permitted implementations, not the numerical order on an accuracy tolerance. The *cardinality signature* constrains only S (and A when explicitly stated). In particular, exact real constants in a fixed program are free only under that signature. All theorems below state which signature they use.

For an encoder admissible at b , let $r_E(Q, d)$ be the infimal value of the declared risk functional over its allowed decoders. Let $\beta_E(d)$ be the infimal risk using the full observation history under the same decision timing. The encoder is selected before the mark; its decoder may depend on the mark. For a set $\mathcal{D}$ of marks define

$$\mathcal{A}_E(b) = \{u \in [0, 1]^{\mathcal{D}} : \text{some admissible } Q \text{ has } r_E(Q, d) \leq u_d \text{ } (d \in \mathcal{D})\}, \quad (2.3)$$

$$\Theta_E(b; \mathcal{D}) = \inf_Q \sup_{d \in \mathcal{D}} \{r_E(Q, d) - \beta_E(d)\}. \quad (2.4)$$

Infima need not be attained. We use the displayed attainable sets, not their product-topology closure: for an infinite task family, taking that closure before a uniform task supremum can change the value. The common encoder and the order of quantifiers in (2.4) are part of the invariant. The collection of attainable sets, baselines, and resource signatures is the *marked decision spectrum*,

denoted $\Theta(E)$. This is the meaning of the symbol Θ in the present paper. It is a profile, not an additional physical parameter. Unbounded squared marks are treated separately without restricting the vectors to $[0, 1]$.

Example 2.1 (The encoder cannot follow the task). Let $X = (X_1, X_2)$ consist of two independent fair bits. A terminal statistic has two values. The two marks ask, respectively, for X_1 and X_2 , with zero-one loss. Each mark separately admits risk zero, but one common encoder has worst risk at least $1/4$. Indeed, for each fixed independent coding seed and each encoder label, the two optimal answers form a word in $\{0, 1\}^2$. Two reconstruction words have average Hamming distance at least $1/2$ from the four equally likely input words: two words may be exact, and each remaining word is at distance at least one. Randomization does not lower the average of the two risks. Hence their maximum is at least $1/4$. Conversely, with a public fair coin, reveal one of the two coordinates and guess the other. Thus the value is $1/4$ when that coin is permitted, whereas interchanging $\inf_Q$ and the task supremum gives zero. No such interchange is made in this paper.

2.2. Typed simulations and invariance. A certified simulation $S : E \Rightarrow F$ consists of initialized causal kernels, a monotone resource transformer ρ_S , and an error bound $\epsilon_S(T)$. The kernels do not know θ or the latent variables. For every target-side admissible policy and every parameter, the simulated marked transcript through horizon T has total variation distance at most $\epsilon_S(T)$ from its law in F . Moreover, composing the simulation with any target encoder of signature b produces a source encoder of signature at most $\rho_S(b)$. These are uniform requirements on one simulator, not permission to choose a different simulator for each task or policy. When marks have different horizons, use the corresponding error for each coordinate; a common ϵ below denotes their supremum.

Two certified simulators represent the same morphism when they have the same transformer and error certificate and induce the same marked execution laws under every admissible continuation. Internal register names are immaterial. We use standard Borel kernels modulo this execution equivalence. The exact morphisms have error zero. A K -state cardinality simulator has the transformer $S \mapsto KS$; its own output is not a second free memory register.

Theorem 2.2 (Spectrum transport and resource equivalence). *For bounded marks and signatures as in (2.2), certified simulations have identities and associative composition up to execution equivalence. If $S : E \Rightarrow F$ and $T : F \Rightarrow G$, the composite has transformer $\rho_S \circ \rho_T$ and error at most $\min\{1, \epsilon_S + \epsilon_T\}$. For every target-admissible Q there is a source-admissible Q' with*

$$r_E(Q', d) \leq r_F(Q, d) + \epsilon_S, \quad \beta_E(d) \leq \beta_F(d) + \epsilon_S. \quad (2.5)$$

If a reverse simulation $R : F \Rightarrow E$ is also supplied, then

$$\Theta_E(\rho_S(b); \mathcal{D}) \leq \Theta_F(b; \mathcal{D}) + \epsilon_S + \epsilon_R. \quad (2.6)$$

In particular, mutually exact finite-state simulations give equal ideal baselines and comparable spectra under the two budget dilations. Whenever a positive power exponent of either cardinality profile exists, the other has the same exponent. These assertions also hold coordinatewise for the attainable risk sets, before taking the task supremum.

Proof. Compose the kernels on the product of the two internal states. Fix one policy for G , regard its composition with T as a policy and continuation for F , and apply the certificate for S . Data processing followed by the certificate for T gives the sum of the two errors. The composition of transformers is forced by the same execution. Associativity follows from iterated kernel integration and the canonical reassociation of the triple product of registers; execution equivalence is preserved under composition because its quantifier includes all continuations. The identity kernel, identity transformer, and zero error give identities. This establishes a category for exact simulations and

the corresponding error calculus for approximate ones; it is not an assertion of a new general randomization theorem.

Run Q on the simulated reports and use its decoder. Every loss in $[0, 1]$ changes in expectation by at most the variation error. Taking infima over decoders proves the first inequality in (2.5); an arbitrarily good full-history policy proves the second. The reverse simulator implies $\beta_F(d) - \beta_E(d) \leq \epsilon_R$. Subtract the baselines, take the common task supremum, and only then take the infimum over Q . This proves (2.6). No compactness or attainment of an optimal encoder is used.

For exact cardinality equivalence with factors K_1, K_2 , the inequalities are $\Theta_E(K_1 M) \leq \Theta_F(M)$ and $\Theta_F(K_2 M) \leq \Theta_E(M)$. Monotonicity in M and $\log(K_i M)/\log M \rightarrow 1$ sandwich any existing limit of $-\log \Theta(M)/\log M$. The same reasoning applies to a single bounded squared prediction mark; multiplication by its loss bound changes no positive power exponent. $\square$

The theorem is an invariance statement for a *typed* equivalence. Ordinary statistical equivalence without a resource certificate need not preserve a causal implementation budget. Conversely, retaining every emitted label in an external transcript changes the signature and thus changes the spectrum. Products of independent observations and products of memories are different operations: the former concern the experiment, the latter the implementation.

Proposition 2.3 (Forgetting causal and resource structure). *Fix a finite horizon and restrict to uncontrolled instruments and parameter-decision marks. Sending an instrument to its full transcript experiment, and an exact causal simulator to its transcript kernel, defines a functor to parameterized statistical experiments with parameter-independent kernels modulo equality on every parameter law. It preserves identities and composition. It forgets the resource certificate; the resulting statistical equivalence alone does not supply an equality of causal-register spectra.*

Proof. Iterating a simulator's initialized kernels at a fixed input transcript gives a parameter-independent Markov kernel from source transcripts to target transcripts. Iterated integration identifies the transcript kernel of a composite with the composite of these kernels. Execution equivalence implies equality on every source parameter law, so this assignment is well defined on the stated classes. The identity gives the identity transcript kernel. A general inverse transcript kernel need not be causal, and a causal inverse may store a nonfinite state; neither datum is reconstructed by the forgetful operation. $\square$

2.3. The terminal decision-complete slice. In this subsection $E = \{P_\theta : \theta \in K\}$ has standard Borel observation space X , K is compact metric, and $\theta \mapsto P_\theta$ is continuous in total variation. A terminal encoder is any Markov kernel $Q : X \rightarrow [S]$. Only its physical alphabet $A \leq S$ is constrained; transient computation and the reconstruction kernel are unrestricted. Let $\mathcal{D}_{\text{all}}$ consist of all finitely supported priors on K and all finite-action losses in $[0, 1]$. For these families we use $\delta(F, E) = \inf_R \sup_{\theta \in K} \|RF_\theta - P_\theta\|_{\text{TV}}$, where R is a parameter-independent reconstruction kernel and $\|P - Q\|_{\text{TV}} = \sup_B |P(B) - Q(B)|$.

Theorem 2.4 (Decision completeness at a fixed common encoder). *For every such Q ,*

$$\sup_{d \in \mathcal{D}_{\text{all}}} \{r_E(Q, d) - \beta_E(d)\} = \delta(QE, E). \quad (2.7)$$

Consequently

$$\Theta_E(S; \mathcal{D}_{\text{all}}) = \inf_{Q: X \rightarrow [S]} \delta(QE, E). \quad (2.8)$$

This is the Blackwell–Le Cam randomization criterion in a compact setting, with the common finite encoder kept outside the decision supremum. In particular, no new priority is claimed for the unconstrained criterion [1, 14, 12].

Proof. A reconstruction R with variation error ϵ transfers each bounded decision rule, proving that the left side is at most the right side. First suppose that both K and X are finite. Put $F = QE$. Finite-dimensional convex minimax applied to the row-stochastic matrices R and the total-variation dual gives

$$\delta(F, E) = \sup_{\pi, h} \left[\sum_{\theta, x} \pi_{\theta} h_{\theta x} P_{\theta}(x) - \sup_R \sum_{\theta, i, x} \pi_{\theta} F_{\theta}(i) R(i, x) h_{\theta x} \right],$$

where π is a prior and $0 \leq h_{\theta x} \leq 1$. To justify convexity of the dual domain, write its variables as $u_{\theta x} = \pi_{\theta} h_{\theta x}$ with $0 \leq u_{\theta x} \leq \pi_{\theta}$ and $\sum \pi_{\theta} = 1$. Regard h as a reward with action set X . The first term is the reward of the identity decision in E , bounded above by its optimal reward; the second is the optimal reward in F . Thus every dual expression is at most one of the optimal reward differences. Changing reward to loss proves equality.

Here is the approximation needed for general X, K . A dense sequence (θ_j) and positive weights with sum one give $m = \sum_j w_j P_{\theta_j}$ dominating every P_{θ} ; domination follows by total-variation continuity. The set of densities $\{dP_{\theta}/dm : \theta \in K\}$ is compact in $L^1(m)$. Given $\eta > 0$, choose a finite L^1 net and approximate its elements by simple functions on a common finite partition $\mathcal{C}$ of X . Conditional expectation onto $\mathcal{C}$ is an L^1 contraction, so the cell reconstruction $R_{\mathcal{C}}(\cdot | C) = m(\cdot | C)$ satisfies

$$\sup_{\theta} \|R_{\mathcal{C}} P_{\theta} - P_{\theta}\|_{\text{TV}} \leq \eta$$

after choosing the net and approximations sufficiently accurately. Zero-mass cells are irrelevant. Hence $\delta(F, E) \leq \delta(F, CE) + \eta$. Now choose a finite parameter net on which both F and CE vary by at most η in total variation. The supremum reconstruction error on K differs from that on the net by at most 2η , uniformly in the reconstruction. Apply the finite result to this net. Its priors are allowed priors on K , and observing X is at least as informative as observing $\mathcal{C}(X)$, so the resulting decision differences for E dominate those for CE . It follows that the left side of (2.7) is at least $\delta(F, E) - 3\eta$. Let $\eta \downarrow 0$. Equation (2.8) now follows by taking the infimum over Q , with no exchange of quantifiers. $\square$

3. COMPACT DUALITY FOR DOMINATED CAUSAL EXPERIMENTS

We begin with a nonfinite experiment. The observation spaces in this section need not be locally compact, and the encoder kernels need not be continuous in the observation. The hypothesis used in place of such regularity is domination of the complete marked observation law.

3.1. The experiment and its information pattern. Fix a horizon T , finite control sets A_t , nonempty finite registers $I_t = [S_t]$, and a finite task set D . Task d has a finite decision set B_d . Let Y_t and W be standard Borel spaces, with probability reference measures μ_t and ν , and write $\mu = \bigotimes_{t=1}^T \mu_t$. The mark W is the actual terminal quantity on the source probability space; it is not a reconstructed sample with the same marginal distribution.

A causal experiment is specified by a consistent family of laws of $(Y_1, \dots, Y_T, W)$ under open-loop action words $a = (a_0, \dots, a_{T-1})$, indexed by a parameter $\theta \in \Theta$. Consistency means that the marginal law of $Y_{1:t}$ is independent of $a_t, \dots, a_{T-1}$, and that feedback laws are obtained by inserting their conditional action probabilities in the open-loop laws. Suppose that these laws have densities

$$p_{\theta, a}(y, w) \quad \text{with respect to } \mu(dy)\nu(dw). \quad (3.1)$$

This hypothesis is satisfied by dominated hidden Markov instruments, including continuous observation models. It excludes, for example, an unnoised repeated real observation supported on a diagonal of product Lebesgue measure. We do not replace domination by an unproved closure claim for arbitrary strategic measures.

The encoder is a single family

$$q_0 \in \mathcal{P}(I_0), \quad q_t(i' \mid i, a, y) \in \mathcal{P}(I_{t+1}), \quad 0 \leq t < T,$$

where $y \in Y_{t+1}$. It is common to every task and every parameter. The task controller and terminal rule are $g_{td}(a \mid i)$ and $v_d(b \mid i)$; neither depends on θ . The task is known to the controller, not supplied as an additional argument to q_t . The current control and observation are transient inputs. Between updates, the only private retained state is $i \in I_t$. Time and the prescribed tables are part of the fixed program. All private draws are fresh and conditionally independent given their arguments. A retained independent public seed, when allowed, is a separate resource. These conventions rule out both a hidden controller transcript and an uncharged retained random tape.

Let $J = D \times \Theta$ be compact metrizable, with D discrete and Θ compact metrizable. More generally the argument works for a compact row set with a continuous map into the finite task set. For $j = (d, \theta)$ and $b \in B_d$, let $0 \leq \ell_j(w, b) \leq 1$ be measurable. Our regularity condition is

$$f_{j,a,b}(y) := \int \ell_j(w, b) p_{\theta,a}(y, w) \nu(dw), \quad j \mapsto f_{j,a,b} \text{ continuous in } L^1(\mu) \quad (3.2)$$

on each task component, for every a, b . A continuous baseline $b_0 \in C(J)$ is fixed. A full-history minimax baseline for task d is constant on its parameter rows, and is one permitted choice; a parameterwise oracle is a different baseline and is not substituted for it.

Write $K_{\mathbf{S}}$ for the space of all such private designs $k = (q_0, q, g, v)$. The risk of a design is

$$R_j(k) = \sum_{i_0, \dots, i_T} \sum_{a, b} q_0(i_0) v_d(b \mid i_T) \prod_{t=0}^{T-1} g_{td}(a_t \mid i_t) \int f_{j,a,b}(y) \prod_{t=0}^{T-1} q_t(i_{t+1} \mid i_t, a_t, y_{t+1}) \mu(dy). \quad (3.3)$$

The sums over b are in B_d . This expression follows by first fixing the control and register paths in the causal law. In particular it does not optimize the parameter rows separately.

3.2. The topology and an executable realization. Each encoder row is a simplex-valued element of a finite product of $L^\infty(\mu_{t+1})$, endowed with its weak star topology. Positivity and the constraint that its coordinates sum to one are weak star closed: test them against nonnegative L^1 functions. Banach–Alaoglu therefore makes the row space compact. Since the probability space is standard Borel, L^1 is separable, and this compact space is metrizable. Give the finite controller, initialization and decoder simplices their usual topologies, and $K_{\mathbf{S}}$ the product topology.

Lemma 3.1 (Tensor testing). *Suppose $u_{tn}, u_t \in L^\infty(\mu_t)$ have absolute value at most one and $u_{tn} \rightarrow u_t$ weak star, for each t . Then*

$$\int F(y) \prod_t u_{tn}(y_t) \mu(dy) \longrightarrow \int F(y) \prod_t u_t(y_t) \mu(dy)$$

for every $F \in L^1(\mu)$. Convergence is uniform over every compact subset of $L^1(\mu)$.

Proof. For $F(y) = \prod_t F_t(y_t)$ with $F_t \in L^1(\mu_t)$ the integrals factor, and each factor converges by the definition of the topology. Finite sums of these tensors are dense in $L^1(\mu)$: rectangular simple functions generate the product sigma field, and a monotone class argument followed by truncation gives the density statement. An L^1 error of ϵ in the test function changes either integral by at most ϵ . This proves convergence for one F . A finite ϵ -net of a compact family proves the uniform assertion. $\square$

Lemma 3.2 (Jointly measurable representatives). *The encoder row space has a choice of representatives $(q, y) \mapsto \hat{q}(y)$ that is jointly Borel, is simplex valued at every y , and represents each equivalence class q almost everywhere. Consequently a Borel probability on $K_{\mathbf{S}}$ can be sampled as one independent public seed and implemented by measurable causal update rules.*

Proof. Choose increasing finite measurable partitions generating the report sigma field modulo completion. The coordinates of $\mathbb{E}[q \mid \mathcal{F}_n](y)$ are finite sums of functions of the form

$$\mathbf{1}_C(y) \frac{\int_C q d\mu_t}{\mu_t(C)}, \quad \mu_t(C) > 0.$$

On each zero-measure partition cell use one fixed probability vector. Their coefficients are weak star continuous in q , so these functions are jointly Borel. For fixed q , the bounded martingale convergence theorem gives coordinatewise convergence to a representative of q almost everywhere. On the jointly Borel set where all coordinates converge use their limit, and elsewhere use one fixed probability vector. Every finite-stage vector is in the simplex, so is the limit. There are only finitely many rows. A sampled k is thus executable by inverse transforms with fresh independent uniform variables. Domination ensures that exceptional report sets have zero probability for every actual action path and parameter. Compact metrizability also ensures that every probability on $K_{\mathbf{S}}$ is the law of a Borel function of a uniform seed. The sampled design is retained publicly; this last retention is not free private memory. $\square$

Theorem 3.3 (Nonfinite common-encoder duality). *Under (3.1)–(3.2), the map $R : K_{\mathbf{S}} \rightarrow C(J)$ is continuous in the uniform norm. The attainable shared-seed risk set and its upper extension are*

$$\mathcal{A}_{\mathbf{S}} = \left\{ \int R(k) \lambda(dk) : \lambda \in \mathcal{P}(K_{\mathbf{S}}) \right\}, \quad \mathcal{U}_{\mathbf{S}} = \mathcal{A}_{\mathbf{S}} + C(J)_+.$$

The first is compact and convex and the second is closed and convex. Set

$$h_{\mathbf{S}}(w) = \min_{k \in K_{\mathbf{S}}} \int_J R_j(k) w(dj), \quad w \in \mathcal{P}(J).$$

Then $u \in \mathcal{U}_{\mathbf{S}}$ if and only if $\int u dw \geq h_{\mathbf{S}}(w)$ for every $w \in \mathcal{P}(J)$, and

$$\min_{\lambda \in \mathcal{P}(K_{\mathbf{S}})} \max_{j \in J} \left\{ \int R_j(k) \lambda(dk) - b_0(j) \right\} = \max_{w \in \mathcal{P}(J)} \left\{ h_{\mathbf{S}}(w) - \int b_0 dw \right\}. \quad (3.4)$$

Both extrema are attained. The private minimax optimum is also attained, but its attainable risk set is $R(K_{\mathbf{S}})$, not its convex hull.

Proof. Compactness of $K_{\mathbf{S}}$ was proved above. Apply Lemma 3.1 to each term of (3.3). The effect functions form finitely many compact subsets of L^1 , by (3.2). Thus convergence of designs gives uniform convergence of all row risks. This proves the first assertion without enumerating deterministic encoders.

For completeness, the barycenter map is continuous from weak convergence of λ to norm convergence in $C(J)$. Approximate the continuous map R uniformly on compact $K_{\mathbf{S}}$ by a finite-valued continuous partition-of-unity interpolation. Integrals of the finitely many continuous coefficients converge; the uniform approximation controls the remainder. The image of compact $\mathcal{P}(K_{\mathbf{S}})$ is therefore compact. Lemma 3.2 identifies this image with executable public mixtures, rather than only a formal relaxed set. Convexity follows by mixing the public seed.

If $u_n = a_n + c_n \rightarrow u$, $a_n \in \mathcal{A}_{\mathbf{S}}$ and $c_n \geq 0$, pass to a convergent subsequence of a_n . Its limit a satisfies $u - a \geq 0$, proving closedness of $\mathcal{U}_{\mathbf{S}}$. A separating continuous functional for a point outside this closed convex upper set must be nonnegative: otherwise translating an attainable risk in a nonnegative direction contradicts its lower separating bound. By the Riesz representation theorem this functional is integration against a finite positive measure, which may be normalized to a probability. Minimizing it over the upper set gives $h_{\mathbf{S}}$. This proves the support characterization.

The bilinear payoff

$$F(\lambda, w) = \int_J \int_{K_{\mathbf{S}}} R_j(k) \lambda(dk) w(dj) - \int_J b_0(j) w(dj)$$

is continuous separately on the two compact convex probability spaces. The classical convex minimax theorem gives equality of the two extrema. For fixed w , a linear functional on probabilities has minimum at a Dirac probability, which gives the inner minimum in (3.4). The left objective is continuous and the right objective, as an infimum of continuous functions of w , is upper semicontinuous. Compactness gives attainment. Finally the private objective is continuous on compact $K_{\mathbf{S}}$. No convexification was used for that assertion. $\square$

The use of policy topologies and static reduction belongs to stochastic team theory; see Saldi [22], especially the weak star construction and continuity/existence results, and Yüksel–Saldi [23], who explain why conditional independence is not generally closed under weak convergence. Here the product domination and finite action-path expansion supply the needed continuity, while the same encoder coordinates remain coupled across all tasks and unknown-parameter rows. Convex separation and minimax are classical tools, not additional novelty claims.

Corollary 3.4 (Effect formulation with arbitrary marks). *Theorem 3.3 still holds when there is no common reference measure for W , provided each open-loop report marginal is dominated by μ and the signed effect measures*

$$C \mapsto \int_{C \times W} \ell_j(w, b) P_{\theta, a}(dy, dw)$$

have $L^1(\mu)$ densities satisfying (3.2). The actual joint marked laws, not arbitrarily selected couplings, are used to define these effects.

Proof. Boundedness of the loss and report domination give absolute continuity of each effect measure. Disintegration of the actual marked law, or simply integration first in that measure, gives (3.3). All compactness and separation arguments in the theorem use only these effects. Report domination still makes the jointly measurable representatives executable. No common measure on W is needed in any of these steps. By contrast, the marked total-variation approximation theorem below retains its stronger joint-density hypothesis. $\square$

Corollary 3.5 (Countably many tasks). *Suppose that the task-row set is countable, every row has the dominated representation above, and every task has finitely many controller and decoder entries. No uniform continuity across rows is required. For a bounded baseline sequence, the shared optimum is attained and*

$$\min_{\lambda} \sup_j \{\mathbb{E}_{\lambda} R_j - b_0(j)\} = \sup_{w \in \mathcal{P}_f(J)} \left\{ \min_k \sum_j w_j R_j(k) - \sum_j w_j b_0(j) \right\},$$

where $\mathcal{P}_f(J)$ denotes the finitely supported probabilities. The right side is a supremum, not an asserted maximum.

Proof. The countable product of the task-controller and decoder simplices with the common encoder space is compact metrizable. Each risk is continuous by the tensor lemma; the representative construction is unchanged. On $\mathcal{P}(K_{\mathbf{S}})$, the objective is a supremum of continuous functions, hence lower semicontinuous and attains its minimum. If every finite subset of row constraints $\mathbb{E} R_j - b_0(j) \leq r$ is feasible, compactness and the finite intersection property make all constraints feasible simultaneously. Finite-row convex minimax therefore gives the displayed identity, by taking r strictly above its right side and then decreasing r to the infimum. This argument makes no uniform finite-seed assertion. $\square$

Corollary 3.6 (The price of a finite mixture). *In Theorem 3.3, put*

$$\omega(\delta) = \sup_{k, d_J(j, j') \leq \delta} |R_j(k) - R_{j'}(k)|.$$

Then $\omega(\delta) \rightarrow 0$. If J has a δ -net of size N , every attainable shared risk can be approximated within $2\omega(\delta)$ in uniform norm by a mixture of at most $N + 1$ private designs. Such a mixture has a private implementation with registers $(N + 1)S_t$, including initialization. If its weights are approximated by a probability vector with denominator 2^p , an additional error at most $(N + 1)2^{-p}$ suffices.

Proof. The compact subset $R(K_S)$ of $C(J)$ is equicontinuous. Carathéodory's theorem applied to its evaluations on the net gives $N + 1$ designs whose mixture agrees there with the prescribed risk. Move an arbitrary row to the net; each of the two risks changes by at most $\omega(\delta)$. Store the chosen design index with the register label for the private implementation. Approximate weights by rounding down the first $L - 1$ coordinates and putting the remainder in the last, where $L \leq N + 1$. Their total variation distance is at most $L2^{-p}$, which bounds the change of every $[0, 1]$ -valued risk. This approximation can be sampled with p fair bits; exact non-dyadic weights do not have an asserted fixed worst-case fair-bit cost. $\square$

3.3. Finite observation approximations, uniformly in memory. Let $\mathcal{F}_{tn}$ be increasing finite partitions generating Y_t , and retain the actual mark W . The finite-report experiment E_n reports only the cell of each observation. Its reverse report channel samples independently from μ_t conditioned on that cell. Cells of zero reference measure are assigned arbitrary outputs and have zero probability in the experiment. Write

$$p_{\theta,a}^{(n)} = \mathbb{E}_{\mu \otimes \nu}[p_{\theta,a} \mid \bigotimes_t \mathcal{F}_{tn} \otimes \mathcal{B}(W)], \quad \delta_n = \frac{1}{2} \sum_a \sup_{\theta} \|p_{\theta,a} - p_{\theta,a}^{(n)}\|_1.$$

Theorem 3.7 (Width-uniform approximation). *If each family $\{p_{\theta,a} : \theta \in \Theta\}$ is compact in $L^1(\mu \otimes \nu)$, then $\delta_n \rightarrow 0$. For every width vector, private or shared randomness, and every task family with losses in $[0, 1]$, the raw minimax risks of E and E_n differ by at most δ_n . The same comparison of excess risks costs at most $2\delta_n$ when the full-history baselines are recomputed. The simulations neither add persistent states nor reveal the mark.*

Proof. Conditional expectations on the generating partitions converge in L^1 for each density. Their norms are at most one as operators on L^1 , so a finite-net argument gives uniform convergence on each compact family. There are finitely many action words.

Reporting the cell is an exact causal channel from E to E_n . After the reverse channel, the open-loop joint density of synthetic reports and the actual W is $p_{\theta,a}^{(n)}$. Under an arbitrary adaptive policy and encoder, expand the joint law by its action and register paths. The conditional rule multiplying each density is between zero and one. Summing the absolute differences over action words, and integrating over the register randomization, bounds the total variation of the marked output laws by δ_n . This bound is uniform over policies, tasks, seeds and widths. Composition of any code with either report channel needs only the original register: cell conversion or within-cell sampling is a current-report operation. Thus every code transfers with the stated risk error in the reverse direction, and with zero error in the forward direction. Taking minima and worst rows proves the raw comparison. The same argument without a width bound compares full-history baselines, and the triangle inequality gives the excess-risk bound. $\square$

4. THE MINIMAL MARKED CONTINUATION QUOTIENT

We now address exact state reduction. A statistic sufficient for the parameter need not be sufficient for a latent decision target. The correct terminal datum is the joint conditional law of the parameter and the designated mark. We give both an intrinsic operational characterization and a finite construction.

Consider a finite hidden instrument with initial law ν_{θ} , transition/report probabilities $K_{\theta,t}(z', y \mid z, a)$, and a terminal mark channel $L_{\theta}(w \mid z_T)$ on a finite set W . Here K uses the new state z' and

the report jointly; no old-state observation channel is silently substituted. A finite collection of path targets can be included by adjoining their necessary finite memory to z . This augmentation, when used, is part of the input size. Fix a reference prior $\pi(\theta) > 0$. For every supported controlled history $h = (a_0, y_1, \dots, a_{t-1}, y_t)$, let $\alpha_t(h)$ be the posterior row on (Θ, Z_t) under the reference prior and the indicated past actions. Control probabilities cancel in Bayes' formula; histories of zero reference probability are assigned an arbitrary existing label. No extra retained label is needed for events that have probability zero under every parameter.

Definition 4.1 (Continuation experiment). For each future open-loop action word $v = a_t \cdots a_{T-1}$, let $\mathcal{C}_t(h; v)$ be the conditional law of $(\Theta, Y_{t+1:T}, W)$ given h , with future actions v . Two supported histories are equivalent, written $h \equiv_t h'$, if $\mathcal{C}_t(h; v) = \mathcal{C}_t(h'; v)$ for every such v . Denote their class by $c_t(h)$ and the number of classes by K_t .

This definition compares experiments rather than lists of selected coordinates. At T it reduces to equality of $\mathcal{L}(\Theta, W \mid h)$, not only equality of $\mathcal{L}(\Theta \mid h)$. Since the parameter set is finite and the prior is strictly positive, changing the reference prior applies an invertible positive reweighting to every continuation law. The classes are consequently independent of this choice.

Lemma 4.2 (Controlled continuation). *The classes satisfy*

$$h \equiv_t h' \implies hay \equiv_{t+1} h'ay$$

for every a, y of positive conditional probability. The probability of y is itself constant on a class. Equality of the open-loop continuation laws implies equality under every randomized feedback policy using the subsequent report history.

Proof. The marginal of the first future report is a marginal of any continuation law beginning with a . It is the same for h, h' . Condition the joint law of $(\Theta, Y_{t+1:T}, W)$ on that report being y . Its positive common denominator is the first-report probability, and its remaining numerator is the continuation law from hay or $h'ay$. Varying the remaining action word gives the successor assertion. For a feedback policy, multiply each open-loop path probability by the product of its conditional control probabilities along that path and sum over action words. These factors depend only on the displayed future reports and on independent randomization. Equality is preserved. Null successors have zero probability under every parameter of positive reference prior. $\square$

4.1. An operational necessity theorem. The category in which a minimality claim is made matters. At a checkpoint t , a *restart preparation* chooses a supported past history H with an arbitrary law η , and then runs its conditional continuation experiment. The candidate statistic is obtained by consuming that same history through a channel $Q_t(dm \mid h)$. It does not resample a new history or replace the actual target W . A restart prediction task fixes a future action word v and an event $A \subseteq \Theta \times Y_{t+1:T} \times W$. Immediately at the checkpoint, before the continuation is observed, the decision maker predicts the indicator of A with an action in $[0, 1]$ and squared loss. The full-history optimal prediction is $p_A^v(h) = \mathcal{C}_t(h; v)(A)$.

Call a source-consuming causal statistic *restart-decision complete* when it preserves the full-history Bayes risk in every one of these tasks, for every preparation η and every checkpoint. The definition requires equality of operational risks. It does not require any posterior coordinate to be explicitly stored.

Theorem 4.3 (Marked operational minimality). *A privately randomized finite-valued statistic M_t is restart-decision complete at t if and only if $c_t(H)$ is recoverable as a function of M_t on every supported history. In particular any complete randomized causal machine has at least K_t retained states at checkpoint t . The deterministic quotient (c_t) is simultaneously complete at all checkpoints, has exactly these widths, and is minimal in this category. For a public seed U , the corresponding*

criterion is recovery of $c_t(H)$ from (U, M_t) , not from M_t alone. The same lower bound on the number of private register labels holds in either signature.

Proof. The optimal Brier risks before and after compression differ by

$$\mathbb{E}_\eta \left[(p_A^v(H) - \mathbb{E}_\eta[p_A^v(H) \mid M_t])^2 \right]. \quad (4.1)$$

Indeed $\mathbb{E}[\mathbf{1}_A \mid H, M_t] = p_A^v(H)$, since the statistic's randomization is independent of the actual continuation given H . Orthogonal projection in L^2 proves the identity. There are only finitely many histories and cylinder events in the present finite experiment. Choose a preparation with positive mass on every supported history. Vanishing of (4.1) for every v, A forces $p_A^v(h) = p_A^v(h')$ whenever both histories give positive probability to a common value m . Thus $h \equiv_t h'$. Each retained value has support in one equivalence class, so the class is a function of that value and there must be at least K_t values.

Conversely, if the class is recoverable, every $p_A^v(H)$ is measurable from M_t , and (4.1) is zero for every preparation. Lemma 4.2 implements the classes by deterministic causal updates. It also proves equality for feedback continuations. For an independent public seed U , apply the same argument to (U, M_t) ; if the declared width bounds only M_t , a free retained U is a genuinely different signature. In an exact complete public scheme with finite widths, zero nonnegative excess also holds for almost every fixed seed, so the lower bound $S_t \geq K_t$ remains valid. One cannot use a seed to merge distinguishable histories while preserving all these zero-loss differences. $\square$

Proposition 4.4 (A quantitative separation). *Suppose a continuation event has probabilities differing by Δ at h, h' . For the equal two-history preparation, any randomized statistic has Brier excess at least*

$$\frac{\Delta^2}{4} (1 - \text{TV}(Q_t(\cdot \mid h), Q_t(\cdot \mid h'))).$$

Proof. Write $x_m = Q_t(m \mid h)$ and $y_m = Q_t(m \mid h')$. Direct evaluation of the conditional variance in (4.1) gives $\frac{\Delta^2}{2} \sum_m x_m y_m / (x_m + y_m)$, with zero terms omitted. Since $xy/(x+y) \geq \min(x, y)/2$ and $\sum_m \min(x_m, y_m) = 1 - \text{TV}(x, y)$, the result follows. $\square$

This is not a minimality assertion for an arbitrary independent simulator with unpriced memory. Nor is it an assertion that a chosen small list of losses separates every latent feature. The restart family is decision complete for the stated marked continuation experiment; its loss tests force the necessary state rather than assuming a posterior coordinate representation in the definition. The continuation and linear-representation aspects have classical antecedents in automata and predictive-state theory [16, 13, 25]; the target mark and the source-consuming operational category are essential here.

4.2. Two exact marked causal channels. Let $k_t^0(y \mid c, a)$ be the reference next-report probability and $F_t(c, a, y)$ the successor class. Define

$$H_t^0(c' \mid c, a) = \sum_{y: F_t(c, a, y) = c'} k_t^0(y \mid c, a), \quad \lambda_\theta(c) = \mathbb{P}_\pi(\Theta = \theta \mid c).$$

On the support of parameter θ , Bayes' formula gives

$$k_{\theta, t}(y \mid c, a) = k_t^0(y \mid c, a) \frac{\lambda_\theta(F_t(c, a, y))}{\lambda_\theta(c)}. \quad (4.2)$$

The marked terminal channel is $G_\theta(w \mid c_T) = \mathbb{P}_\pi(W = w \mid \Theta = \theta, c_T)$. It is part of the quotient; it is not discarded.

Theorem 4.5 (Exact marked reconstruction). *The original instrument and the class instrument admit causal channels in both directions preserving the joint law of the reported transcript and the actual terminal mark, under every parameter and admissible feedback policy. The reverse report kernel is*

$$L_t(y \mid c, a, c') = \frac{\mathbf{1}_{\{F_t(c, a, y) = c'\}} k_t^0(y \mid c, a)}{H_t^0(c' \mid c, a)} \quad (4.3)$$

when the denominator is positive. It is independent of θ . The class instrument uses transitions $H_{\theta, t}(c' \mid c, a) = H_t^0(c' \mid c, a) \lambda_\theta(c') / \lambda_\theta(c)$ and terminal channel G_θ . Thus the channels preserve every bounded loss on the declared marked experiment, not just parameter likelihoods.

Proof. The forward channel consumes each actual report and updates c by F_t ; its current class is a finite register. For the reverse channel, first observe the new class c' , then draw y by (4.3). Multiplying this kernel by $H_{\theta, t}$ gives exactly (4.2). The terminal conditional distribution of W given the full original history and θ is $G_\theta(\cdot \mid c_T)$ by the definition of the terminal class. Consequently the product of transition factors, report reconstruction factors and this terminal mark factor is the original joint density of $(Y_{1:T}, W)$, for each open-loop action word. The feedback factors can be inserted before summing action words, as in Lemma 4.2. This proves the claimed joint, parameter-uniform identity. Cases with zero λ_θ or zero transition probability lie outside that parameter's support and may be assigned arbitrary kernels. The positive reference prior makes these conventions consistent across parameters.

A reverse transducer needs the previous class, the new class and the current action while forming its report. Between updates it retains only the previous class. Composing it with an S_t -state machine is therefore possible with at most $K_t S_t$ retained labels at time t ; the new input is transient. These resource costs are separate from the bare exact-simulation identity. $\square$

Example 4.6 (A parameter-sufficient statistic that loses its mark). Let $\mathbb{P}(\Theta = 0) = 9/10$, let Y be an independent fair bit, and put $W = Y \oplus \Theta$. The parameter posterior is constant in Y . A terminal quotient that keeps only that posterior has one class. Yet the Bayes error in guessing the actual W is $1/10$ with Y and $1/2$ with that class alone. The marked terminal laws $\mathcal{L}(\Theta, W \mid Y = 0)$ and $\mathcal{L}(\Theta, W \mid Y = 1)$ differ, so the marked quotient has two classes. Its reverse channel preserves the relation $W = Y \oplus \Theta$, not merely the marginal fair-bit law of Y . This example explains why an exact reconstruction theorem for reports alone cannot be used for arbitrary latent-target losses.

5. SYNTHESIS FROM CONTINUATION SPACES

The continuation quotient can be computed without first expanding all controlled histories. Its linear dimension and its number of positive states are different quantities; the algorithm uses the former to test equivalence and the latter to bound its output.

Assume the finite marked instrument of Section 4 has rational data. At time t write $n_t = |\Theta| |Z_t|$ and $n = \max_t n_t$. Its block-diagonal transition matrices are

$$M_{t, a, y}((\theta, z), (\theta', z')) = \mathbf{1}_{\{\theta = \theta'\}} K_{\theta, t}(z', y \mid z, a).$$

The terminal columns are $e_{\theta, w}(\theta', z) = \mathbf{1}_{\{\theta = \theta'\}} L_\theta(w \mid z)$. Define rational continuation spaces backwards:

$$V_T = \text{span}\{e_{\theta, w} : \theta, w\}, \quad V_t = \text{span}\{M_{t, a, y} v : a, y, v \in V_{t+1}\}. \quad (5.1)$$

They have dimensions at most n_t , and contain the all-ones column. For the latter assertion sum the terminal columns, and then use $\sum_y M_{t, a, y} \mathbf{1} = \mathbf{1}$ for any fixed a .

Lemma 5.1 (Finite linear test for a marked class). *Two normalized posterior rows $\alpha_t(h), \alpha_t(h')$ determine the same marked continuation class if and only if they agree on V_t . It suffices to compare their evaluations on any rational basis B_t of that space.*

Proof. The probability of a specified future report word and terminal pair (θ, w) under an action word is the row $\alpha_t(h)$ applied to the column

$$M_{t,a_t,y_{t+1}} \cdots M_{T-1,a_{T-1},y_T} e_{\theta,w}.$$

These columns span exactly V_t , by induction in (5.1). Equality of all these probabilities is precisely Definition 4.1. Equality on their span is equivalent to equality on a basis. $\square$

Theorem 5.2 (Output-sensitive exact synthesis). *Let B be the total binary length of the rational source description, including any finite target augmentation and the positive reference prior. The marked quotient, one representative posterior per class, and its deterministic transition tables can be constructed in time polynomial in*

$$B + T + n + \sum_t |A_t| |Y_{t+1}| + \sum_{t=0}^T K_t.$$

No factor equal to the number of expanded histories is required. The stored transition table has $\sum_{t < T} K_t |A_t| |Y_{t+1}|$ entries, including null entries. A bit-serial implementation retains $\lceil \log_2 K_t \rceil$ state bits, with finite transient workspace and a finite table-lookup running-time bound determined by this table. This is an exact-state synthesis theorem, not a polynomial-time algorithm for optimal lossy common-encoder design.

Proof. Compute the spaces (5.1) backwards by rational elimination. At each stage there are at most $|A_t| |Y_{t+1}| n$ candidate columns. Retain independent columns, rather than repeatedly replacing the basis by arbitrary normalized linear combinations. Each retained column is then a product of at most T input matrices with a terminal column. A common denominator for all input probabilities bounds the binary length of its entries by a polynomial in B, T, n . Fraction-free elimination, or determinant bounds for rational elimination, therefore takes polynomial bit time.

Start at the initial posterior $\alpha_0 = (\pi(\theta)\nu_\theta(z))$. At stage t maintain one posterior representative α for each class found. For every a, y , form

$$r = \alpha M_{t,a,y} \mathbf{1}.$$

If $r = 0$, record the null transition. Otherwise form $\alpha' = \alpha M_{t,a,y}/r$, evaluate $\alpha' B_{t+1}$, and compare this vector with the stored keys at $t + 1$. Reuse its label if equal; otherwise store a new label and the representative α' . By Lemma 5.1, the key is the marked continuation class. The successor result ensures that all histories represented by α have this same transition, even though their full hidden posteriors can differ. Thus retaining one representative loses no reachable class. Induction over t proves that precisely the K_t classes are generated.

There are only the displayed number of transitions to examine. Every representative is the normalized posterior of some history of length at most T . The numerator and denominator are finite sums of products of rational input probabilities. Their binary lengths are polynomial in B, T, n , even if their magnitudes are very small. Exact comparison of keys, either by pairwise search or rational sorting, and all matrix operations consequently take polynomial bit time in the stated input and output parameters.

The executor needs only the class label and a fixed time-indexed table. A sequential scan of that finite table, with binary counters and the current action/report held transiently, is one explicit implementation. It has program length bounded by the table length times the binary length of an entry, and polynomial transient workspace and scan time in the table description. A more efficient random-access convention may improve this bound, but is not needed for finite effectivity. Report acquisition remains a separate interface cost. $\square$

The invariant-space computation is the classical weighted-automaton equivalence method; see Kiefer [25], Proposition 2.1 and Theorem 2.3, and Tzeng [28]. What is being synthesized here is a

positive, normalized, marked causal quotient, not a minimal signed linear representation. Linear dimension alone would not give its persistent-state cost.

Example 5.3 (Small linear rank and growing positive width). Let $\Theta \in \{0, 1\}$ have positive prior and, conditional on it, let the reports be independent Bernoulli variables with parameters $p \neq q$ in $(0, 1)$. Retain Θ as a terminal mark. The hidden linear dimension is two at every time. After t reports, the posterior odds for a history with k successes are the initial odds multiplied by

$$(q/p)^k ((1-q)/(1-p))^{t-k}.$$

These $t + 1$ values are distinct since $p \neq q$. The parameter posterior is a marginal of the marked continuation law, and conditional future laws depend only on this posterior. Hence $K_t = t + 1$, whereas the expanded report tree has 2^t leaves. The algorithm constructs the quotient in polynomial time in T , without mistaking rank two for width two.

6. A COMMON-ENCODER OBSTRUCTION WITH AN INTERIOR DUAL WITNESS

The following example puts the task-coupling and randomization issues in one experiment. It is not a difference between distinct source families or between incompatible loss normalizations.

The source reveals $X = \theta \in \{1, 2, 3\}$ without noise. Task $d \in \{1, 2, 3\}$ asks whether $\theta = d$, with binary output and zero-one loss. The encoder has two labels, is not told d , and is common to the nine rows (d, θ) . The task is supplied to the decoder. All full-history task baselines are zero. A task-specific two-label encoder would answer its task exactly.

Theorem 6.1 (Three tasks, two labels). *For the common encoder, the best deterministic worst-row risk is one, the best private-randomized risk is $1/2$, and the best shared-seed risk is $2/9$. The uniform probability on all nine rows is a dual optimizer for the shared problem. Every row receives positive dual weight, and all nine risks of the displayed optimal shared scheme are equal.*

Proof. A deterministic two-label encoder merges at least two source symbols. For the membership task of either symbol, the decoder makes an error on at least one of them. Thus its worst-row risk is one. Moreover, under the uniform distribution on the nine rows every deterministic design makes at least two errors: each of the two membership tasks for a merged pair has an error. The uniform risk is at least $2/9$. Freezing all encoder and decoder randomization gives this same lower bound for every shared mixture of complete designs.

For the matching scheme choose a singleton k uniformly from the three symbols using the public seed. Encode whether $X = k$. If $d = k$, answer the task exactly. If $d \neq k$, answer zero on the singleton label; on the pair label answer one with probability $2/3$. For a positive row $d = \theta$, error occurs when $k \neq d$ and the decoder answers zero, with probability $(2/3)(1/3) = 2/9$. For a negative row $d \neq \theta$, error occurs only when k is the third symbol and the decoder answers one, with probability $(1/3)(2/3) = 2/9$. This establishes optimality and the asserted strictly positive dual witness. The additional decoder coin may be fresh private randomness; it may also be frozen into the public mixture of deterministic designs.

For a private encoder write $p_x = \mathbb{P}(M = 1 \mid X = x)$. For each task d , the probability of answering one is an affine function $F_d(p_x)$, since it is obtained by applying two decoder probabilities to the two encoder labels. One p_d lies between the other two values, including the possibility of equality. An affine function cannot be greater than $1/2$ at this middle value and less than $1/2$ at both end values. Hence some row of task d has error at least $1/2$. Ignoring the message and making a fair binary guess attains $1/2$. A common retained coin shared with the encoder would invalidate this affine argument, which is exactly why it is a different resource. $\square$

For a single task, independent randomization cannot improve a scalar expected-loss minimum, as in the strategic-measure results of [23]. The worst-row criterion here is a maximum of linear risks,

not one linear risk. Convexification can therefore strictly improve it. There is no contradiction with that classical irrelevant-randomization principle.

7. CONTROLLED COMMON ENCODERS AND THE MINIMAX RISK BODY

The causal constraint is a constraint on the encoder, not on each task separately. We now give a variational characterization with this constraint inside the feasible set. The distinction between a shared random seed and private fresh randomization is essential to the statement.

7.1. Finite instruments and the information pattern. Fix a horizon T , finite parameter set K , finite hidden spaces Z_t , report alphabets Y_t , and a finite control alphabet A . The instrument is specified by probability vectors ν_θ and stochastic matrices

$$K_{\theta,t}(z', y \mid z, a), \quad 0 \leq t < T.$$

The hidden variable is not available to either the encoder or the controller. A finite collection D of tasks has finite terminal action sets B_d and losses $\ell_{d\theta}(z, b) \in [0, 1]$. Any bounded loss on a finite trajectory, including losses at several decision times, is of this form after adjoining that trajectory to the hidden state. This enlargement is a description of the experiment, not memory supplied to its controller.

The encoder has widths $\mathbf{S} = (S_0, \dots, S_T)$, with $S_0 \geq 1$. A parameter-independent initialization law chooses the initial label before the first control; this label is visible to the controller and is charged in S_0 . Deterministic initialization can be labelled 1 by relabelling states. Before transition t , the task- d controller chooses a from its current label i . The encoder then observes (a, y) and updates the label. At the last time, a task- d terminal rule chooses b . The encoder's update table is the *same* for all tasks and parameters; the controller's tables may depend on d but not on θ . The current control is an input to the ensuing update, not a retained action history. Task-dependent restrictions on legal controls may be imposed on the controller tables. The controller has no additional persistent register.

For a minimax task retain one row $j = (d, \theta)$ per parameter. For a Bayesian task retain one row j with its specified prior κ_j on K . More generally a row has a task $d(j)$ and a probability vector κ_j ; point masses give the minimax rows. Let J be this finite row set, $M = |J|$, and let β_d be the full-history optimal risk for the *whole* task d . In particular, the baseline of a minimax task is not a separate parameterwise optimum. Write $b_j = \beta_{d(j)}$.

A *deterministic design* ω consists of one family of updates $f_t(i, a, y) \in [S_{t+1}]$, together with all controller tables $a_t^d(i) \in A$ and terminal tables $b^d(i) \in B_d$. Let $\Omega_{\mathbf{S}}$ be the finite set of these designs. Its raw risk vector is denoted $R(\omega) = (R_j(\omega))_{j \in J}$, where

$$R_j(\omega) = \sum_{\theta} \kappa_j(\theta) \mathbb{E}_{\theta}^{\omega, d(j)} \ell_{d(j)\theta}(Z_T, b^{d(j)}(I_T)). \quad (7.1)$$

The expectation uses the actual closed-loop law. In particular, controls are not optimized separately inside hidden-parameter components.

There are two randomization conventions. In the *shared-seed* convention an independent variable is available to the encoder and the task controller throughout the run. It is chosen independently of the task and parameter; the risk averages its realization before taking the supremum over tasks or parameters. The adversary does not select a new parameter after seeing that realization. In the *private* convention there is no such retained shared variable. Each declared kernel uses fresh randomness, with conditional independence given its allowed inputs. Any seed remembered across updates is charged in the label. We write Θ_{sh} and Θ_{pr} for these two specified slices of (2.4). The term shared seed here has the precise meaning of the public independent seed permitted in Section 2; it never contains data from the experiment.

7.2. The variational theorem. Define the upper risk body and its lower support function by

$$\mathcal{U}_E(\mathbf{S}; J) = \text{co}\{R(\omega) : \omega \in \Omega_{\mathbf{S}}\} + \mathbb{R}_+^J, \quad (7.2)$$

$$h_E(\mathbf{S}; w) = \min_{\omega \in \Omega_{\mathbf{S}}} w \cdot R(\omega), \quad w \in \Delta_J. \quad (7.3)$$

Here Δ_J is the probability simplex. The body is defined from controlled execution risks, not from a presumed predictive quotient.

Theorem 7.1 (Controlled common-encoder minimax theorem). *For the finite instrument and information pattern just specified:*

- (1) *The shared-seed attainable upper risk set is exactly $\mathcal{U}_E(\mathbf{S}; J)$. The common-encoder excess satisfies*

$$\Theta_{\text{sh}}(\mathbf{S}; D) = \min_{\lambda \in \Delta(\Omega_{\mathbf{S}})} \max_{j \in J} \left\{ \sum_{\omega} \lambda_{\omega} R_j(\omega) - b_j \right\} \quad (7.4)$$

$$= \max_{w \in \Delta_J} \{h_E(\mathbf{S}; w) - w \cdot b\}. \quad (7.5)$$

An optimum uses at most $M + 1$ deterministic designs. Thus the encoder is chosen once, before the task, in both formulae.

- (2) *The functions $h_E(\mathbf{S}; \cdot)$ determine the whole upper risk body. For two matched task/row systems,*

$$\mathcal{U}_F(\mathbf{S}'; J) \subseteq \mathcal{U}_E(\mathbf{S}; J) \iff h_E(\mathbf{S}; w) \leq h_F(\mathbf{S}'; w) \quad (w \in \Delta_J).$$

In particular equality of all these support functions is equivalent to equality of the attainable upper bodies. Relabelling histories, hidden states or machine labels changes neither assertion.

- (3) *For constant width S , cardinality alone being charged,*

$$\Theta_{\text{pr}}((M + 1)S; D) \leq \Theta_{\text{sh}}(S; D) \leq \Theta_{\text{pr}}(S; D). \quad (7.6)$$

The first inequality is implemented by retaining the design index in the actual register, including the permitted randomized initialization. If a different signature forbids randomized initialization, this inequality refers to its explicitly initialization-enlarged signature. For a fixed finite row family, any existing positive cardinality exponent is consequently the same under these two randomization conventions.

Proof. First include the decoder and controller tables in the design. For one fixed common encoder their choices for distinct tasks are independent optimization variables, whereas the same choice for one task must work for all its parameter rows. Compactness of the finite products of probability simplexes implies attainment. Hence minimizing jointly over the common encoder and all its task rules gives exactly the original order $\inf_Q \max_d (r_E(Q, d) - \beta_d)$; it does not give a decoder indexed by the unknown parameter.

Realize every update and controller kernel by a uniform variable at its specified time, independent of the instrument. For a shared-seed scheme, condition on the entire shared seed and these variables. One obtains a deterministic common update family and one deterministic controller for each task. Although different tasks then generate different observation laws, the frozen maps are defined on *all* their possible inputs. They therefore form one element of $\Omega_{\mathbf{S}}$. Its conditional risk is computed under the corresponding controlled instrument, not under an incorrectly frozen uncontrolled history law. Averaging gives a convex combination of the finite vectors $R(\omega)$.

Conversely sample a design index with probabilities λ , expose that independent index to encoder and controller, and execute its tables. The encoder law is independent of which task will be requested. Its risk vector is exactly $\sum \lambda_{\omega} R(\omega)$. Taking coordinatewise upper bounds proves (7.2) and (7.4). Finite-dimensional convex minimax gives (7.5). Equivalently, the primal is the linear

programme minimizing v subject to $\lambda \geq 0$, $\sum \lambda = 1$, and $\sum \lambda_\omega R_j(\omega) \leq v + b_j$ for every row. Carathéodory's theorem in $\mathbb{R}^M$ gives support at most $M + 1$ without changing the optimum risk vector. This argument freezes a tape to form its *distribution of designs*; it never moves a worst-task supremum through that distribution.

The body in (7.2) is closed and convex. If a point lies outside it, finite-dimensional separation supplies a linear functional whose infimum on the body exceeds its value at the point. The functional must be nonnegative on every coordinate direction, because the body contains every positive ray starting at one of its points. Normalizing its coefficients places it in Δ_J . Its infimum is (7.3). This proves the comparison assertion, including both directions of equality. A relabelling gives a bijection of the same execution laws and of their deterministic designs.

Every private scheme is also a shared-seed scheme that ignores its extra input. Conversely an optimal shared scheme can use $k \leq M + 1$ designs. Sample its index privately before the first observation and retain (r, I_t) , which has at most kS values. Each subsequent update, control and readout uses this actual pair. This has the same law as the shared scheme; no external transcript or hidden seed is used. This proves (7.6). Monotonicity, together with $\log((M + 1)S)/\log S \rightarrow 1$, proves the exponent assertion. The constant $M + 1$ is not uniform over growing or infinite task families. $\square$

The theorem identifies a finite controlled spectrum by one variational functional. It is more than the transport of a supplied simulation: its feasible set is constructed from the instrument and the prescribed memory, and the converse recovers every admissible execution. Convex minimax, separation and finite mixed strategies are classical tools. Their use here does not reassign their priority. The substantive information restriction is that all tasks share f , while all parameters within a task share its controller, and every retained design index is accounted for when the shared-seed convention is removed.

7.3. Controlled continuation partitions. Let $\mathcal{H}_t$ be the full finite tree of action–report histories, including histories null under a particular parameter or task. Histories null under every possible execution can be assigned arbitrary updates. A family of partitions $\mathcal{P}_t$ is controlled successor compatible when

$$h \sim_t h' \implies hay \sim_{t+1} h'ay \quad (a \in A, y \in Y_{t+1}). \quad (7.7)$$

Proposition 7.2 (Intrinsic feasible partitions). *Deterministic common encoders with widths $\mathbf{S}$ are, up to label names and unused states, exactly the partition families satisfying (7.7) and $|\mathcal{P}_t| \leq S_t$. A deterministic design adds, for each task, a control constant on every current cell and a terminal decision constant on every terminal cell. Thus the risk body and both sides of (7.5) can equivalently be defined on controlled continuation partitions with these cellwise rules.*

Proof. Equal machine states have equal successors on the same action and report, proving necessity. Conversely label the cells and send (C, a, y) to the cell containing hay for any $h \in C$. Condition (7.7) makes this independent of h . The cellwise controller rule supplies the legal control, and induction identifies the actual label with the actual history's cell for every parameter and task. Defining the maps on the whole tree ensures that an action absent under one task has not made its update task dependent. $\square$

7.4. Private randomization is a different finite problem. For completeness there is also an exact, nonconvex description without a shared seed. Write $g_t^d(a \mid i)$ for the private controller, $q_t(i' \mid i, a, y)$ for the common private update, and $v^d(b \mid i)$ for its terminal rule. Let q_0 be a

parameter-independent initialization law when randomized initialization is permitted. Then

$$\begin{aligned} p_0^{d\theta}(z, i) &= \nu_\theta(z)q_0(i), \\ p_{t+1}^{d\theta}(z', i') &= \sum_{z, i, a, y} p_t^{d\theta}(z, i)g_t^d(a \mid i)K_{\theta, t}(z', y \mid z, a)q_t(i' \mid i, a, y). \end{aligned} \quad (7.8)$$

The risk of row j is

$$\sum_{\theta, z, i, b} \kappa_j(\theta) p_T^{d(j)\theta}(z, i) v^{d(j)}(b \mid i) \ell_{d(j)\theta}(z, b).$$

Proposition 7.3 (Private flow characterization). *Minimizing the largest baseline-subtracted row risk over the probability simplexes in (7.8) gives exactly $\Theta_{\text{pr}}(\mathbf{S}; D)$. The minimum is attained. With rational input this is a finite polynomial optimization problem, not in general the linear programme (7.4).*

Proof. Induction on the actual execution gives the displayed occupancies and risk. Conversely their kernels implement precisely that law with no extra retained variable. The variables form a compact product of finite simplexes, and every row risk is a polynomial in them. Continuity gives attainment. The shared occurrences of q_t cannot be split by task or parameter; doing so would change the information pattern. $\square$

Example 7.4 (A strict cost for hiding the seed). Observe $X = \theta \in \{1, 2, 3\}$ without error, retain two labels, and finally guess θ with zero-one loss under worst-parameter risk. The full-information baseline is zero. With a shared seed, choose uniformly one of the three two-element subsets and encode the two selected symbols without error. The omitted symbol is necessarily wrong. Each parameter is omitted with probability $1/3$, so the risk is $1/3$. For any fixed seed, the sum of the three correct-reconstruction probabilities of a two-label experiment is at most two; averaging proves the matching lower bound.

Without a shared seed let r_1, r_2 be the two decoder distributions on three symbols. Correctness for symbol x is at most $\max\{r_1(x), r_2(x)\}$. If all three correctness probabilities exceeded $1/2$, one of two probability vectors would exceed $1/2$ at two coordinates, an impossibility. Thus the private risk is at least $1/2$. It is attained by assigning symbol 1 to the first label and symbols 2, 3 to the second, whose decoder tosses a fair coin. Therefore

$$\Theta_{\text{sh}}(2) = \frac{1}{3}, \quad \Theta_{\text{pr}}(2) = \frac{1}{2}.$$

The gap is not removed by allowing randomized private encoding. A single unknown parameter is selected before the seed, as required above.

8. ONE VARIATIONAL PRINCIPLE: OCCUPANCIES, BAYES ENVELOPES AND PRESSURE

The support function in Theorem 7.1 can be computed without first constructing a quotient or listing full decision trees. The relevant state for this calculation is an *offline* joint occupancy. It is not an additional observation available during execution.

8.1. An exact support-function recursion. For $w \in \Delta_J$ put

$$z_0^{d\theta}(z, i) = \mathbf{1}_{\{i=1\}} \nu_\theta(z) \sum_{j: d(j)=d} w_j \kappa_j(\theta).$$

For arbitrary nonnegative arrays $z = (z^{d\theta}(x, i))$, a prescription consists of actions $a^d(i)$ and a *single* table $f(i, a, y)$. Define its linear occupancy update by

$$(\mathcal{T}_{a, f} z)^{d\theta}(x', i') = \sum_{x, i, y} z^{d\theta}(x, i) K_{\theta, t}(x', y \mid x, a^d(i)) \mathbf{1}_{\{f(i, a^d(i), y)=i'\}}. \quad (8.1)$$

The time t is understood from the two spaces in this formula. Define

$$V_T(z) = \min_{b^d(i) \in B_d} \sum_{d, \theta, x, i} z^{d\theta}(x, i) \ell_{d\theta}(x, b^d(i)), \quad (8.2)$$

$$V_t(z) = \min_{a, f} V_{t+1}(\mathcal{T}_{a, f} z), \quad t = T - 1, \dots, 0. \quad (8.3)$$

Theorem 8.1 (Occupancy representation of the spectrum). *For the finite controlled instrument,*

$$h_E(\mathbf{S}; w) = V_0(z_0(w)), \quad \Theta_{\text{sh}}(\mathbf{S}; D) = \max_{w \in \Delta_J} \{V_0(z_0(w)) - w \cdot b\}. \quad (8.4)$$

Every V_t is a positively homogeneous, concave, piecewise-linear function on its nonnegative occupancy cone. A minimizing sequence of prescriptions compiles into an actual common-encoder design with the specified widths. No realized task index or hidden parameter is supplied to f .

Proof. Fix a deterministic sequence of prescriptions. Induction in (8.1) shows that $z_t^{d\theta}$ is the joint law of (Z_t, I_t) under that task and parameter, multiplied by its initial weight. Its terminal linear cost is therefore $w \cdot R(w)$. At the last time the terminal tables in (8.2) give precisely all legal terminal rules: $b^d(i)$ is common to every θ in task d .

At an earlier time, choosing $a^d(i)$ and one common f determines the entire next occupancy. Conversely every possible first prescription of a deterministic continuation has this form. Backward induction therefore proves that (8.3) minimizes over exactly all legal future designs. This proves the first equality in (8.4); Theorem 7.1 proves the second. For any fixed continuation the cost is linear in the initial occupancy. There are finitely many continuations, so V_t is the minimum of finitely many linear forms. The stated geometric properties follow.

The occupancy is propagated during design, starting at the known array $z_0(w)$. Once a sequence of minimizing tables is chosen, its future occupancies are deterministic functions of that sequence. Only the finite tables and the actual label are used online. In particular, one must not store the occupancy array without charging its size, nor allow a different prescription for the unobserved realized parameter. $\square$

This recursion is a designer's dynamic programme with coupled information constraints. Common-information prescriptions and belief-state dynamic programming have substantial antecedents [20, 21]. In the cited common-information preprint, Corollary 5 already treats absent common memory and finite local memories; its unconditional information state is an offline occupancy. The additional assertion here is the exact equality with the *common-encoder minimax spectrum*, including its dual adversarial weights, controlled continuation constraint and explicit seed cost. The recursion does not assert polynomial complexity: the number of linear pieces and the number of finite designs can be very large.

8.2. The common quadratic task family. There is a particularly transparent intrinsic form when reports are uncontrolled. Let the finite history law $p_t(h)$ be fixed, let U_{dt} be square-integrable Hilbert-valued targets, and let $\alpha_{dt} \geq 0$. Task d has loss $\sum_t \alpha_{dt} \|U_{dt} - \tilde{U}_{dt}\|^2$. The decoder's answers do not affect later observations or updates. Set $m_{dt}(h) = \mathbb{E}[U_{dt} \mid H_t = h]$. For a successor-compatible partition family $\mathcal{P}$, define

$$D_d(\mathcal{P}) = \sum_t \alpha_{dt} \sum_{C \in \mathcal{P}_t} \sum_{h \in C} p_t(h) \|m_{dt}(h) - \bar{m}_{dt, C}\|^2, \quad \bar{m}_{dt, C} = \frac{\sum_{h \in C} p_t(h) m_{dt}(h)}{p_t(C)}. \quad (8.5)$$

Zero-mass cells are omitted. Define the single weighted geometry

$$M_{w, t}(h) = (\sqrt{w_d \alpha_{dt}} m_{dt}(h))_{d \in D}$$

to lie in the Hilbert direct sum of the target spaces.

Corollary 8.2 (Common-encoder variance duality). *For a shared independent seed and the declared widths,*

$$\begin{aligned}\Theta_{\text{sh}}(\mathbf{S}; D) &= \min_{\lambda \in \Delta(\mathcal{P})} \max_d \sum_{\mathcal{P}} \lambda_{\mathcal{P}} D_d(\mathcal{P}) \\ &= \max_{w \in \Delta_D} \min_{\mathcal{P}} \sum_t \mathbb{E} \|M_{w,t}(H_t) - \mathbb{E}[M_{w,t}(H_t) \mid \mathcal{P}_t]\|^2.\end{aligned}\quad (8.6)$$

The minima range over one successor-compatible family of width at most S_t at each time. This is an exact characterization of the maximum over tasks, not a statement only about one scalarized loss. Its private version lies between the values given by this formula at S and by its implementation with $(|D| + 1)S$ states.

Proof. Conditional expectation gives the orthogonal decomposition of each squared risk into its full-history Bayes risk and prediction of $m_{dt}(H_t)$. On a deterministic state cell its unique optimal readout is the displayed cell mean. All tasks can use their respective means simultaneously without changing the common encoder. Thus the coordinatewise best risk vector of a deterministic partition is exactly $D(\mathcal{P})$. There are finitely many history partitions, although the target Hilbert spaces need not be finite dimensional.

Conditioning on an independent seed gives a distribution of these partitions; conversely their mixtures are executable under the shared convention. The proof of Theorem 7.1 now applies to this finite vector set. The identity $\sum_d w_d D_d(\mathcal{P}) = \sum_t \mathbb{E} \|M_{w,t} - \mathbb{E}(M_{w,t} \mid \mathcal{P}_t)\|^2$ is the direct-sum norm formula. Storing the mixture index proves the last assertion exactly as in (7.6). $\square$

For $|D| = 1$, (8.6) reduces to the earlier conditional-variance characterization, Theorem 17.1. For several tasks it supplies its missing minimax convexification. The outer maximizing weight is common to every time; selecting a separate favourable weight or encoder at each checkpoint is not permitted. Compatible predictive trees are one class of feasible partitions of this same functional. Their energy estimates in the supporting theory supply quantitative bounds for that class; they are not hypotheses of Theorems 7.1 or 8.1.

Corollary 8.3 (A1: maximum over finite query and checkpoint tasks). *Consider the actual finite-command accepted-cell/failure protocol of Corollary 17.2, with its fixed prior, finite horizon and parameter-independent command laws. Fix any finite family of weighted quadratic query/checkpoint tasks. Its optimal shared-seed common causal encoder is given by (8.6), using the accepted and summed failure likelihoods of that protocol. At most $|D| + 1$ partition designs are needed. Retaining their index gives the same optimum with at most $(|D| + 1)S$ private states.*

Proof. The observation likelihoods and posterior means are derived in Corollary 17.2 from $g_j k_j(\theta)$ on acceptance and $\sum_j (1 - g_j) k_j(\theta)$ on the one failure report. These give one finite history law for all tasks. Insert its posterior query means in (8.5) and apply Corollary 8.2. The unseen failed raw-cell label is never used in a partition update. Unlike the earlier scalarization, the present formula retains the maximum over the whole prescribed task family. $\square$

Proposition 8.4 (A1 with private randomness at unchanged width). *In the same finite-command protocol, let $p_t(h)$ be its actual history law, and let $m_{dt}(h)$ be the finite vector of posterior query means for task d , with query weights absorbed into that vector. Write $\eta_t^q(i \mid h)$ for the conditional label law generated recursively by one common private encoder q . The private maximum-over-tasks excess at unchanged width is exactly*

$$\min_{q,v} \max_{d \in D} \sum_{t,h,i} \alpha_{dt} p_t(h) \eta_t^q(i \mid h) \|m_{dt}(h) - v_{dt}(i)\|^2. \quad (8.7)$$

The readouts can be restricted to the compact boxes containing the weighted query means. This is a finite polynomial optimization problem in the stochastic update rows and readout coordinates, with

coefficients given by the actual likelihood integrals. It has a minimum. No finite arithmetic claim is made when those coefficients are arbitrary reals.

Proof. Given a supported history, the label law is a finite sum of products of the encoder's transition probabilities. The same row is used whenever the preceding label and observed command/report agree, in particular on the common failure report. Conditional orthogonality subtracts the full-history Bayes risk, leaving precisely the displayed expression. Conversely every listed stochastic row and readout is an admissible private machine, so both infima are equal. Projecting readouts into the coordinatewise range of the query means cannot increase squared error. The feasible set is compact, and its objective is the maximum of finitely many polynomials. Continuity proves attainment. $\square$

8.3. Exponential criteria and hidden-phase control. The same support calculation has a multiplicative form. Let $\gamma > 0$ and let $C_{d\theta}$ be a bounded cumulative cost on the finite trajectory, including a terminal decision. Put

$$M_j(\omega) = \sum_{\theta} \kappa_j(\theta) \mathbb{E}_{\theta}^{\omega, d(j)} e^{\gamma C_{d(j)\theta}}.$$

For task d the risk is $\gamma^{-1} \log \max_{j: d(j)=d} M_j$; a Bayesian task has its single prior row. Let $\beta_d^{\exp}$ denote its corresponding full-history baseline.

Corollary 8.5 (Risk-sensitive common-encoder duality). *The shared-seed excess under these exponential risks equals*

$$\Theta_{\text{sh}}^{\exp}(\mathbf{S}; D) = \frac{1}{\gamma} \log \left[\max_{w \in \Delta_J} \min_{\omega \in \Omega_{\mathbf{S}}} \sum_j w_j e^{-\gamma \beta_{d(j)}^{\exp}} M_j(\omega) \right]. \quad (8.8)$$

For additive costs, the inner minimum is computed by (8.1)–(8.3) after replacing the transition by $K_{\theta,t} e^{\gamma c_{d\theta,t}}$ and the terminal loss by $e^{\gamma c_{d\theta,T}}$. The initial row weights are multiplied by $e^{-\gamma \beta_d^{\exp}}$. These are nonnegative occupancies, not normalized posterior probabilities.

Proof. For a mixture of designs, the exponential moment is $\sum_{\omega} \lambda_{\omega} M_j(\omega)$. It is *not* the exponential of an average of logarithmic risks. Since logarithm is increasing, subtracting the baseline and maximizing rows gives

$$\frac{1}{\gamma} \log \max_j \left[e^{-\gamma \beta_{d(j)}^{\exp}} \sum_{\omega} \lambda_{\omega} M_j(\omega) \right].$$

Minimize over λ and apply finite minimax to the bracketed positive moment table. This proves (8.8). Expanding the product of transition weights proves the Feynman–Kac occupancy assertion by the same induction as before. Positivity, not normalization, is what that induction requires. $\square$

For example, let an unobserved phase $\theta \in \{0, 1\}$ have probabilities $1/2, 1/2$, provide no report, and charge $C = \mathbf{1}_{\{a \neq \theta\}}$. Every observation-admissible action, randomized or not, has moment $(1 + e^{\gamma})/2$ and risk $\gamma^{-1} \log((1 + e^{\gamma})/2) > 0$. Optimizing separately within each phase would instead give zero. The single $a^d(i)$ in the occupancy recursion forbids precisely that exchange. This supplies a finite-model resolution of the common-policy issue arising in phase mixtures; it does not presume that a latent phase decomposition of a microscopic system has already been constructed.

9. MINIMAL POSTERIOR-AND-REPORT-PREDICTIVE COORDINATES

The variational theorem does not require a sufficient statistic. When such a statistic is desired, the finite instrument admits a direct construction. It distinguishes the cost of prediction from the cost of preserving the whole parameter experiment.

Fix the finite instrument of Section 7. For an action–report history $h = (a_0, y_1, \dots, a_{t-1}, y_t)$ define the unnormalized forward vector by

$$\begin{aligned} u_{\theta,0}(z) &= \nu_\theta(z), \\ u_{\theta,t+1}(hay, z') &= \sum_z u_{\theta,t}(h, z) K_{\theta,t}(z', y \mid z, a), \quad L_\theta(h) = \sum_z u_{\theta,t}(h, z). \end{aligned} \quad (9.1)$$

Choose a reference prior $\pi_\theta > 0$. A history is supported if $\bar{L}(h) := \sum_\theta \pi_\theta L_\theta(h) > 0$. On supported histories put

$$\begin{aligned} \lambda_\theta(h) &= \pi_\theta L_\theta(h) / \bar{L}(h), \\ k_t(y \mid h, a) &= \bar{L}(hay) / \bar{L}(h). \end{aligned} \quad (9.2)$$

The likelihoods in these formulae do not contain action-selection probabilities. Under any parameter-independent observation-based policy, those probabilities cancel from the posterior and from likelihood ratios. Thus the formulae apply to every policy on its supported histories, not only to a policy used to explore the history tree.

A history coordinate $s_t(h)$ is called *saturated and predictive* if it determines $\lambda(h)$, determines every vector $k_t(\cdot \mid h, a)$, and has a deterministic update from (s_t, a, y) on supported continuations. At time T only the first condition is imposed. This is the finite, terminal-parameter version of the three conditions in Section 16. Additional latent-target marks require the corresponding conditional kernels as further decorations; parameter sufficiency alone does not imply their sufficiency. Sections 4–5 supply the marked reconstruction theorem and an output-sensitive construction.

Construct partitions backwards. At level T group histories with the same vector $\lambda(h)$. Having constructed the class map c_{t+1} , let $c_t(h)$ be the class determined by the finite tuple

$$\left(\lambda(h), (k_t(y \mid h, a), c_{t+1}(hay))_{a,y} \right). \quad (9.3)$$

For an unsupported continuation its class entry is one fixed cemetery symbol. Let K_t be the number of classes at level t .

Theorem 9.1 (Minimal posterior-and-report-predictive coordinate). *For every finite controlled instrument the partitions (9.3) form a saturated predictive coordinate. Every other such deterministic history coordinate refines them at each level. Consequently K_t is its least possible width, simultaneously at every level. The partitions, as equivalence relations on supported histories, are independent of the choice of the strictly positive reference prior.*

There is a parameter-independent finite reconstruction kernel $L_t(y \mid c, a, c')$ which, from successive quotient classes and the chosen controls, reconstructs the report stream under every admissible policy. The original and quotient parameterized report experiments have mutually exact causal simulations. This assertion concerns reports and parameter likelihoods only; latent-target preservation is the separate marked statement of Theorem 4.5. Their additional persistent state is at most $\max_t K_t$ labels, so composing either simulation with an S -state encoder costs at most $(\max_t K_t)S$ states.

If the instrument and prior have rational entries, the classes and reconstruction kernels can be computed by exact finite arithmetic. The calculation is polynomial in the size of the expanded finite history tree and its rational input length. That tree, and the least width K_t , may grow exponentially with the horizon.

Proof. Equality of the keys at level t implies equality of the posterior vectors, of all reference predictive probabilities, and of the class of every common positive-probability successor. Sending (c, a, y) to that successor class therefore defines an update. Unsupported continuations may be assigned arbitrarily. This proves the three required properties.

For minimality, at time T any saturated coordinate separates unequal posterior vectors, so its fibres refine c_T . Suppose the assertion has been proved at $t + 1$. Equal coordinates at time t have equal posterior and predictive vectors. Their supported successors under any fixed (a, y) have equal next coordinates by recursion, and hence equal c_{t+1} by the induction hypothesis. Their complete keys (9.3) coincide. Backward induction proves the claimed simultaneous refinement, rather than just a bound on a terminal statistic.

To check prior independence, first observe that equality of posterior vectors under a positive prior is equivalent to proportionality of the nonnegative likelihood vectors $(L_\theta(h))_\theta$. This relation is independent of the prior. For a fixed parameter with $\lambda_\theta(h) > 0$, the actual next-report probabilities are

$$P_\theta(y \mid h, a) = k_t(y \mid h, a) \frac{\lambda_\theta(hay)}{\lambda_\theta(h)}. \quad (9.4)$$

In an existing class the right side is constant on histories for every supported successor, because the successor class determines its posterior. The set of parameters with positive likelihood is also constant on the class. Under a different positive prior, its new posterior weights are functions of its old posterior weights; averaging these common parameter-specific predictive kernels therefore gives a common new reference predictive kernel. The existing partition is saturated, predictive and recursive for the new prior. Minimality in both directions proves equality of the two partitions.

For reconstruction, write $F_t(c, a, y)$ for the class update and define

$$H_t^0(c' \mid c, a) = \sum_{y: F_t(c, a, y) = c'} k_t(y \mid c, a),$$

$$L_t(y \mid c, a, c') = \frac{\mathbf{1}_{\{F_t(c, a, y) = c'\}} k_t(y \mid c, a)}{H_t^0(c' \mid c, a)} \quad \text{if } H_t^0(c' \mid c, a) > 0.$$

Choose an arbitrary probability vector when the denominator is zero. The posterior at the next class is constant over the summed reports. Equation (9.4) thus factors the actual joint law of report and next class into this same L_t and the parameter-specific class-transition probability. Conditioning further on the next class cancels the likelihood factor. This gives a single parameter-independent report reconstruction, including when parameters have different supports. Histories having zero likelihood under a parameter are not visited under that parameter.

Starting from the initial class, the reconstruction stores the previous class, receives the next class after the actual chosen action, samples its report using L_t , and replaces its stored class. A target policy then chooses its next action from the reconstructed reports. Induction in time proves equality of the resulting action–report laws for every parameter and policy. In the other direction, apply F_t to the original reports. Product registers give the displayed storage bound; neither simulation presumes a cost-free transcript.

Finally, (9.1) enumerates the rational forward vectors on a finite tree. Formulae (9.2) and (9.3) use only finite sums, products, divisions by positive numbers, and equality tests of rational tuples. Class labels can be assigned by sorting those tuples backwards. To see the stated bit-complexity bound, take a common denominator for all rational input entries. Its bit length is at most their total input length. At level t , a forward entry can be represented with the t th power of this denominator times the initial denominator; sums add at most the bit length of the number of summands at each step. Thus every intermediate integer has bit length polynomial in horizon and input length. Posterior divisions and tuple comparisons preserve a polynomial bound. There are only as many forward and backward steps as nodes and edges of the expanded tree, times its finite hidden and parameter dimensions. This proves effectivity without any oracle for exact real comparisons. $\square$

The theorem constructs the object assumed by the causal factorization in this finite class. Its minimality is among the *specified* likelihood-saturated, raw-report-predictive coordinates. It is not

a claim that every exact simulator, or every task-specific lossy encoder, must use these coordinates. The variational principle optimizes the latter directly and may use substantially less memory.

Example 9.2 (Saturation can have an exponential price). Let the parameter range over $\{0, 1\}^d$, and reveal it noiselessly one bit at a time during the first d observations. Thereafter all reports are independent fair bits. At time d all future-report predictions coincide, so the future-report predictive quotient has one class. Nevertheless the posterior on the parameter is a different point mass for each word. Every saturated coordinate at that time has at least 2^d classes, and the construction above has exactly 2^d . Predictive closure alone therefore supplies no controlled-cost parameter saturation. This is a cost calculation in a concrete model, not an assumption that the missing saturation has bounded size.

Example 9.3 (Likelihoods alone need not close prediction). There is one parameter. Let Y_1 be a fair bit and put $Y_2 = Y_1$. The constant coordinate is recursive and contains every parameter likelihood, but its one-step predictive law at time 1 is not a function of that coordinate. The least saturated predictive widths are $(K_0, K_1, K_2) = (1, 2, 1)$. The last quotient class can discard the bit, because reconstruction of Y_2 still uses the previous class. This example separates the hypotheses of the factorization theorem; it does not assert impossibility of other simulators with other charged resources.

10. FINITE COMPILATION AND SEVERAL CHARGED RESOURCES

The finite variational principle also has an effective implementation. This is a separate source of multi-resource control from the energy oracle used in the predictive-tree construction. The price of synthesis is explicit and can be large; it is not moved into a free online oracle.

Use the finite instrument and finite decision alphabets of Section 7, with rational probabilities and losses. Let $S = \max_t S_t$, $Y = \max_t |Y_t|$, $B = \max_d |B_d|$, and $m = |D|$. A table implementation of one deterministic design has description length bounded, up to an absolute coding constant, by

$$\begin{aligned} P_0 = & \sum_{t=0}^{T-1} S_t |A| |Y_{t+1}| \lceil \log_2(S_{t+1} + 1) \rceil \\ & + m \sum_{t=0}^{T-1} S_t \lceil \log_2(|A| + 1) \rceil + m S_T \lceil \log_2(B + 1) \rceil \\ & + O((T + m) \log_2(T + S + |A| + Y + B + 2)). \end{aligned} \quad (10.1)$$

Unused states and forbidden controller actions can simply be retained as redundant table entries. The externally specified time index is the same clock convention as in (2.1). If no external clock is permitted, its $T + 1$ values must also be included in the register.

Theorem 10.1 (Effective finite-source realization). *An optimal shared-seed common-encoder scheme for the rational finite instrument can be synthesized by a finite exact calculation. It can use $k \leq M + 1$ deterministic tables with rational mixing weights. For every integer $p \geq 1$ there is a dyadic implementation whose excess is at most*

$$\Theta_{\text{sh}}(\mathbf{S}; D) + k 2^{-p}. \quad (10.2)$$

Its fixed description has $O(kP_0 + kp)$ bits. Initialization uses exactly p independent fair bits; no random bits are required subsequently. With an explicitly permitted read-only shared seed its persistent register has width at most S . Without that convention the same scheme has width at most kS , including the stored design index.

Each update reads one supplied finite-alphabet report. On a bit-serial table implementation, transient space can be bounded by $O(p + \log(P_0 k + 2) + \log(S + |A| + Y + B + T + m + 2))$ bits and the running time of an update or initialization by $O(kP_0 + kp + p + 1)$ bit operations, after

increasing the universal coding constant. No unit-cost access to an arbitrary real constant is required. Reading the physical report itself remains an input cost, rather than a claim about how an underlying continuous measurement was acquired.

Proof. Enumerate the finite deterministic designs. Their risks are rational numbers obtained from finite products and sums of rational transition entries and losses. Full-history baselines can likewise be computed by enumerating full-history policies and solving their finite minimax linear programmes; for a Bayesian task their minimum is already a finite linear optimization. Thus the primal programme (7.4) has rational data and has a rational optimal basic solution. Rational linear algebra and the usual elimination of affine dependence among its risk vectors give a rational optimal mixture of at most $M + 1$ designs. This gives a finite synthesis algorithm, not a bound polynomial in the horizon or the number of reports. Integer determinants of the finite rational constraint matrix give finite bit bounds for all weights, so synthesis does not require a probability oracle.

Let its weights be $\lambda_1, \dots, \lambda_k$. For $r < k$ set $\tilde{\lambda}_r = 2^{-p} \lfloor 2^p \lambda_r \rfloor$ and let the last weight be the nonnegative remainder. Then

$$\|\lambda - \tilde{\lambda}\|_{\text{TV}} = \sum_{r < k} (\lambda_r - \tilde{\lambda}_r) \leq (k - 1)2^{-p}.$$

For every row, its risk lies in $[0, 1]$, so the difference of its mixed risks is at most this total variation. Taking the maximum and subtracting the unchanged baseline proves (10.2).

A uniform p -bit integer, compared to the cumulative integer weights, selects the dyadic design exactly. Its deterministic tables then implement the scheme. There are k copies of the tables counted in (10.1) and k integer weights of at most $p + 1$ bits, which proves the description bound. In the private implementation the index is part of (r, I_t) and hence multiplies the width by k . In the shared implementation the seed is explicitly allowed throughout the run; its index can instead be reread from that permitted input. A modified or data-dependent seed would not qualify for this exception.

For a conservative machine model, store the immutable description as a sequential binary table with self-delimited addresses. Locate an update, a controller action, or a terminal answer by scanning it. Counters need only logarithmic space in its length and in the finite alphabet sizes. Comparing the p -bit initialization integer with its cumulative weights requires $O(p)$ extra space and $O(kp)$ time by a sequential scan. Copying the selected finite answer and updating the label has the same bounds. Thus the displayed time and space bounds are valid without a random-access assumption. They are deliberately not optimized. $\square$

Proposition 10.2 (Stability of finite synthesis). *Suppose two finite instruments have matched alphabets, losses and timing. Their initial hidden laws have total variation distance at most η_0 for every parameter, and the corresponding transition–report rows have distance at most η_t uniformly in hidden state and control. Then, for every admissible common-encoder/controller design, the difference of each bounded-loss risk is at most $\varepsilon = \min\{1, \eta_0 + \sum_{t < T} \eta_t\}$. Their full-history baselines differ by at most ε , and their shared or private excess spectra at the same signature differ by at most 2ε . An additional uniform error ζ in the loss adds 2ζ to this last bound.*

Proof. Couple the initial hidden states maximally. Until the two trajectories first disagree, choose the same controls, report-driven updates and random bits in both executions. Couple the corresponding hidden/report transition rows maximally. The probability of a disagreement by the last time is at most $\eta_0 + \sum \eta_t$. A bounded loss changes in expectation by at most that probability. This argument is uniform over every parameter and every legal controller, including full-history controllers. Infima over controllers and common encoders, and maxima over tasks or parameters, are nonexpansive in uniform error. Apply that observation once to the constrained risk and once to its baseline. A uniformly perturbed loss changes each of these by at most ζ . $\square$

This provides a multi-resource finite theorem with computable synthesis, finite randomization, finite description and finite execution cost. Together with Theorem 9.1, it also makes finite sufficient reconstruction effective whenever the expanded finite source is given rationally. It does not compute multiplicative energies for an arbitrary infinite predictive tree, nor does it turn a merely measurable sufficient statistic into a finite algorithm. Those are different input classes with different resource signatures.

Exact rational reconstruction tables need not have dyadic probabilities. Their computability does not imply exact sampling with a fixed number of fair bits. A fixed-bit implementation rounds their rows and uses Proposition 10.2 to charge the resulting path error; exact non-dyadic sampling requires a different randomization-time convention.

11. A GAUSSIAN EXPERIMENT WITH A GENUINE TEMPORAL PENALTY

The append-register Gaussian theorem in the supporting development concerns a terminal vector experiment. The following experiment instead forces information to survive until the question that uses it arrives. Its final target is scalar, but its causal exponent depends on the number of coordinates acquired before that question.

Fix $d \geq 2$ and $\sigma > 0$. Let $\theta_1, \dots, \theta_d$ be independent and uniform on $[-1, 1]$, and observe sequentially

$$X_j = \theta_j + \sigma Z_j, \quad 1 \leq j \leq d,$$

where the Z_j are independent standard Gaussians. After these observations have been discarded, an independent uniform query $J \in \{1, \dots, d\}$ is revealed. The machine then updates its register and estimates θ_J . There is no external observation transcript. The query is an input to that final update, not an additional persistent decoder register. The cost is squared loss, bounded by four after projecting the answer onto $[-1, 1]$. Write

$$m(x) = \mathbb{E}[\theta_1 \mid X_1 = x], \quad \beta = \mathbb{E} \text{Var}(\theta_1 \mid X_1).$$

Let $\mathcal{E}_S$ be the infimal excess above β when the total register has at most S states at every stage. Independent shared randomness is permitted; the same conclusions hold without it.

Definition 11.1 (Delayed-query information signature). There are d sequential scalar-acquisition updates $I_j = F_j(I_{j-1}, X_j, U_j)$, followed by one query update $I_{d+1} = F_{d+1}(I_d, J, U_{d+1})$ and a terminal output $\hat{\theta} = D(I_{d+1}, U_{d+2})$. Each register has at most S values. The U_j are independent private draws; an optional independent retained public seed may be an argument of every rule. The query is independent of the pre-query history, and neither J nor an old report is available to the terminal decoder except through I_{d+1} . The cardinality signature permits arbitrary measurable real arithmetic. Proposition 12.2 gives a distinct finite-acquisition and finite-table implementation.

Theorem 11.2 (Delayed Gaussian query law). *There are constants $0 < c_{d,\sigma} \leq C_{d,\sigma} < \infty$ such that for every $S \geq 2^d$,*

$$c_{d,\sigma} S^{-2/d} \leq \mathcal{E}_S \leq C_{d,\sigma} S^{-2/d}. \quad (11.1)$$

The order holds for the maximum over the d individual coordinate query risks, with a common encoder before the query.

In contrast, an encoder which receives the query together with the full observation vector at its sole checkpoint has excess of order S^{-2} . Thus the scalar terminal target has a strictly worse cardinality exponent under the delayed-query causal constraint. The lower bound uses the register before the query and is not a final-output dimension bound.

Proof. The posterior density on $[-1, 1]$ is proportional to $\exp(-(x - u)^2/(2\sigma^2))$. Differentiation under the integral on this compact interval gives

$$m'(x) = \frac{\text{Var}(\theta_1 \mid X_1 = x)}{\sigma^2} > 0. \quad (11.2)$$

The posterior variance is positive because its density is positive on an interval. The function m is smooth, odd, and strictly increasing. Choose $b > 0$ and put $a = m(b) > 0$. The density of X_1 is positive and continuous, and m' is positive and continuous on $[-b, b]$. The change-of-variables formula therefore gives a density for $m(X_1)$ which is bounded below by a positive constant on $[-a, a]$. Independence implies that $M = (m(X_1), \dots, m(X_d))$ has density bounded below by a positive constant on the cube $Q = [-a, a]^d$.

We record an elementary volume bound. If $c_1, \dots, c_S$ are arbitrary points in $\mathbb{R}^d$, and v_d is the volume of the unit ball, choose $r = (|Q|/(2Sv_d))^{1/d}$. The union of their radius- r balls has volume at most $|Q|/2$. Consequently

$$\int_Q \min_{i \leq S} \|x - c_i\|^2 dx \geq \frac{|Q|}{2} \left(\frac{|Q|}{2Sv_d} \right)^{2/d}. \quad (11.3)$$

This bound does not require cells to be convex or centres to lie in Q .

Consider any admissible machine and condition on its independent shared random seed, when present. Its pre-query register I_d has at most S values. Subsequent private updates and readouts can do no better than being given (I_d, J) and using its conditional posterior mean. Let $c_i = (\mathbb{E}[\theta_j \mid I_d = i])_{j=1}^d$, with the seed included in all these conditional expectations. The Markov relation from the full observation to the register, and independence of the random seed, imply $c_i = \mathbb{E}[M \mid I_d = i]$. Orthogonality of conditional expectation shows that the resulting excess for the uniform query is

$$\frac{1}{d} \mathbb{E} \|M - c_{I_d}\|^2 \geq \frac{1}{d} \mathbb{E} \min_{i \leq S} \|M - c_i\|^2.$$

For each fixed seed the law of M is unchanged. Its density lower bound on Q and (11.3) give $c_{d,\sigma} S^{-2/d}$. Averaging the seed preserves this lower bound. The argument permits randomized private encoding: replacing its selected centre by the nearest centre is only a lower bound, not an assumption that the encoding is deterministic. The maximum coordinate risk is at least the uniform average, proving its lower bound as well.

For the upper bound put $L = \lfloor S^{1/d} \rfloor \geq 2$ and partition $[-1, 1]$ into L equal intervals. At observation j compute the interval label of $m(X_j)$ and append it in base L to the stored labels. At that time there are at most $L^j \leq S$ states. After the query, replace the whole label by the interval label of the requested coordinate; the terminal decoder outputs its midpoint. This last update leaves only L possible labels and retains no separate query. Its squared excess in every coordinate is at most $L^{-2} \leq 4S^{-2/d}$. Conditional orthogonality again subtracts the same full-observation baseline β . The construction is deterministic and works with or without a shared seed.

For the noncausal checkpoint comparison, the encoder sees (X, J) and quantizes only $m(X_J)$ into S equal intervals. This has excess at most S^{-2} . The one-dimensional version of the same density lower bound and (11.3) gives the matching lower bound, even with randomized encoding and a shared seed. Its S terminal centres are arbitrary real answers in $[-1, 1]$, so no additional discreteness assumption has entered. This proves all assertions. $\square$

The theorem is a Bayesian slice of the parameterized Gaussian experiment with the specified product prior, not a claim about uniform deficiency over the whole Gaussian family. Its constants depend on d and σ . It controls persistent cardinality and finite acquisition count, not the arithmetic cost of evaluating the real posterior mean. Under a finite numerical signature, an approximation of m requires a separate accuracy and computation bound. The obstruction nevertheless holds for every such implementation, since it was proved for the larger unrestricted-arithmetic class.

The lower bound is the same conditional-variance geometry as Corollary 8.2, now evaluated directly for a continuous history law. In the finite theorem, successor compatibility determines which cells can be executed. Here the late query forces all coordinate readouts to factor through the same pre-query cell, and the volume bound computes its cost. This explains both the lossless

append-only regime and a regime where forgetting before a future question changes the exponent, without identifying either one with static terminal dimension.

12. MARKED PROJECTIONS AND FINITE NUMERICAL ACQUISITION

The common object behind the decision problems is the attainable family of joint laws under one admissible encoder, with its task controllers and its actual target marks. Evaluating a bounded loss is a linear projection of this family. A risk body for one selected task family is therefore a projection, not by itself a classification of the whole experiment. We record the exact links to the quantitative regimes used later in the article.

Proposition 12.1 (Quadratic and metric projections). *For an uncontrolled standard Borel history H and a square-integrable Hilbert-valued target U , write $m(H) = \mathbb{E}[U \mid H]$. Any statistic I formed from H with independent randomization has optimal quadratic excess*

$$\mathbb{E} \|m(H) - \mathbb{E}[m(H) \mid I]\|^2. \quad (12.1)$$

For finitely many such tasks the common-encoder weighted support problem is the conditional variance of the direct sum of their posterior means multiplied by the square roots of their task weights. On finite histories this is precisely the compatible-partition functional in Corollary 8.2. On the predictive trees of Theorem 23.1, its within-cell variances are comparable to the retained tree-energy sums under the hypotheses of that theorem.

For a revealed law $\pi \in \mathcal{P}([0, 1])$, take as actual target π and as output a realized probability law $\hat{\pi}$. The loss $W_1(\pi, \hat{\pi})$ yields exactly the law-presentation profile of Theorem 19.1. Replacing the output by a sample, and testing only its marginal, is a different task and does not preserve this profile.

Proof. Conditional on (H, I) the target mean is $m(H)$. Orthogonal projection gives

$$\mathbb{E} \|U - a(I)\|^2 = \mathbb{E} \|U - m(H)\|^2 + \mathbb{E} \|m(H) - a(I)\|^2.$$

The second term is minimized by $a(I) = \mathbb{E}[m(H) \mid I]$, proving (12.1). Weighted sums are squared norms in the Hilbert direct sum. For a deterministic finite partition, conditioning on I is averaging over its cell, giving the stated finite functional. The lower and upper comparisons with tree energy are exactly the separation and diameter estimates of Theorem 23.1; no new equality between energy and variance is asserted. The continuous Gaussian volume argument is another evaluation of (12.1), not an invocation of the finite-history partition theorem on a continuous history space.

For the second assertion, the joint output loss of an encoder and its random decoder is, by definition, $\mathbb{E} W_1(\pi, D(Q(\pi)))$. The decoder must output a law on each run. This is the profile in (19.1), whose minimax order is $1/\log(S+1)$ under its stated common-code convention. The single-marginal probe in Proposition 19.3 has a different output and a different rate. Integrating out the realized law before applying the distance changes the loss, so does not commute with this projection. $\square$

The revealed-law family is generally nondominated. It belongs to the marked-spectrum definition, but is not being smuggled into the product-dominated compactness theorem. Conversely the latter has a proved closed attainable body, whereas a general infinite family of tasks need not have one without hypotheses. Terminal statistical deficiency uses the decision-complete task family and compactness specified in Theorem 2.4; a finite selection of task supports is not sufficient to infer Blackwell equivalence. These distinctions allow the tree, Wasserstein, finite Gaussian and count bounds to be used together without equating their output spaces.

12.1. A finite table for the delayed Gaussian problem. Retain the notation and information pattern of Definition 11.1. The following bound separates acquisition, numerical evaluation and persistent storage.

Proposition 12.2 (Finite-precision Gaussian implementation). *Let $L = \lfloor S^{1/d} \rfloor \geq 2$. An acquisition interface returns one of the interior bins of a grid of width at most h on $[-R, R]$ and two tail symbols, where $R > 1$. Let the stored table approximate $m(x)$ at each interior bin midpoint to error at most ε , with values clipped to $[-1, 1]$. Quantize those approximations into L equal intervals and store the resulting labels in the base- L register. Then the squared excess in every coordinate is at most*

$$\left(L^{-1} + \frac{h}{2\sigma^2} + \varepsilon\right)^2 + 8 \exp\left(-\frac{(R-1)^2}{2\sigma^2}\right). \quad (12.2)$$

For fixed computable $\sigma > 0$, there are rational uniform-grid choices with $1/(2L) \leq h \leq 1/L$, $\varepsilon \leq 1/L$, and $1 + \sigma\sqrt{6\log L} \leq R \leq 2 + 2\sigma\sqrt{6\log L}$ that give excess $O_\sigma(L^{-2})$. The number of acquisition symbols is $O_\sigma(L\sqrt{\log L})$, the table has $O_\sigma(L(\log L)^{3/2})$ bits, and each scalar acquisition uses $O_\sigma(\log L)$ report bits. For rational σ^2 the table is computable by a terminating certified rational procedure. Its offline evaluation cost is separate from the bit-serial lookup and register-update costs.

Proof. Equation (11.2) and $|\theta| \leq 1$ imply $0 < m'(x) \leq \sigma^{-2}$. An interior-bin midpoint differs from X by at most $h/2$, so its posterior mean differs by at most $h/(2\sigma^2)$. Table evaluation and the final equal-interval quantization add at most ε and $1/L$. On a tail event, any clipped output differs from $m(X)$ by at most two. Since $X = \theta + \sigma Z$ and $|\theta| \leq 1$,

$$\mathbb{P}(|X| > R) \leq 2 \exp(-(R-1)^2/(2\sigma^2)).$$

The conditional-variance identity subtracts the same full-observation baseline and gives (12.2). This estimate holds coordinatewise; no union bound over all coordinates is needed for a single queried coordinate. The pre-query and post-query register updates are exactly the base- L construction of Theorem 11.2.

There are $O(R/h)$ interior bins. Only their final labels in $[L]$ need be stored after certified preprocessing; thus the table uses $O((R/h) \log L)$ bits. One may include the rational grid endpoints and the two prescribed tail labels in the fixed program. Choosing rational upper and lower approximations with constant-factor slack gives the asserted bounds without requiring an exact logarithm or square root in the online machine. A sequential lookup scans the finite table; binary arithmetic then appends or extracts a base- L digit. It uses $O(d \log L + \log(R/h))$ transient bits besides the read-only program and polynomial time in the table length and the register bit length. The physical analog-to-bin acquisition is an explicit interface operation, not claimed to be implemented for free by the register machine.

For completeness, certified values of m are available without an exact real oracle when σ^2 and the grid are rational. They are ratios of the integrals

$$\int_{-1}^1 u e^{-(x-u)^2/(2\sigma^2)} du, \quad \int_{-1}^1 e^{-(x-u)^2/(2\sigma^2)} du.$$

The denominator is at least $2e^{-(R+1)^2/(2\sigma^2)} > 0$ for $|x| \leq R$. Rational Taylor bounds for the exponential, combined with Riemann-sum error bounds from uniform derivative bounds on the compact rectangle, give rational intervals for both integrals of arbitrarily small width. Refine until the quotient interval has width at most ε , and choose its rational midpoint. If it lies near an equal-interval threshold, either adjacent label still has error at most $1/L + \varepsilon$ after enlarging the certified tolerance by a fixed factor, or the rational midpoint may simply be compared exactly to the rational thresholds. Termination follows from the denominator lower bound. The same construction for fixed computable σ uses certified intervals for it. No polynomial bound in the unspecified description of a general computable real is asserted. $\square$

The cardinality converse remains valid for this finite numerical signature because its implementations form a subclass of the unrestricted measurable machines used in that converse. Finite acquisition therefore preserves the delayed-query exponent when its error parameters are scaled as above, while displaying the additional resources that the cardinality theorem alone does not price.

13. POSITIVE OBSERVATION MODELS AND HISTORY REDUCTION

We give explicit model consequences in which the marked law, continuity and contraction hypotheses can be checked from primitive data. These results concern the models specified here. They do not identify a finite-state model with an unconstructed microscopic billiard or kinetic model.

Let $Z_t \in [n]$, $a \in A$ with A finite, and suppose

$$\epsilon \leq P_a(z, z') \leq M, \quad Y_{t+1} = m_{a, Z_{t+1}} + \sigma_{a, Z_{t+1}} \xi_{t+1}, \quad \xi_t \sim N(0, 1), \quad (13.1)$$

where $0 < \epsilon < M$, $|m_{a,z}| \leq m_*$ and $0 < \sigma_- \leq \sigma_{a,z} \leq \sigma_+$. All primitive coefficients may depend continuously on a compact unknown parameter, with the same bounds. The initial hidden probabilities and any finite marking kernel also depend continuously on this parameter. The observation is of the *new* state. Its joint transition/report kernel is

$$K_a(z', dy \mid z) = P_a(z, z') g_{a,z'}(y) dy.$$

Finite terminal or path marks are permitted.

Theorem 13.1 (A continuous-report consumer). *For every fixed horizon, the instruments (13.1) satisfy the compact dominated hypotheses of Theorem 3.3 for bounded losses continuous in the primitive parameter. Their common-encoder minimax problems have attaining private designs and attaining shared mixtures.*

Partition $[-R, R]$ into intervals of length at most h , where $R > m_$, and use two additional tail cells. Reconstruction is uniform on interior intervals and uses any fixed probability density on each tail. For every memory width and every admissible controller, the resulting marked finite-report approximation has risk error at most $Te(h, R)$, where*

$$e(h, R) = \frac{h}{2\sigma_-} \sqrt{\frac{2}{\pi}} + 2 \exp\left(-\frac{(R - m_*)^2}{2\sigma_+^2}\right). \quad (13.2)$$

The corresponding excess-risk error is at most $2Te(h, R)$ when baselines are recomputed.

Proof. Use a positive Gaussian probability as the reference measure on each report space and counting probability on finite marks. With an open-loop action word fixed, the joint marked density is a finite sum over hidden paths of products of transition probabilities, Gaussian densities and terminal marking probabilities. Gaussian location-scale densities depend continuously in L^1 on their parameters when the variance is bounded away from zero. This follows, for example, by pointwise convergence and equality of their integrals, or directly by truncating uniformly Gaussian tails and using uniform continuity on a compact interval. Products of probability densities are continuous in L^1 by telescoping, and so are finite mixtures. Compactness of the primitive parameter set gives compactness of the path-density family in L^1 . Multiplication by bounded continuously parameterized losses preserves the required effect continuity. Theorem 3.3 applies.

If I is an interval of length at most h and $\bar{g}_I$ is its average density, the fundamental theorem of calculus gives

$$\int_I |g - \bar{g}_I| \leq h \int_I |g'|.$$

Indeed average $|g(x) - g(y)|$ over $y \in I$ and bound the difference by the integral of $|g'|$ between x, y . For a Gaussian density, $\int |g'| = \sqrt{2/\pi}/\sigma$. On the two tails, replacing a conditional density changes total variation by at most their total probability, bounded by the exponential term in (13.2). Thus

each conditional observation channel has total variation error at most $e(h, R)$ uniformly in hidden state, control and parameter. Couple the hidden transition identically and maximally couple the original and reconstructed observations. As long as observations and random tapes have matched, both controllers make the same decisions. The probability of a first mismatch during T steps is at most $Te(h, R)$. On its complement the entire hidden path, all reports and the actual mark agree. Bounded losses change by at most this probability. The report conversions are transient operations, so the memory width is unchanged. Comparing the unrestricted baselines in the same way finishes the proof. $\square$

13.1. A proved forgetting estimate. For positive probability rows u, v , put

$$d_H(u, v) = \max_i \log(u_i/v_i) - \min_i \log(u_i/v_i).$$

Multiplying both rows by the same positive diagonal matrix and then normalizing leaves d_H unchanged. We use the elementary positive matrix contraction in its explicit form.

Lemma 13.2 (Positive-matrix contraction). *If the entries of a matrix P belong to $[\epsilon, M]$, then*

$$d_H(uP, vP) \leq \rho d_H(u, v), \quad \rho = \frac{M - \epsilon}{M + \epsilon} < 1.$$

For arbitrary nonnegative probability rows, the first images satisfy $d_H(uP, vP) \leq 2 \log(M/\epsilon)$.

Proof. Set $x_i = \log(u_i/v_i)$ and interpolate $v_i \exp(sx_i)$, $0 \leq s \leq 1$. The derivative of the logarithm of output coordinate j is the expectation of x_i under weights proportional to $v_i \exp(sx_i) P_{ij}$. For two output coordinates j, k , the likelihood ratio of their normalized weight distributions has maximum divided by minimum at most $(M/\epsilon)^2$. If two probability distributions have likelihood ratios in an interval $[a, b]$ with $b/a \leq C$, their total variation is at most $(\sqrt{C} - 1)/(\sqrt{C} + 1)$. To verify this bound, put all ratios below one at the lower endpoint and all ratios above one at the upper endpoint; the mass constraint makes the resulting variation at most $(1 - a)(b - 1)/(b - a)$. Maximizing over $b/a \leq C$, $a \leq 1 \leq b$, gives $a = C^{-1/2}$, $b = C^{1/2}$ and the displayed value. Differences of the two derivative expectations are therefore at most $\rho \text{osc}(x)$. Integrate in s and maximize over j, k . At $s = 0$ all output log ratios vanish, proving the first claim. For the second, every coordinate of either probability row times P lies in $[\epsilon, M]$; its coordinatewise ratio lies in $[\epsilon/M, M/\epsilon]$. $\square$

Theorem 13.3 (Uniform finite-history recovery). *Fix the primitive parameter. Two filters for (13.1), started with arbitrary initial probabilities and fed the same actions and observations for $L \geq 1$ steps, satisfy*

$$\text{TV}(\pi_L, \tilde{\pi}_L) \leq \frac{1}{2} \log(M/\epsilon) \rho^{L-1}.$$

In particular a last- L -history filter, initialized with a fixed probability at its left endpoint, approximates the full filter with this error uniformly in the earlier history. With N report cells, a literal suffix implementation uses at most $(|A|N)^L$ labels, besides a finite beginning-of-stream convention. For any $[0, 1]$ -valued terminal loss on the current hidden state, an optimal suffix-based action has risk at most twice this filtering error above the full-filter optimum.

Proof. Prediction multiplies a posterior row by P_a . Conditioning on the current report multiplies it by the positive diagonal matrix with entries $g_{a,z'}(y)$ and normalizes. The latter operation preserves d_H , and Lemma 13.2 contracts the former. The first prediction gives diameter at most $2 \log(M/\epsilon)$ and the next $L - 1$ predictions give the factor ρ^{L-1} . For probability rows, $\text{TV}(u, v) \leq \tanh(d_H(u, v)/4) \leq d_H(u, v)/4$; the first inequality follows from the same two-endpoint likelihood-ratio bound in the lemma. Restart the truncated filter with the fixed prior at time $t - L$ and apply this result to obtain the suffix estimate. Its raw suffix is updated by shifting and appending the current control/report cell. Finally every $[0, 1]$ loss expectation changes by at

most the total variation of the two posteriors. Compare the optimal action under each posterior through the other posterior, which costs at most twice that distance. $\square$

This contraction is not asserted for arbitrary past-path target marks: augmenting the hidden state to retain such a mark can destroy strict positivity. The general marked duality retains these marks without using the forgetting assertion.

The parameter must be fixed in this forgetting assertion. A static unknown phase need not be forgotten, and applying the assertion to its posterior would be false. The compact robust theorem applies to the unknown-parameter experiment without that extra assertion. Also, summable variations alone have not been used to infer exponential decay: here it follows from the explicit positive-matrix contraction. The distinction is important for history-kernel applications.

14. NONLINEAR FINITE-STATE SEMIGROUPS

Let Q^a be conservative generators on $[n]$, $a \in A$ finite, and let v_i^a be bounded potentials. Define on $\mathbb{R}^n$

$$H_i(u) = \max_{a \in A} \left\{ v_i^a + \sum_{j \neq i} q_{ij}^a (e^{u_j - u_i} - 1) \right\}. \quad (14.1)$$

This is a nonlinear Hamiltonian. The following proof does not apply a linear resolvent identity to it.

Theorem 14.1 (Global semigroup and nonlinear resolvent). *The equation $\dot{u} = H(u)$ has a global solution for every initial condition and defines a semigroup S_t that is order preserving, additively homogeneous and nonexpansive in the sup norm. If $V = \max_{i,a} |v_i^a|$, then $\|S_t u\|_\infty \leq \|u\|_\infty + tV$. For every $\lambda > 0$ the equation*

$$u - \lambda H(u) = f$$

has a unique solution $J_\lambda f$. The resolvent is order preserving, additively homogeneous and nonexpansive. In particular H is maximal dissipative with this sign convention.

Proof. The map H is locally Lipschitz as the maximum of finitely many smooth functions. If $u_i - v_i = \max_j (u_j - v_j)$, then $u_j - u_i \leq v_j - v_i$ for every j , and hence $H_i(u) \leq H_i(v)$. The upper right derivative of the maximum coordinate of the difference of two solutions is therefore nonpositive. Applying the same argument to the reverse difference proves sup-norm contraction and comparison. Since $H(u + c\mathbf{1}) = H(u)$, uniqueness gives additive homogeneity. Comparing with the time-dependent constant barriers $(\max_i u_i + tV)\mathbf{1}$ and $(\min_i u_i - tV)\mathbf{1}$ gives the norm bound. It prevents finite-time escape, so the locally defined solution is global; uniqueness gives the semigroup law.

For the resolvent, consider the damped equation $\dot{u} = f - u + \lambda H(u)$. The interval of constant barriers

$$(\min_i f_i - \lambda V)\mathbf{1} \leq u \leq (\max_i f_i + \lambda V)\mathbf{1}$$

is invariant. At a maximal difference between two solutions the upper derivative is at most the negative of that difference. Thus its flow contracts distances by e^{-t} . At time one this is a strict contraction of a complete invariant rectangle into itself, so it has a unique fixed point. Commutation with the flow shows that the fixed point is stationary: every later image is another fixed point of the time-one map. It therefore solves the resolvent equation. The same maximal-coordinate comparison applied to two resolvent equations gives $\max_i (u_i - \tilde{u}_i) \leq \max_i (f_i - \tilde{f}_i)$, which proves uniqueness, comparison and nonexpansiveness. Translation by constants proves homogeneity. $\square$

Theorem 14.2 (Marked mesh approximation and stability). *Define the log-moment Bellman step*

$$(F_\Delta u)_i = \max_a \log \left(e^{\Delta(Q^a + \text{diag } v^a)} e^u \right)_i.$$

For fixed t and bounded initial u , with $\Delta = t/m$, $F_\Delta^m u \rightarrow S_t u$ with error $O(\Delta)$, uniformly on bounded sets. Each mesh problem, after acquisition of the initial state, has finite report and control spaces and a bounded actual path-reward mark. Its bounded moment effects therefore have the common-encoder risk-body duality of Corollary 3.4, for compact continuously parameterized primitive data. The seed average is taken on these moments before taking a logarithm.

If $\tilde{Q}^a, \tilde{v}^a$ are another such family and

$$\delta_Q = \max_{i,a} \sum_{j \neq i} |q_{ij}^a - \tilde{q}_{ij}^a|, \quad \delta_v = \max_{i,a} |v_i^a - \tilde{v}_i^a|,$$

then on a common oscillation bound B for the trajectories,

$$\|S_t u - \tilde{S}_t u\|_\infty \leq t\{\delta_v + (e^B + 1)\delta_Q\}.$$

For the resolvents the same bound holds with t replaced by λ and B bounding their solutions.

Proof. The matrix exponential is a positive Feynman–Kac operator. On a bounded set of u , its Taylor expansion and the logarithm give $F_\Delta u = u + \Delta H(u) + O(\Delta^2)$, uniformly for the finite family of generators. Positivity makes F_Δ order preserving, and $F_\Delta(u + c\mathbf{1}) = F_\Delta u + c\mathbf{1}$. These two properties imply sup-norm nonexpansiveness. Its iterates and the exact flow remain in the same bounded region up to time t , by the constant barriers. The local flow expansion has the same first-order term and a uniform $O(\Delta^2)$ remainder. Telescope the errors over m steps using nonexpansiveness; the bound is $mO(\Delta^2) = O(t\Delta)$.

For a fixed control during one interval, the positive kernel $e^{\Delta(Q^a + \text{diag } v^a)}$ is the expectation of the exponential integrated reward with the terminal state indicator. Keeping the actual bounded reward integral as a mark, rather than interpreting this nonstochastic matrix as a transition probability, gives its proper probabilistic representation. Concatenation gives the mesh instrument. Finite sampled states and controls have a finite report representation; continuous reward marks can be integrated into its bounded effect functions. For this reward-maximizing problem, use a decreasing affine normalization of the bounded moment as a loss in $[0, 1]$ when applying the risk theorem. This reverses the optimization in the required way. For finite state/action paths this integration needs no assumption that reward marks have a common density: Corollary 3.4 applies directly. For finite parameter and task sets, one may also augment the finite hidden state by the sampled state/control path and replace the moment loss by its conditional expectation given that path. This gives a finite instrument with exactly the same moment risks, to which Theorem 7.1 applies. It is an equivalence for these moment effects, not a finite-state realization of every continuous reward mark. Continuity of the effects on a compact primitive family follows from continuity of the finite matrix exponentials and products.

On $\text{osc}(u) \leq B$, direct subtraction of (14.1) gives $\|H(u) - \tilde{H}(u)\|_\infty \leq \delta_v + (e^B + 1)\delta_Q$. The maximal-coordinate comparison used above now bounds the upper derivative of the solution difference by this constant. Integration proves the flow estimate. Subtracting the two resolvent equations at a maximizing coordinate proves its resolvent counterpart. The logarithm of an average exponential moment is not an average of logarithms; none of these steps changes that ordering. $\square$

The semigroup construction is a finite-state instance of the logarithmic control-semigroup method of Nisio [26] and risk-sensitive control. Its role here is an explicit nonlinear consumer with a proved range condition and a marked finite-mesh interface, not a claim that finite-dimensional comparison settles the domain and coercivity questions of an infinite-dimensional kinetic generator.

15. HIDDEN PHASES AND RISK-SENSITIVE CONTINUATION

Let C be a finite latent phase chosen once with positive initial weights, and let Z_t follow a controlled hidden model conditional on C . Its joint transition/report densities are $K_{c,a}(z', y \mid z)$

with respect to a probability measure $\mu(dy)$. Let $r_{c,a}(z, z')$ and the terminal reward $h_c(z)$ be bounded, and fix $\gamma > 0$. One controller must choose each action without observing C . The objective in this section is to minimize the positive exponential moment. A logarithm may be applied afterwards.

The exponential information-state mechanism is classical; see Bäuerle and Rieder [27]. We use the following explicit unnormalized version so that no phasewise policy is hidden in the notation. Define a nonnegative row η on (c, z) and the tilted update

$$(\eta K_a^\gamma(y))(c, z') = \sum_z \eta(c, z) K_{c,a}(z', y | z) e^{\gamma r_{c,a}(z, z')}. \quad (15.1)$$

Unlike an ordinary phase posterior, η includes the accumulated reward weight. Its total mass must not be discarded without a corresponding multiplicative value factor.

Theorem 15.1 (One-policy phase recursion). *For full-history controls, the minimum remaining exponential moment has the recursion*

$$V_T(\eta) = \sum_{c,z} \eta(c, z) e^{\gamma h_c(z)}, \quad V_t(\eta) = \min_{a \in A} \int V_{t+1}(\eta K_a^\gamma(y)) \mu(dy).$$

The value is positively homogeneous and Borel, and an optimal measurable selector exists. The minimum is outside the phase sum: replacing it by separate phasewise minima is in general incorrect. For finite-register common encoders, the exact moment risk body is instead obtained from the coupled design of Theorem 3.3 (or its finite specialization), not by providing η as free online memory. The model satisfies that theorem whenever its marked path densities and bounded effects satisfy the stated domination and continuity hypotheses; in particular finite phases with positive Gaussian observations and bounded finite-path rewards do so for fixed primitives, or for compact continuous primitive families as in Theorem 13.1, using Corollary 3.4 when only effect domination is needed.

Proof. For a fixed report/action history, $\eta_t(c, z)$ is the joint density of phase, current state and that report history, multiplied by the exponential accumulated reward and summed over past hidden states. This interpretation is true initially and propagates exactly by (15.1). At the terminal time summing the remaining exponential terminal factor gives V_T . At time t the action is one measurable choice from the current history; it cannot depend on an unobserved phase. For a fixed choice a , integrate the remaining value of the updated density against the reference observation law. Backward induction gives the recursion. The minimum of finitely many Borel integrals is Borel and the least-index minimizer is a measurable selector. Induction also gives positive homogeneity. This is the unnormalized information-state form of risk-sensitive dynamic programming; it accounts for the probability and reward weights together.

For a finite-register design, fix its action and register paths and expand the true marked path law. Its exponential reward is a bounded positive loss after affine rescaling. The same encoder kernels appear in every task and phase row, so the compact risk theorem applies without giving the controller the unbounded-dimensional history or an uncharged exact real vector. The Gaussian model's domination and compactness follow from the finite mixture proof in Theorem 13.1. This proves the last assertion. $\square$

Example 15.2 (Strict failure of phasewise optimization). There is no observation. The equally likely phase is $C \in \{0, 1\}$, the action is $a \in \{0, 1\}$, and the cost is $\mathbf{1}_{\{a \neq C\}}$. The best actual exponential moment is $(1 + e^\gamma)/2$, for any mixture of the two actions. Its logarithmic value is $\gamma^{-1} \log((1 + e^\gamma)/2)$. Choosing $a = C$ separately in each hidden phase would give moment one, but is not an admissible policy. The tilted recursion and the common-encoder risk body both retain the actual value.

16. A PRIOR-INDEPENDENT EXPERIMENT BEHIND PREDICTIVE COORDINATES

A future-test quotient relative to one Bayesian preparation need not preserve a parameterized experiment. The additional requirement is *likelihood saturation*. This section gives both a terminal construction and the extra recursion needed for a causal reconstruction. All assertions about measurable coordinates are modulo the stated common null sets.

Theorem 16.1 (A saturated posterior quotient). *Let K be compact metric, let X be standard Borel, and let $E = \{P_\theta : \theta \in K\}$ be continuous in total variation. Choose a dense sequence (θ_j) and weights $w_j > 0$ with $\sum_j w_j = 1$. Put $m = \sum_j w_j P_{\theta_j}$ and define*

$$T(x) = \left(w_j \frac{dP_{\theta_j}}{dm}(x) \right)_{j \geq 1} \in [0, 1]^{\mathbb{N}}. \quad (16.1)$$

There is one parameter-independent kernel R such that

$$R(T_{\#}P_\theta) = P_\theta \quad (\theta \in K). \quad (16.2)$$

Thus E and TE are statistically equivalent. The completed σ -field generated by T is minimal among sufficient statistics. A different dense sequence and strictly positive weights give the same minimal sufficient σ -field modulo m and the same experiment equivalence class, not necessarily the same coordinates. For the terminal cardinality signature, their marked decision spectra are identical for every budget and every family of bounded terminal parameter-decision marks, whose losses depend on the parameter and the action. An additional latent-target mark requires its own joint-law sufficiency condition. No computability or finite recursive representation of T is asserted.

Proof. If $m(B) = 0$, then $P_{\theta_j}(B) = 0$ for all j ; continuity and density imply $P_\theta(B) = 0$ for every θ . Choose Borel versions of the countably many densities. They give $\sum_j T_j = 1$ m -almost surely; modify them on one common null set if necessary. Disintegrate m given T to obtain the standard Borel conditional kernel $R(t, dx)$. For every j , the density of P_{θ_j} is T_j/w_j , so

$$P_{\theta_j}(B) = \int \frac{t_j}{w_j} R(t, B) T_{\#}m(dt) = R(T_{\#}P_{\theta_j})(B).$$

For arbitrary θ , choose $\theta_{j_k} \rightarrow \theta$. Both applying T and applying R contract total variation. Taking limits proves (16.2).

For minimality, let S be sufficient, in the conditional-law sense that one kernel R_S is a conditional law of X given S under every P_θ . Mixing shows that it is also the conditional law under m . The factorization

$$\frac{dP_{\theta_j}}{dm}(x) = \frac{d(S_{\#}P_{\theta_j})}{d(S_{\#}m)}(S(x))$$

follows by integrating both sides on arbitrary measurable sets using R_S . Thus every coordinate of T is $\sigma(S)$ -measurable. Conversely the preceding disintegration proves sufficiency of T . Mixtures obtained from different dense sequences are mutually absolutely continuous: each dominates every member of E , hence also the other mixture. Minimality then proves equality of the completed sufficient σ -fields.

Finally, any terminal encoder from TE is an encoder from E by composition with T . Conversely, an S -label encoder Q from E is implemented from TE by QR , with exactly the same label laws for all parameters. This uses only transient reconstruction randomization and at most S terminal labels. It is a kernel assertion, not an algorithm that stores a sampled infinite object in a finite register. The two sets of attainable terminal risks, and hence their spectra, coincide. $\square$

The theorem makes explicit a classical dominated-sufficiency mechanism [14, 12]. Its role here is to identify the precise terminal slice on which the prior-relative and parameter-uniform notions have the same resource profile. A predictive-test quotient has this property when its generated

σ -field contains the likelihood coordinates (16.1). Without that inclusion the conclusion is false, even for a two-point parameter set: observe $Y = \theta \in \{0, 1\}$ once, and allow only constant future tests. Their quotient is a point although the observed experiment distinguishes both parameters exactly. Predictive equivalence always refers to a declared family of future questions.

16.1. The causal factorization. We give a finite-horizon formulation to avoid any assertion that a terminal sufficient statistic is automatically a causal sufficient state. Let $H_t = (A_0, Y_1, \dots, A_{t-1}, Y_t)$ be the observed history. All policies have the same parameter-independent action kernels. Under a reference prior on θ , write $\bar{P}$ for the mixture law. Suppose Borel coordinates $S_t = S_t(H_t)$ satisfy the following *saturation conditions*, with versions fixed for every admissible history and action where they are used:

$$S_t = F_t(S_{t-1}, A_{t-1}, Y_t), \quad (16.3)$$

$$\bar{P}(dy \mid H_{t-1}, a) = k_t(dy \mid S_{t-1}, a), \quad (16.4)$$

$$\frac{dP_\theta|_{H_t}}{d\bar{P}|_{H_t}} = \ell_{\theta,t}(S_t). \quad (16.5)$$

The likelihood identity is required for each parameter, initially on a dense reference family and then for all parameters by finite-horizon variation continuity. These conditions hold uniformly over the policies under consideration. Alternatively one may assume domination and the identity directly for every parameter. The reference prior is not allowed to change with the policy during a reconstruction.

Theorem 16.2 (Causal reconstruction of a saturated quotient). *Under (16.3)–(16.5), there are parameter-independent kernels $L_t(dy \mid s, a, s')$ which reconstruct the observation stream from its successive quotient states, with the correct joint law under every admissible policy. Hence the original and quotient instruments have mutually exact causal simulations. If S_t has at most K_0 values at each time, both simulations may be implemented with K_0 internal states and the cardinality spectra obey the budget dilations of Theorem 2.2. If the quotient is not finite, the theorem asserts statistical and causal reconstruction only, with the storage of S_{t-1} explicitly charged; it does not assert a finite-register resource equivalence.*

Proof. Let $H_t^0(ds' \mid s, a)$ be the pushforward of $k_t(\cdot \mid s, a)$ by $F_t(s, a, \cdot)$. Disintegration, including the Borel variables (s, a) , gives

$$k_t(dy \mid s, a) = \int L_t(dy \mid s, a, s') H_t^0(ds' \mid s, a), \quad (16.6)$$

with $F_t(s, a, y) = s'$ under the resulting joint kernel. On histories with $\ell_{\theta,t-1}(s) > 0$, change of measure gives the actual joint conditional law of (Y_t, S_t) as

$$L_t(dy \mid s, a, s') \frac{\ell_{\theta,t}(s')}{\ell_{\theta,t-1}(s)} H_t^0(ds' \mid s, a).$$

In particular, conditioning further on s' removes the likelihood factor. On zero-likelihood histories the kernel may be chosen arbitrarily; such histories are not visited under that parameter. For parameters outside a countable reference support, equality of the constructed finite-horizon laws extends by total variation, or follows directly from the assumed domination identity.

Start at the declared initial state. Given a target policy on the reconstructed reports, feed its chosen action into the quotient instrument; on receiving s' , draw y with $L_t(\cdot \mid s, a, s')$, then replace s by s' . Induction proves equality of every finite parameter-indexed report/action law. This induction is essential: it reconstructs responses after the same actions, not after actions selected with hidden parameter information. In the forward direction, apply F_t recursively to the real reports. Both executions retain only the previous quotient state in addition to the target

encoder's own register. This costs $K_0 S$ states if that state has K_0 possible values. The general standard Borel version has no such finite bound, exactly as stated. $\square$

A likelihood statistic alone supplies (16.5), not (16.4). Conversely, a Bayesian predictive state supplies future report predictions, not necessarily all parameter likelihoods. The conjunction is the bridge. Sequential comparison of filtered experiments already has a substantial decision-theoretic history [15]; the present statement fixes the additional register cost of the reconstruction rather than claiming that sequential comparison itself is new.

For marks involving an additional latent target rather than the parameter, its conditional kernel must also be included in the predictive saturation condition. The causal reconstruction theorem does not infer equality of latent-target laws from equality of report laws alone.

17. A NECESSITY THEOREM FOR CAUSAL REALIZATION

We now characterize the causal constraint without imposing a tree geometry. The time dependence of a machine is retained, so the result does not replace a free clock by an uncharged stationary-state assumption. Fix a horizon T , finite report alphabets, one Bayesian law, and square-integrable Hilbert targets U_t . There is no control in this section. Let $\mathcal{H}_t$ be the finite set of positive-probability histories, $p_t(h) = \mathbb{P}(H_t = h)$, and $m_t(h) = \mathbb{E}[U_t \mid H_t = h]$. Let $\beta_t \geq 0$ be fixed weights.

A family of partitions $(\mathcal{P}_t)_{t=0}^T$, with $\mathcal{P}_0 = \{\emptyset\}$, is *successor compatible* if

$$h, h' \in C \in \mathcal{P}_t, \quad hy, h'y \in \mathcal{H}_{t+1} \implies hy, h'y \text{ belong to the same cell of } \mathcal{P}_{t+1}. \quad (17.1)$$

Only common positive-probability continuations enter this condition. For $C \in \mathcal{P}_t$ put $\bar{m}_C = p_t(C)^{-1} \sum_{h \in C} p_t(h) m_t(h)$ and define

$$\mathcal{V}(\mathcal{P}) = \sum_{t=1}^T \beta_t \sum_{C \in \mathcal{P}_t} \sum_{h \in C} p_t(h) \|m_t(h) - \bar{m}_C\|^2. \quad (17.2)$$

Theorem 17.1 (Recursive partition characterization). *For the cardinality signature, allowing randomized time-dependent updates and decoders and independent public randomness, the optimal weighted excess over the full-history Bayes risk equals*

$$\inf_{\#I_t \leq S} \sum_{t=1}^T \beta_t \mathbb{E} \|m_t(H_t) - \hat{U}_t\|^2 = \min_{\substack{\mathcal{P} \text{ successor compatible} \\ |\mathcal{P}_t| \leq S}} \mathcal{V}(\mathcal{P}). \quad (17.3)$$

Thus same-order realization of a sequence of static checkpoint optima is equivalent to the existence of successor-compatible partitions with the corresponding widths and variance bounds. This is a necessary as well as a sufficient condition. It is invariant under relabelling of histories; suffix closure is one sufficient construction of these partitions, not a necessary choice of coordinates.

Proof. A deterministic initialized machine partitions $\mathcal{H}_t$ according to I_t . Its update depends only on (t, I_t, y) , so its partitions satisfy (17.1). Given a state cell, the mean $\bar{m}_C$ minimizes its squared error by Hilbert orthogonality. The resulting excess is precisely (17.2).

Conversely, label each partition's cells by at most S labels. For each cell C and symbol y , send its label to the cell containing hy for any supported $h \in C$ with supported extension hy . Compatibility makes the answer independent of the choice of h . If there is no such extension, define the update arbitrarily. At time t decode the label as $\bar{m}_C$. Induction identifies the actual label with the cell of the actual history. There are finitely many partition families, so the minimum exists.

To include randomization, realize the finitely many update and decoder kernels by independent uniforms, and condition on their entire tape. For each tape, the rule is a deterministic time-dependent S -state machine. Independence of the tape and the experiment preserves the history law

and conditional means. The average of its weighted costs cannot be smaller than the deterministic minimum. Public independent randomness is handled in the same way. This freezing argument concerns one fixed Bayesian sum; it does not interchange a minimax task supremum with the average over tapes.

Unrestricted checkpoint encoders correspond to arbitrary partitions at each time, without (17.1). Hence they have cost $\sum_t \beta_t e_S^2(m_t(H_t))$. Formula (17.3) gives exactly the asserted equivalence of order comparisons, including the case when a static cost is zero. $\square$

Corollary 17.2 (Finite-command cell instruments). *Let $k_1, \dots, k_J$ be Borel functions on a parameter space with $k_j \geq 0$ and $\sum_j k_j = 1$, and fix a prior μ . At each time the raw cell is drawn independently conditional on the parameter. A command g from a prescribed finite menu in $[\eta, 1 - \eta]^J$, $0 < \eta < 1/2$, has a prescribed parameter-independent law ν_t . Cell j is reported as j with probability g_j and otherwise as the single failure symbol $\dagger$. Only the command and the reported symbol are observed. Fix a finite horizon, finite future-query menus with known likelihoods $H_{tq} \in [0, 1]$, independent query weights ω_{tq} , and nonnegative checkpoint weights β_t .*

For the persistent-cardinality signature, the optimal common causal encoder for the weighted sum of quadratic prediction excesses is exactly (17.3), with the finite history law and query means computed from the actual cell likelihoods. This applies to the finite-command specialization of the A1 finite-cell protocol [19]. It is an exact scalarized criterion, not an asserted solution of the different maximum-over-checkpoints criterion with private randomization.

Proof. For the observed pair (g, x) define

$$\lambda_{g,x}(\theta) = \begin{cases} g_x k_x(\theta), & x \in \{1, \dots, J\}, \\ \sum_j (1 - g_j) k_j(\theta), & x = \dagger. \end{cases}$$

For a history $h = ((g_s, x_s))_{s \leq t}$ its probability is $\prod_{s \leq t} \nu_s(g_s)$ times $\int \prod_{s \leq t} \lambda_{g_s, x_s} d\mu$. On positive-probability histories the mean of query q is

$$m_{tq}(h) = \frac{\int H_{tq}(\theta) \prod_{s \leq t} \lambda_{g_s, x_s}(\theta) \mu(d\theta)}{\int \prod_{s \leq t} \lambda_{g_s, x_s}(\theta) \mu(d\theta)}.$$

Take the Hilbert prediction vector with coordinates $\sqrt{\omega_{tq}} m_{tq}(h)$ and apply Theorem 17.1. All histories are finite, and the law used there is exactly the actual acquisition law above. A failure history contains $\dagger$, not the unobserved cell j ; the successor compatibility condition therefore requires one common update for that report. Integration of the given Borel likelihoods may require arbitrary real constants or offline work. Neither is claimed finite under the cardinality signature. Zero-evidence histories are omitted rather than given artificial posteriors. $\square$

Corollary 17.3 (Continuation obstruction). *Let g be a bounded readout on all finite words over a finite alphabet. For stationary deterministic machines required to reproduce g on every word, an exact S -state realization exists if and only if the relation*

$$h \equiv_g h' \iff g(hw) = g(h'w) \text{ for every finite } w \quad (17.4)$$

has at most S classes. More generally put $d_(h, h') = \sup_w \|g(hw) - g(h'w)\|$. Every stationary deterministic realization with uniform error at most ϵ has state fibres of d_* -diameter at most 2ϵ , and thus at least as many states as any 2ϵ -separated set of histories.*

Proof. Histories in one machine state have the same state after every common continuation. Exact output equality implies (17.4); for approximate outputs the triangle inequality gives the diameter bound. The relation (17.4) is a right congruence. Its classes, updated by appending the new symbol, form an exact machine, whose output is $g(h)$, well defined by the empty continuation. This is the output version of the classical finite-automaton congruence argument [16]; the finite-horizon, weighted-distortion characterization in Theorem 17.1 is the form needed here. $\square$

Example 17.4 (A two-valued checkpoint requiring exponentially many persistent states). At the first time observe a uniformly distributed word $X \in \{0, 1\}^d$. At the second time observe an independent uniform index $J \in \{1, \dots, d\}$ and predict $U = X_J$. The raw first acquisition costs d bits; it is not called a one-bit observation. The full-history conditional mean at the second checkpoint has only two values, so its static two-label error is zero. Nevertheless a causal exact predictor requires $S \geq 2^d$. Two different first-stage words in one state differ in a coordinate which can be queried with positive probability. The second-stage update cannot separate them. Retaining the whole word is sufficient.

There is also a quantitative obstruction. If e is the smallest error probability of an S -state predictor, $2 \leq S \leq 2^d$, then

$$e \geq h_2^{-1} \left(1 - \frac{\log_2 S}{d} \right), \quad (17.5)$$

where h_2 is the binary entropy and its inverse takes values in $[0, 1/2]$. Indeed, condition also on every independent public coding variable R . Since the first register has at most S values, $H(X | I_1, R) \geq d - \log_2 S$. Predicting each coordinate optimally from (I_1, R) gives errors $e_j \leq 1/2$, and the binary entropy inequality and concavity give

$$H(X | I_1, R) \leq \sum_{j=1}^d H(X_j | I_1, R) \leq \sum_{j=1}^d h_2(e_j) \leq d h_2 \left(d^{-1} \sum_j e_j \right).$$

The average on the right is a lower bound for any actual queried-bit error. This proves (17.5). If squared loss is used instead, its optimal value is at least $e/2$, since $p(1-p) \geq \min(p, 1-p)/2$. Thus the discrepancy persists for the Hilbert-loss slice of the spectrum. It is a temporal obstruction, not poor scalar quantization. The query must be answered after the update; J is not an additional persistent decoder input.

18. FINITE-PREFIX, FINITE-DESCRIPTION REALIZATION

The recursive partition theorem isolates a necessity condition. Predictive-energy trees provide a verifiable sufficient construction. The full tree theorem and its geometric hypotheses are proved in Section 23 below. We now strengthen its implementation: exact ordering of nearly tied real energies is unnecessary, and the raw acquisition need not expose an infinite word.

Use the factorial legal language, stationary cylinder masses p_v , energies a_v , scales $D_v = (a_v/p_v)^{1/2}$, and constants a_-, ρ, U, C_0, c_0 of that theorem. In particular $a_\emptyset = 1$, $a_{vi} \leq \rho a_v$ with $\rho < 1$, $a_{vi} \geq a_- a_v$, $D_{vi} \leq D_v$, and $a_{uv} < a_v$ for nonempty u . Legality and the child list are decidable. Suppose an oracle returns a positive rational upper estimate u_v satisfying, for a fixed $\kappa \geq 1$,

$$a_v \leq u_v \leq \kappa a_v. \quad (18.1)$$

No uniform cost for this oracle is hidden in a cardinality assertion. Set

$$\bar{a}_v = \min\{u_w : w \text{ is a contiguous factor of } v, \text{ including } \emptyset\}. \quad (18.2)$$

For $0 < \eta < 1$, split precisely the legal vertices for which $\bar{a}_v > \eta$. Retain all their legal children. Denote the resulting full finite tree by $\mathcal{T}_\eta$, its leaf cut by $\mathcal{C}_\eta$, and its number of leaves by l_η .

Theorem 18.1 (Robust tree realization with charged acquisition). *The rational-threshold construction is finite, suffix closed, and supports both deletion and prepend-and-parse insertion. Its depth and state count satisfy*

$$D_\eta \leq 1 + \left\lceil \frac{\log(\kappa/\eta)}{|\log \rho|} \right\rceil, \quad (18.3)$$

$$V_\eta = 1 + \sum_{v \text{ internal}} \deg^+(v) \leq (U+1)(2l_\eta - 1). \quad (18.4)$$

For constants depending only on the displayed structural bounds and κ ,

$$\sum_{v \in \mathcal{C}_\eta} a_v \asymp e_{l_\eta}^2(Z) \asymp G_{l_\eta}. \quad (18.5)$$

In the deletion experiment, at most D_η symbols of a snapshot acquisition suffice; no later access to that snapshot is used. The same average-risk order as the exact greedy construction follows. For every sufficiently large state budget S , a dyadic threshold can be selected with $V_\eta \leq S$, $l_\eta \geq cS$, and $D_\eta = O(\log(S+1))$.

If the online readouts are in a fixed bounded subset of $\mathbb{R}^d$ and representatives can be computed to prescribed accuracy, the selected machine has a finite rational presentation. Readout precision $p = O(\log(S+1))$ suffices to retain the risk order. A word list, transition table and readout table have total description

$$P = O(S \log(S+1) + Sdp) \quad (18.6)$$

for a fixed alphabet. Its online persistent storage is one of at most S states; a transition-table implementation uses transient storage $O(\log(S+1) + dp)$ bits. An acquisition uses $O(\log(S+1))$ symbol queries. Online table accesses can be implemented in word-RAM time proportional to these query counts; a sequential bit-table implementation gives the conservative bound $O(S(\log(S+1) + dp) \log(S+1))$ bit operations per acquisition. Offline oracle time is separate. The Hilbert-space cardinality assertion does not, by itself, encode an infinite-dimensional readout in finitely many bits.

Proof. If w is a factor of v , removing a prefix and then a suffix cannot decrease energy. Consequently

$$a_v \leq \bar{a}_v \leq \kappa a_v, \quad \bar{a}_{iv}, \bar{a}_{vi} \leq \bar{a}_v. \quad (18.7)$$

The second assertion follows by inclusion of factor sets and requires no accurate comparison between two distinct true energies. Thus the internal vertices form a prefix-closed and suffix-closed set. Each internal vertex is split into all its legal children. A proper suffix of a leaf is a legal child of a suffix of its parent, hence is present; repeating this argument proves full suffix closure. If iv had no leaf prefix when v is a leaf, it would be internal, forcing its suffix v to be internal, a contradiction. These are exactly the two update operations required.

At depth n , $\bar{a}_v \leq \kappa \rho^n$, which proves finiteness and (18.3). Every nonroot leaf has an internal parent, so

$$a_{-}\eta/\kappa < a_v \leq \eta. \quad (18.8)$$

The exact vertex count follows by counting edges; contracting unary chains gives (18.4) as in the full-tree lemma below.

Here is a lower bound not depending on exact greedy ordering. Refine all leaves by $h = 3(U+1)$ further legal levels. Each leaf has at least 2^3 descendants at that level, because there are at most U consecutive unary vertices. The resulting incomparable cylinders number at least $8l_\eta$. Every one has energy at least $a_-^{h+1}\eta/\kappa$. The separation hypothesis and monotonicity of D make the open tubes of radii δD_w about their images disjoint for $2\delta < c_0$. At most l_η tubes can contain one of l_η arbitrary proposed Hilbert centres. Each unoccupied cylinder contributes at least $\delta^2 a_w$ to the squared error. Hence $e_{l_\eta}^2(Z) \geq cl_\eta \eta$. One representative per original leaf gives $e_{l_\eta}^2(Z) \leq C_0^2 \sum a_v$, and (18.8) proves the first comparison in (18.5). The second is the static conclusion of Theorem 23.1, whose proof does not use the present theorem. Its oscillation estimate in the deletion orientation, or its cylinder diameter estimate in the insertion orientation, also proves the stated risk bounds for this cut.

A prefix parser stops on reaching a leaf, after at most D_η queries. It never reads or stores the rest of the snapshot. This is a finite-latency acquisition convention: the snapshot is available during that update and the prediction deadline permits these microsteps. No claim is made for a

moving physical state observed after an unmodelled read delay. Insertion and deletion then use the already proved closure.

For budget selection, use $\eta_j = 2^{-j}$ and take the last threshold with $V_{\eta_j} \leq S$. There is a constant bound on $V_{\eta_{j+1}}/V_{\eta_j}$. In fact, beneath a leaf at threshold η , all descendants at depth k have estimated energy at most $\kappa\rho^k\eta$, and so the next threshold adds at most k levels once $\kappa\rho^k \leq 1/2$. The ratio is bounded by a fixed geometric sum of powers of the alphabet size. It follows that the chosen V_η is at least a fixed multiple of S , and then (18.4) gives $l_\eta \geq cS$. For completeness, a leaf at depth n has $a_v \geq a_-^n$, so $n \geq \log(1/\eta)/|\log a_-|$. Every legal path branches at least once in $U + 1$ levels. The complete cut therefore has at least $2^{\lfloor \log(1/\eta)/((U+1)|\log a_-|) \rfloor}$ leaves. Since $l_\eta \leq S$, this bounds $\log(1/\eta)$ by $O(\log(S + 1))$. Combine with (18.3). Fixed-dilation comparison for G then converts (18.5) into the same- S order.

Finally replace each finite-dimensional readout by a rational vector within Euclidean distance δ . Pointwise squared loss is at most twice its previous value plus $2\delta^2$. Take $\delta^2 \leq \eta$ and use $\sum a_v \geq a_- l_\eta \eta / \kappa$. Bounded readouts require $p = O(\log(1/\eta) + \log(d + 1))$ binary digits per coordinate. Store all vertex words, their legal child/deletion/insertion transitions, and these rational vectors. There are $O(S)$ transitions for a fixed alphabet, each state index costs $O(\log(S + 1))$ bits, and the word-list depth is $O(\log(S + 1))$. This proves (18.6). A streaming parser uses only its current vertex index during the acquisition; no prefix string survives the update. The stated time bounds follow by random access, or by a full sequential scan for each table access. The number and precision of the offline queries can be listed, but (18.1) alone gives no uniform running-time bound for answering them. This is why offline time is not absorbed into (18.6). $\square$

The use of all factors in (18.2) is essential. Independent relative approximations u_v need not respect suffix monotonicity, even when the true energies do. Thresholding the raw approximations can therefore produce a tree that cannot execute its own suffix update. The envelope repairs that defect while changing energies by at most the fixed factor κ .

19. THE MEASURE-VALUED SLICE: SHARP PRESENTATION ENTROPY

The decoder's output type is part of a mark. In particular, outputting a probability law close in expected Wasserstein distance is stronger than simulating one randomly chosen observation from that law. This separation is important for a genuinely measure-valued predictive state.

Let $\mathcal{M} = \mathcal{P}([0, 1])$ with W_1 . A *uniform law presentation* receives $\pi \in \mathcal{M}$ as a transient input and returns one of S persistent labels. Its decoder outputs a law $\hat{\pi}$. The fixed code, including its representative laws, is chosen once for all π . In particular the unknown input is not inserted into the decoder's free program constants. Let

$$H_S = \inf_{Q, D} \sup_{\pi \in \mathcal{M}} \mathbb{E} W_1(\pi, D(Q(\pi))). \quad (19.1)$$

Both maps may use independent private or public randomness. The criterion is the expected distance of the *realized* decoded law. Only persistent cardinality is charged here; the input-law interface and general real-law computation are unrestricted. Regard π as the unknown parameter and the uncompressed observation as π itself. Then H_S is the spectrum of the single bounded minimax law-presentation mark, with full-observation baseline zero. It is not a supremum of point-prior Bayes risks: such a prior would reveal the unknown law to the decoder and change the problem.

Theorem 19.1 (Uniform presentation of a probability measure). *There are absolute $c, C > 0$ such that*

$$\frac{c}{\log(S + 1)} \leq H_S \leq \frac{C}{\log(S + 1)}, \quad S \geq 1. \quad (19.2)$$

Moreover no continuous map from $\mathcal{M}$ into a fixed $\mathbb{R}^k$ can be an exact coordinate sufficient to evaluate every bounded Lipschitz integral on all of $\mathcal{M}$. This statement does not exclude noncontinuous Borel encodings of a standard Borel space into one real coordinate.

Proof. For the upper bound use the explicit net in Lemma 25.1 below: equally weighted n -point empirical laws with points on the grid $\{0, 1/n, \dots, 1\}$, in increasing order. There are $\binom{2n}{n} \leq 4^n$ such laws. Quantile discretization followed by grid rounding gives covering radius $3/(2n)$. Taking $n = \lfloor \log S / \log 4 \rfloor$ proves the upper bound for $S \geq 4$; increase C for smaller S .

For the lower bound, partition $[0, 1]$ into n equal intervals. In each interval put mass $1/n$ at either its quarter point or its three-quarter point. Denote the resulting law by π_ω , $\omega \in \{0, 1\}^n$. Their quantile functions give the exact identity

$$W_1(\pi_\omega, \pi_{\omega'}) = \frac{d_H(\omega, \omega')}{2n^2}. \quad (19.3)$$

Greedily choose binary words and delete Hamming balls of radius $\lfloor n/4 \rfloor$. The selected words have mutual distance greater than $n/4$, and their number is at least $2^{c_0 n}$, where $c_0 = 1 - h_2(1/4) > 0$. Indeed $\sum_{j \leq n/4} \binom{n}{j} \leq 2^{nh_2(1/4)}$, as follows by bounding these binomial terms using the binomial expansion at parameter $1/4$. Thus the selected measures have mutual W_1 distance greater than $1/(8n)$.

Choose $n \leq C_0 \log(S + 1)$ sufficiently large that the packing has at least $2S$ members. Give it the uniform prior. For any S proposed law centres, their open balls of radius $1/(16n)$ contain at most S packing members in total. At least half the prior therefore incurs distance at least $1/(16n)$, no matter how it is encoded. Fixing all independent coding randomness gives this same bound for each tape; integration preserves it. The worst input risk is at least the uniform prior risk, which proves the lower bound in (19.2).

For the last assertion, bounded Lipschitz integrals separate probability measures, so a sufficient coordinate map must be injective. The measures supported on any $k + 2$ distinct points form a $(k + 1)$ -simplex. A small sphere about its barycentre lies in its relative interior and is homeomorphic to S^k . By the Borsuk–Ulam theorem every continuous map of this sphere to $\mathbb{R}^k$ identifies two antipodal points [18]. Hence no such injective coordinate exists. For $k = 0$ two distinct measures already give the contradiction. $\square$

The entropy mechanism is consistent with the power-exponential geometry of Wasserstein spaces studied in [17]; the packing argument is supplied here because the exact presentation loss, including randomization, matters for the result.

Corollary 19.2 (A nonlinear posterior class with a matching bound). *Consider the nonlinear binary instrument of Corollary 25.3 below, with*

$$h_0(x) = x/5 + x^2/50, \quad h_1(x) = 3/5 + h_0(x), \quad \Phi_y \pi = \frac{3}{4}(h_y)_\# \pi + \frac{1}{4}(h_{1-y})_\# \pi.$$

For each fixed report y ,

$$\frac{1}{5} W_1(\pi, \nu) \leq W_1(\Phi_y \pi, \Phi_y \nu) \leq \frac{6}{25} W_1(\pi, \nu). \quad (19.4)$$

Consequently, for every fixed t , a single finite-register scheme that must present the posterior for every input prior has worst expected W_1 error at least $c5^{-t}/\log(S + 1)$ at time t , even if the decoder is additionally told the reports. A uniform recursive presentation has error $O(1/\log(S + 1))$ at all times, when the initial law is encoded at the same accuracy and the update error is of that order. No fixed finite-dimensional continuous exact predictive coordinate exists on this all-priors posterior class. Constants in the lower bound are not asserted uniform as $t \rightarrow \infty$ for one fixed preparation.

Proof. The images of h_0 and h_1 lie in two disjoint ordered intervals. The mixture weights for a fixed y agree for π and ν . Quantile coupling therefore gives

$$W_1(\Phi_y \pi, \Phi_y \nu) = \frac{3}{4} W_1((h_y)_\# \pi, (h_y)_\# \nu) + \frac{1}{4} W_1((h_{1-y})_\# \pi, (h_{1-y})_\# \nu).$$

The derivatives of both increasing maps lie between $1/5$ and $6/25$, proving (19.4). Iterate it along any length- t report word and apply the same packing as in the preceding proof. The reports of this instrument are independent fair bits, with a law independent of the initial prior. Conditioning on the report word thus does not change the uniform packing prior. The centre-counting lower bound survives even with this word supplied freely to the decoder; integrating it over words and coding tapes proves the assertion for the actual, less informed decoder.

The recursive upper bound follows from Theorem 25.2 with $\rho = 6/25$, initializing by a net point rather than by an arbitrary fixed law. The initial error is at most $3/(2n)$ and the accumulated error is bounded by $[3/(2n) + \eta]/(1 - 6/25)$. Every fixed-word map on laws is continuous and injective by the lower bound in (19.4). Its restriction to the compact simplex used above is a homeomorphism onto its image. Applying the same sphere argument proves the coordinate obstruction. $\square$

Proposition 19.3 (Marginal-probe simulation is a different presentation problem). *Suppose only one Bernoulli probe is required: after compression a function $g : [0, 1] \rightarrow [0, 1]$ with Lipschitz constant at most one is specified, and the desired output law is $\text{Bern}(\pi g)$. The probe is fixed independently of the realized encoder randomness. For $S \geq 2$ there is a common S -label encoder whose output-law error is at most $1/[2(S-1)]$, uniformly in π, g . Thus this marginal simulation mark cannot be substituted for the realized-law mark in (19.1).*

Proof. Sample $X \sim \pi$ transiently, store the nearest point Z of the S -point uniform grid, and discard X . On receiving g , output a Bernoulli variable with parameter $g(Z)$. Its unconditional law is Bernoulli with parameter $\mathbb{E}g(Z)$, and $|\mathbb{E}g(Z) - \pi g| \leq \mathbb{E}|Z - X| \leq 1/[2(S-1)]$. The encoder is chosen before g . It does not produce a realized law close to π in W_1 : its natural decoded law is δ_Z . Nor does the argument cover probes chosen after observing Z , or an unlimited transcript of independent samples. Those are different tasks and interfaces. This output-law simulation criterion is not identified with a realized-law decision loss. $\square$

20. A DYNAMIC FINITE-EXPERIMENT CONSEQUENCE

The decision-complete slice of the same spectrum has a different geometry. We first make its temporal register convention explicit, and then connect it to the physical count experiment.

Fix $n, r \geq 1$, a compact $K \subset \mathbb{R}^r$ with nonempty interior, and a positive definite covariance Γ . At successive times observe independent

$$Y_t \sim N(\theta_t, \Gamma), \quad \theta = (\theta_1, \dots, \theta_n) \in K^n. \quad (20.1)$$

Each Y_t is discarded after the update. No earlier output labels are retained outside the register. At the terminal time only its final label is supplied to a parameter-independent reconstruction kernel. This is not an S -symbol channel whose entire transcript is available.

Theorem 20.1 (Sequential accumulation under a total register budget). *For the cardinality signature with persistent S states and at most S terminal outcomes, the decision-complete terminal spectrum of (20.1) satisfies*

$$c_n S^{-2/(nr)} \leq \Theta_{G^{(n)}}(S; \mathcal{D}_{\text{all}}) \leq C_n S^{-2/(nr)}. \quad (20.2)$$

The infimum is over causal encoders obeying the total state budget at every time. Lower bounds include arbitrary randomized, time-dependent encoders. A deterministic causal encoder attains the upper order. Constants are for fixed n, r, K, Γ , not uniform in a growing horizon. The readout/reconstruction randomness is external and fresh and private, while the physical alphabet

and the persistent register remain finite. No public tape or other side information is supplied alongside the final label in this theorem.

Proof. The full observation vector is a Gaussian location experiment of dimension nr , with compact mean set K^n and covariance $I_n \otimes \Gamma$. Every final register gives an experiment with at most S outcomes, regardless of its causal implementation. The lower bound of Theorem 26.1, proved below, applies to all such experiments. Because a register is a garbling of the full observation, its forward deficiency is zero. Theorem 2.4 therefore identifies its reverse deficiency with its worst normalized Bayes regret, proving the lower bound in (20.2).

For the upper bound put $J = \lfloor S^{1/(nr)} \rfloor$. At time t compactify each coordinate of Y_t by $\tau(x) = 1/2 + \arctan(x)/\pi$, quantize into J cells per coordinate, and append the resulting r -tuple of cell indices to the current register word. At that time there are $J^{rt} \leq J^{rn} \leq S$ possible words. The known absolute time determines their length; it carries no data-dependent information. Thus a single S -value register, not n separately uncharged registers, implements the rule. At the final time apply the positive tensor-hat reconstruction of Lemma 26.3 in dimension nr . For $J \geq 2$ its error is $O(J^{-2}) = O(S^{-2/(nr)})$. Enlarge the constant for the remaining finite budget range. This proves the upper bound using a genuine causal encoder. Its real-valued input computation has no finite-time claim under this signature. $\square$

If instead the sequence of emitted cell labels were supplied to the terminal decoder, its alphabet would be their product. In that different model a small current register could leave a large data-bearing transcript. The lower bound above is about the total retained experiment, not the size of the most recent communication symbol. This is one reason zero-delay channel coding and total-register coding require separate resource signatures [3, 4].

20.1. A theorem-level downstream arrow. We now give the adapter in the coordinates of the retained known-mark exposure protocol, rather than identifying all of the algebraic paper with this statistical subexperiment.

Corollary 20.2 (Known-mark exposure adapter). *Fix positive exposure centre λ^0 , known marks, a derivative matrix D of full row rank, and $\Lambda = \text{diag}(\lambda^0)$. Put $\Sigma = D\Lambda D^\top$ and $B_* = \Lambda D^\top \Sigma^{-1}$. Let L be a fixed normal-compression map of rank $r \geq 1$, let F be an orthonormal frame for its range, and set*

$$C = F^\top L D, \quad A = F^\top L, \quad \Gamma = C A C^\top.$$

*On a fixed compact index set H , observe independent counts of means $N\lambda_N(h)$, where $\lambda_N(h) = \lambda^0 + N^{-1/2}B_*h$. If AH contains an r -dimensional ball, the deterministic unjittered statistic $T_{N,J}$ of Section 27, applied to the score $C(N - N\lambda^0)/\sqrt{N}$, has at most $J^r \leq S$ outputs. For $\mathcal{G} = \{N(Ah, \Gamma) : h \in H\}$, it satisfies*

$$cS^{-2/r} \leq \Delta(\mathcal{E}_{N,J}, \mathcal{G}) \leq C_K S^{-2/r}, \quad J = \lfloor S^{1/r} \rfloor, \quad 2^r \leq S \leq KN^{r/6}. \quad (20.3)$$

Only the physical output cardinality and the stated joint sample/register window are bounded here. Reconstruction randomization is external, parameter independent and private; no finite-bit complexity of the raw real score computation is asserted.

Proof. $DB_* = I$, so $CB_* = A$. Since $F^\top L$ has row rank r and D is surjective, C has row rank r and $\Gamma > 0$. The compact local alternatives have uniformly positive exposures for all sufficiently large N . Thus every hypothesis of Theorem 27.1 is verified with these actual score weights, mean set and covariance. That theorem supplies both inequalities in (20.3); its lower bound allows every experiment with at most S outcomes, not only this statistic. Theorem 2.4 gives the all-decision interpretation of the limiting obstruction, and Theorem 2.2 transports each bounded task through the comparison kernels. No algebraic primary-decomposition theorem is a premise of this argument. $\square$

The left inequality concerns every experiment with at most S terminal outputs approximating this specified Gaussian family, and therefore also the physical statistic. The upper inequality concerns that explicit statistic and its positive reconstruction kernel. In particular, achieving Gaussian-family distance at most $CN^{-1/5}$ forces $S \geq cNr^{10}$, and budgets of that order suffice for this statistic. These are label-budget statements in the displayed joint window; they are not a sample-optimality theorem over all finite- N physical objectives.

Theorem 2.4 supplies a decision interpretation of the limiting experiment's finite-output obstruction, while Theorem 2.2 transports each bounded task through the Poisson comparison kernels. These are genuine arrows between specified experiments. They neither require nor assert a Sinai local limit theorem, a particle path large-deviation principle, or a physical phase construction.

SUPPORTING THEORY AND MODEL VERIFICATION

The following sections give the complete proofs used above. Unless a more detailed signature is specified, they use the cardinality signature of Section 2: only persistent states or physical terminal outcomes are bounded, with arbitrary transient computation and independent reconstruction randomness. In the tree application, Theorem 18.1 additionally replaces the full-word acquisition by a charged finite-prefix interface. In the measure application the new matching lower bound is for the uniform realized-law presentation task, not for a single marginal probe. The Gaussian and Poisson inequalities concern the indicated terminal experiment family. These local conventions are part of their use here.

21. POSITIVE EXPERIMENTS AND PREDICTIVE QUOTIENTS

21.1. Actual histories. All spaces of states, observations and actions are standard Borel. For an unknown parameter λ , a controlled experiment has a probability kernel

$$K_t^\lambda(dx', dy, dc \mid x, a), \quad (21.1)$$

where c is the cost of the actual step. Failure, rejection and stopping are outcomes, not missing probability mass. A policy chooses a_t from the observed past and independent randomization. Preparation has a specified law and cost. With an absorbing symbol after stopping, repeated kernel integration constructs one probability law on the infinite history space. Its finite marginals are consistent because every kernel in (21.1) has mass one. This is the usual kernel extension theorem [2]; conditioning on success would define a different experiment unless its preparation and waiting costs were separately charged.

For a fixed Bayesian preparation, let π be a reachable belief state on the hidden state, including the parameter when it is random. The positive instrument associated with an action and observation event is

$$\mathsf{T}_{a,B}\pi(E) = \int K(E, B, dc \mid x, a) \pi(dx).$$

After an observed event of positive probability, normalization gives the updated belief. For nonatomic reports we use a specified disintegration, with all equalities interpreted outside the appropriate null sets. A formula for an event of probability zero is not an additional observation channel.

21.2. The future-test quotient. A future test consists of a finite observation-based policy, a stopping rule, and a bounded measurable terminal payoff. Its conditional expectation at a belief π is written $\pi(v)$, where v is the corresponding backward test function. Fix a family $\mathcal{V}$ containing constants and closed under the one-step pullbacks of the allowed tests. It may, for example, contain all bounded finite-horizon executable tests. Define

$$\pi \sim_{\mathcal{V}} \pi' \iff \pi(v) = \pi'(v) \quad (v \in \mathcal{V}). \quad (21.2)$$

The set of classes is the predictive quotient. The choice of tests is part of the experiment specification: identifying a hidden state is not required if none of the designated tests distinguishes it.

Proposition 21.1 (Minimality and descent). *The quotient (21.2) is the coarsest set-valued statistic that determines all designated predictions. If the test family is one-step closed, normalized Bayesian updates descend to the quotient where their normalization is positive. If, in addition, the reachable belief domain B is compact metrizable, a countable family of continuous tests determines (21.2), and each fixed-report update is continuous on its positive-probability domain, the quotient has a canonical metrizable realization as a subset of $\mathbb{R}^{\mathbb{N}}$, and each such update descends continuously on compact subdomains on which its denominator is bounded away from zero.*

Proof. Map π to the vector of all its test expectations. Its fibres are exactly (21.2). Any statistic determining the expectations has fibres contained in these fibres, so the vector factors through that statistic as a set map. For the update, the numerator of a future prediction is a one-step pulled-back test and the denominator is the pulled-back constant. Equivalent states have equal numerators and denominators, hence equal normalized predictions.

In the countable continuous case use $q(\pi) = (\pi(v_j))_{j \geq 1}$. After scaling the tests to be bounded by one, its range is a compact subset of $[-1, 1]^{\mathbb{N}}$, with its product metric. A continuous map from a compact space into a Hausdorff space is a quotient map. The fibre-respecting continuous update therefore factors continuously on the stated compact subdomain. No assertion of a Borel cross-section, or of a smooth quotient for arbitrary Borel tests, is used. The minimality statement in the first paragraph alone would not imply either assertion. $\square$

The proposition is relative to the prepared Bayesian experiment. An identity for one prior is not a parameter-uniform comparison of the family $\{K^\lambda\}_\lambda$. All experiment distances below explicitly take the supremum over the parameter when that stronger meaning is intended.

21.3. Attainable Bayes geometry. Let U be square integrable in a separable Hilbert space H , let $\mathcal{H}$ be the actual information, and put $m = \mathbb{E}[U \mid \mathcal{H}]$. Define

$$e_M^2(m) = \inf_{z_1, \dots, z_M \in H} \mathbb{E} \min_i \|m - z_i\|^2.$$

Independent encoder/decoder randomization is permitted, but a retained observation is not a free input after encoding.

Proposition 21.2 (Checkpoint geometry and information loss). *The optimum squared loss of a single M -label checkpoint statistic is*

$$\mathbb{E}\|U - m\|^2 + e_M^2(m). \quad (21.3)$$

For any sub- σ -field $\mathcal{C} \subset \mathcal{H}$, posterior laws of λ satisfy the barycentre identity $\pi_{\mathcal{C}} = \mathbb{E}[\pi_{\mathcal{H}} \mid \mathcal{C}]$. Consequently convex posterior functionals decrease under coarsening. Whenever the quantities are finite,

$$I(\lambda; \mathcal{H}) - I(\lambda; \mathcal{C}) = \mathbb{E} D_{\text{KL}}(\pi_{\mathcal{H}} \parallel \pi_{\mathcal{C}}). \quad (21.4)$$

In a differentiable dominated family, the score of a statistic S is $\mathbb{E}[s_\lambda \mid S]$, and the lost Fisher information matrix is $\mathbb{E} \text{Cov}(s_\lambda \mid S)$, provided differentiation and conditioning are justified in L^2 .

Proof. Conditional orthogonality gives $\mathbb{E}\|U - z\|^2 = \mathbb{E}\|U - m\|^2 + \mathbb{E}\|m - z\|^2$ for every $\mathcal{H}$ -measurable output, and also after adding independent randomness. Freeze that randomness. There are at most M decoder outputs, so nearest-centre assignment is no worse than the given encoding. Conversely every finite codebook has a measurable nearest assignment, with deterministic tie-breaking. Taking infima proves (21.3), without assuming attainment of an optimal codebook. Conditional expectation against bounded parameter tests gives the barycentre identity. Conditional

Jensen proves the convex assertion. The chain rule for the logarithm of posterior densities gives (21.4); finite partitions followed by monotone approximation give the same assertion under the stated finiteness condition. Differentiating the marginal density of S gives the score identity. The matrix law of total variance gives the Fisher identity. These are the classical decision and information identities in [1, 11]; their role here is to specify the geometry being implemented, not to certify recursive closure. $\square$

21.4. Normalized response and observation modules. Let b_θ be a differentiable finite positive measure with $b_\theta(1) > 0$, and let $\pi_\theta = b_\theta/b_\theta(1)$. Differentiating this actual normalization gives

$$\partial_\theta \pi_\theta(f) = \frac{\partial_\theta b_\theta(f) - \pi_\theta(f) \partial_\theta b_\theta(1)}{b_\theta(1)}. \quad (21.5)$$

If the derivative has score density s relative to b_θ , the right side is $\text{Cov}_{\pi_\theta}(f, s)$. Thus a response direction is invisible exactly when its centered score annihilates the linear span of the designated tests. If the observation protocol consists of products from two declared linear test spaces V, W , its null space is $(VW)^\perp$, with $VW = \text{span}\{vw : v \in V, w \in W\}$. This last statement follows immediately from linearity, but it requires that those products really are the observed probes. It does not identify an algebraic multiplicity with a physical noise variance. Formula (21.5) and this annihilator are the normalization and observation-module inputs used to type a contact experiment.

22. CAUSAL MORPHISMS AND CHARGED APPROXIMATION

A resource signature specifies ordered coordinates, for example persistent cardinality, numerical precision, transient workspace, program description, preparation count and physical time. A budget transformer is a monotone map between two such signatures, accompanied by an actual implementation. A coordinate may explicitly be unlimited. The persistent-cardinality results below leave fixed programs, fixed real constants, time and transient computation unlimited unless a more specific implementation is given. There is no claim that these resources are equivalent.

A causal simulation from an experiment $\mathcal{E}$ to $\mathcal{F}$ consists of a source controller, a causal transducer and its initialized state. For every target policy the joint construction must produce the target observation/cost history within a declared parameter-uniform total-variation error. The transducer never sees the unknown parameter or the unobserved physical state. Resource bounds are imposed on this joint construction, not just on the terminal observation kernel. We use $\|P - Q\|_{\text{TV}} = \sup_A |P(A) - Q(A)|$.

Theorem 22.1 (Typed simulation and recursive approximation). *The following statements hold.*

(i) *Serial causal simulations of errors ϵ_1, ϵ_2 and budget transformers ρ_1, ρ_2 compose with error at most $\epsilon_1 + \epsilon_2$ and transformer $\rho_1 \circ \rho_2$. A simulator with K persistent states and a target observer with M states can be run with KM persistent states. For losses in $[0, L]$, the risk changes by at most $L(\epsilon_1 + \epsilon_2)$. Other resource coordinates are transformed according to the supplied implementations. The identity simulation has zero error and the identity transformer. Independent parallel simulations have additive TV error and the product register, with preparations and other costs charged by their joint implementation.*

(ii) *Suppose the true predictive state lies in a metric space (S, d) and updates by $s_t = F_t(s_{t-1}, y_t, a_t)$. For all reports and actions suppose F_t is ρ -Lipschitz in its state, $\rho < 1$. Let an implementable approximate update $\tilde{F}_t$ have local error at most η_t , and let $Q_t : S \rightarrow C_t$ be a measurable finite quantizer with radius ϵ_t . The register*

$$\hat{s}_t = Q_t \tilde{F}_t(\hat{s}_{t-1}, y_t, a_t) \quad (22.1)$$

has at most $|C_t|$ states at time t and, on every common observation path,

$$d(s_t, \hat{s}_t) \leq \rho^t d(s_0, \hat{s}_0) + \sum_{j=1}^t \rho^{t-j} (\epsilon_j + \eta_j) =: b_t. \quad (22.2)$$

A data-dependent persistent numerical residue must be part of C_t ; it cannot be placed in an uncharged side register. If the report kernels are L_0 -Lipschitz in total variation as functions of the predictive state, the length- n simulated history differs by at most $L_0 \sum_{t \leq n} \mathbb{E} b_t$. The same statement holds up to a bounded stopping time, with the stopped sum. If the true conditional mean is $g(s_t)$ with g L_1 -Lipschitz into a Hilbert space, its implemented prediction has Bayes excess at most $L_1^2 \mathbb{E} b_t^2$.

Proof. Compose the two transducers on their joint initialized state. Applying a Markov kernel contracts total variation, so the triangle inequality gives the sum of errors, uniformly over the policy and parameter for which the two certificates were supplied. The joint register is the Cartesian product. This proves the cardinality bound, and integration of a loss with range $[0, L]$ proves the risk bound. The identity is the unchanged history. For independent parallel laws, insert one product factor at a time and use TV contraction to obtain the same sum bound. The composition order of the budget maps follows from first realizing the target in the intermediate experiment and then realizing that implementation in the source.

For (ii), insert $F_t(\hat{s}_{t-1}, y_t, a_t)$ between the two updates. Contraction, the local defect and the quantization radius give $d(s_t, \hat{s}_t) \leq \rho d(s_{t-1}, \hat{s}_{t-1}) + \eta_t + \epsilon_t$. Induction proves (22.2). For the law comparison, telescope the products of the history kernels, replacing one kernel at a time. The prefix before the replaced kernel is distributed according to the true experiment; its integrated local TV error is at most $L_0 \mathbb{E} b_t$. All suffix kernels are contractions in TV. Summing proves the bound. An absorbing stopped state proves the stopped version by the identical calculation. The policy is evaluated on the same history in each local comparison: no continuity of a possibly discontinuous policy is presumed. Finally apply conditional orthogonality as in Proposition 21.2 and the Lipschitz bound for g . $\square$

Corollary 22.2 (Resource transport of an attainable profile). *If an M -state target observer has risk R_M and a K -state causal simulation has error ϵ for its bounded decision problem, then the source has a KM -state observer of risk at most $R_M + L\epsilon$. Two simulations give the corresponding two directed inequalities; they need not preserve the same cardinality. A K -valued decoder-side event can always be absorbed into a register at a factor K , but a continuous side observation admits no such finite-factor assertion in general.*

Proof. Use the product construction. For the finite event store its current value as the second register coordinate. For the last assertion, an exact uniform real observation supplied directly to a decoder has zero squared estimation error with one otherwise empty state. If it is available only before an M -label encoding, scalar quantization has error $1/(12M^2) > 0$. Thus no finite factor simulates that side interface for all accuracies. $\square$

The theorem joins quotient approximation to experiment comparison. Contractivity is a property to be verified on the true quotient, and Q_t is an executable representation of it. A finite-dimensional formula for the conditional mean is sufficient for one application, but is not a hypothesis of the theorem. In Section 25 the true state is an entire probability measure. In the next section a different structure replaces metric contraction by an exact finite-tree update.

23. A STRUCTURAL THEOREM FOR PREDICTIVE TREES

23.1. Local hypotheses and the finite profile. Let $\mathcal{L}$ be a pruned factorial language on a finite alphabet of size $b \geq 2$. Thus prefixes and suffixes of legal words are legal, every word has

an infinite continuation, and only legal children occur in its rooted tree. Write $[v]$ for a cylinder in the boundary Ω and $\emptyset$ for the empty word. Assume that along any branch there are at most U consecutive vertices with only one legal successor. This condition allows unary vertices but excludes an infinite deterministic branch. Let μ be a stationary probability on Ω , positive on every legal cylinder, and put $p_v = \mu[v]$.

A predictive energy is a positive family a_v , with $a_\emptyset = 1$. Put $D_v = (a_v/p_v)^{1/2}$. The following are the hypotheses of the structural theorem. There are constants $a_- > 0$, $\rho < 1$, and $c_0, C_0 > 0$ such that

$$a_- a_v \leq a_{vi} \leq \rho a_v, \quad \sum_{i: vi \in \mathcal{L}} a_{vi} \leq \rho a_v, \quad (23.1)$$

$$a_{uv} < a_v \quad (u \neq \emptyset), \quad D_{vi} \leq D_v. \quad (23.2)$$

These statements concern legal words only. A bounded Borel map Z from Ω into a separable Hilbert space satisfies

$$\text{diam } Z([v]) \leq C_0 D_v, \quad (23.3)$$

$$\|Z(\omega) - Z(\omega')\| \geq c_0 D_u \quad \text{if } u \text{ is their longest common prefix.} \quad (23.4)$$

The image may be infinite dimensional. Separation is an explicit geometric hypothesis, not a consequence of the word notation.

Starting with the root, repeatedly split a leaf of maximal energy into *all* its legal children. Fix a tie rule. Let $\mathcal{T}_N$ be the last tree in this sequence with at most N leaves, $\mathcal{P}_N$ its leaf cut, and

$$G_N = \sum_{v \in \mathcal{P}_N} a_v. \quad (23.5)$$

A split at a unary vertex is an actual step. Finiteness of $\mathcal{T}_N$ will follow below. The construction is a mathematical finite algorithm; computable implementation of arbitrary real energies is an additional input, not part of their Borel specification.

There are two causal interpretations of these data.

Deletion model. Independent acquisitions $\omega \sim \mu$ occur at renewal times of independent positive integer length L , with $\mathbb{E}L < \infty$. The acquired sequence is observed exactly. At age k the desired output is $g(\sigma^k \omega)$, where g is bounded and Hilbert valued. Write

$$w_k = \mathbb{P}(L > k)/\mathbb{E}L, \quad W(n) = \sum_{k \geq n} w_k, \quad Z(\omega) = (\sqrt{w_k} g(\sigma^k \omega))_{k \geq 0}.$$

In addition to (23.1)–(23.4), assume the local oscillation bound

$$\sum_{k < |v|} w_k (\text{osc}_{[v_{k:}]} g)^2 + 4\|g\|_\infty^2 W(|v|) \leq C_1 D_v^2. \quad (23.6)$$

A register sees the renewal event only during its update; the decoder receives neither that event nor a stored raw observation for free.

Insertion model. Let $(Y_t)_{t \in \mathbb{Z}}$ be a stationary finite-alphabet observed process whose reversed past $\omega_t = (Y_t, Y_{t-1}, \dots)$ has law μ and language $\mathcal{L}$. For a square-integrable stationary Hilbert target U_t , suppose

$$m_t = \mathbb{E}[U_t \mid Y_s : s \leq t] = Z(\omega_t), \quad B = \mathbb{E}\|U_t - m_t\|^2. \quad (23.7)$$

The finite register may start at zero with a fixed state; it is not initialized with the infinite past. Infinite stationary pasts here specify the reference prediction and its law. A finite warm-up is included in the limiting average. No renewal or independent-observation assumption is imposed in this interpretation.

For either model let R_M be the infimum, over all M -value registers with causal standard Borel update and decoding kernels, of the upper limiting expected average squared loss. Absolute time and independent coding randomization are permitted. In the insertion model put $E_M = R_M - B$.

Theorem 23.1 (Predictive-tree realization in two orientations). *Under (23.1)–(23.4),*

$$cG_M \leq e_M^2(Z) \leq CG_M. \quad (23.8)$$

Every complete greedy tree is suffix closed. A tree with L leaves has exactly $1 + \sum_{v \text{ internal}} d^+(v)$ vertices, and at most $(U+1)(2L-1)$ vertices; all of them may be used as designated states. For every fixed $B_0 \geq 1$, $G_{\lceil B_0 M \rceil} \asymp_{B_0} G_M$.

In the deletion model, with (23.6),

$$cG_M \leq e_M^2(Z) \leq R_M \leq CG_M. \quad (23.9)$$

In the insertion model,

$$cG_M \leq e_M^2(Z) \leq E_M \leq CG_M. \quad (23.10)$$

The converses apply to the lower limiting average of each randomized, time-dependent observer. The upper bounds are achieved in order by stationary deterministic updates. In particular a logarithmic age or history multiplier is not required. Constants depend on the displayed structural hypotheses, not on the register budget.

23.2. Balance, dilation and exact closure.

Lemma 23.2 (Greedy balance). *For all sufficiently large N , $N/b \leq |\mathcal{P}_N| \leq N$. If θ_N is the largest energy of a current leaf, then*

$$a_- \theta_N \leq a_v \leq \theta_N \quad (v \in \mathcal{P}_N). \quad (23.11)$$

The trees are finite and their profiles have the fixed-dilation property in Theorem 23.1.

Proof. Suppress unary vertices in any finite complete tree. A rooted tree with L leaves and no unary internal vertex has at most $2L - 1$ vertices, since its number of edges is the sum of its internal outdegrees. Each vertex of the suppressed tree accounts for itself and at most U unary vertices on its incoming path, including the initial root path. Thus the original tree has at most $(U+1)(2L-1)$ vertices. An infinite sequence of splits with bounded leaf number is impossible, proving finiteness. The split following $\mathcal{T}_N$ crosses the budget and adds at most $b-1$ leaves. Hence $|\mathcal{P}_N| > N - b + 1$.

The maximal leaf energy never increases. Every present leaf was born with energy at least a_- times an earlier maximal energy. This proves (23.11), including any unary splits. Set $\ell = U + 1$. Every vertex has at least two distinct descendants at depth ℓ , and at least 2^j at depth ℓj . Choose j with $2^j > b(B_0 + 2)$. If the later maximal energy at budget $\lceil B_0 N \rceil$ were less than $a_-^{\ell j + 1} \theta_N$, every original leaf would have been split through all these ℓj levels: all intervening vertices have energy at least that threshold. The resulting number of leaves would exceed $\lceil B_0 N \rceil$, a contradiction. Therefore the later maximum is at least a fixed multiple of θ_N . Applying (23.11) at the later budget gives $G_{\lceil B_0 N \rceil} \geq c_{B_0} N \theta_N \geq c_{B_0} G_N$. Splitting decreases the energy sum by (23.1), giving the reverse inequality. Adjusting constants covers the finite initial range. $\square$

Lemma 23.3 (Two exact operations on one tree). *Every proper suffix of an internal greedy-tree vertex is internal, and every suffix of any vertex is present. Consequently deletion of the first letter stays in the tree. If v is a leaf and iv is legal, then iv has a leaf prefix in the same tree. It can therefore be updated by prepending i and reading only the letters of v until that leaf is reached.*

Proof. Suppose an internal word uv , $u \neq \emptyset$, is about to be split. If its suffix v has not been split, some current leaf t is a prefix of v . Child decrease and strict prefix-attachment decrease give $a_t \geq a_v > a_{uv}$, contrary to maximality. Thus the suffix is already internal. For a leaf $i_1 \cdots i_n$ with

$n \geq 2$, its parent is internal; the suffix $i_2 \cdots i_{n-1}$ of that parent is internal, so its legal child $i_2 \cdots i_n$ is present. The one-letter case ends at the root. Iteration gives closure of every vertex.

If there were no leaf prefix of iv , the complete tree would contain iv as an internal vertex. Its suffix v would then be internal by the first assertion, contradicting that it is a leaf. This proves the insertion assertion. Only legal children were used in either proof. $\square$

23.3. Static lower bound and the causal converse.

Lemma 23.4 (Arbitrary centres). *The bounds (23.8) hold for arbitrary Hilbert-space centres, not only centres on $Z(\Omega)$.*

Proof. Choose one image point in each leaf cylinder to obtain $e_N^2(Z) \leq C_0^2 G_N$. For the lower bound use $\mathcal{P}_{4bM}$, which has at least $4M$ leaves for large M . The open tubes about $Z([v])$ of radii δD_v are disjoint for a sufficiently small fixed $\delta > 0$. Indeed for incomparable v, v' their common prefix u has $D_u \geq \max(D_v, D_{v'})$, and (23.4) separates the two images by at least $c_0 D_u$. Take $2\delta < c_0$.

At most M of the tubes contain one of M proposed centres. On every unoccupied cylinder the squared error is at least $\delta^2 D_v^2$. Its integrated contribution is $\delta^2 a_v$. By balance, the at least three-quarters unoccupied leaves carry at least $3a_-/4$ of the total energy sum. Thus $e_M^2(Z) \geq cG_{4bM} \geq c'G_M$ by Lemma 23.2. The finite initial range is harmless: (23.4) and positive cylinder masses at unbounded branching depths give infinitely supported image law, so every fixed finite-centre error is positive. Alternatively the same tube proof at a larger fixed cut proves that positivity directly. $\square$

Lemma 23.5 (Renewal converse). *In the deletion experiment every randomized, nonstationary M -state observer has lower limiting average loss at least $e_M^2(Z)$.*

Proof. A standard Borel probability kernel is realized as a measurable function of its input and an independent uniform random variable [2]. Applying this at the countably many update and decode times gives an independent coding sequence. A data-dependent persistent position in such a sequence, if used, belongs to the charged state; the representation itself supplies no history channel.

Condition on an acquisition at a fixed time, its pre-acquisition history, and the independent coding variables. The acquired ω still has law μ . Along a hypothetical continuation with no new acquisition, each of the at most M post-acquisition labels determines one output string. For a truncation at age n , the survival-weighted loss is therefore at least $\mathbb{E}L e_M^2(Z^{<n})$, where $Z^{<n}$ is the first n coordinates of the weighted orbit. At a fixed physical time, events specifying the last acquisition are disjoint. Summing this bound over acquisition times at least n before the end of a horizon and using the Cesàro renewal count gives lower limiting average at least $e_M^2(Z^{<n})$. For completeness, the elementary renewal theorem gives expected renewals per unit time tending to $1/\mathbb{E}L$; a fixed truncation removes at most n terminal time indices, each with bounded contribution. No stationary coding rule is presumed.

Finally $e_M^2(Z^{<n}) \uparrow e_M^2(Z)$. Monotonicity follows by projection. The reverse limiting inequality follows by appending zero tails to an arbitrarily good truncated code: the extra mean squared norm is at most $\|g\|_\infty^2 W(n) \rightarrow 0$. This completes the proof. $\square$

Proof of Theorem 23.1. The static assertion, finite count, dilation and suffix closure are proved above. In the deletion experiment retain every tree vertex as a designated state. At acquisition store the unique leaf prefix of ω . Between acquisitions delete its first letter until the empty word is reached, and then keep that word. Choose a representative in each nonempty cylinder and output its g value; use a fixed bounded output at the empty word. For an acquired leaf v the weighted error is bounded by the left side of (23.6). Averaging and applying renewal reward gives at most $C_1 G_N$. Lemma 23.5 gives the unrestricted converse.

For insertion, after reaching a leaf use the prepend-and-parse rule of Lemma 23.3. During initialization store the observed reversed word while it is internal, or its first leaf prefix when one appears. This uses only vertices of the same tree and after at most its maximum depth gives the correct leaf prefix of the full observed past. Decode a representative Z value in that leaf cylinder. Its stationary excess loss is at most $C_0^2 G_N$ by (23.3) and conditional orthogonality. The finite warm-up has zero limiting average contribution. The converse holds at each time: freeze independent coding randomness, apply orthogonality against m_t , and use the at most M decoder values to lower bound excess by $e_M^2(Z)$. Stationarity makes this the same number at every time, even for time-dependent decoders.

In either orientation choose N a fixed sufficiently small multiple of M , so that $(U+1)(2|\mathcal{P}_N|-1) \leq M$. Dilation converts G_N into $O(G_M)$ without a time-depth multiplier. One fixed state handles the remaining bounded range of budgets. This proves the two risk assertions. $\square$

The static part of the proof uses familiar separated-cylinder quantization ideas [9, 6]. Maximal-energy prefix splitting is likewise a classical coding device [7]. The additional conclusion is that a strict *predictive* energy inequality makes the adaptive tree executable in both orientations, with its whole persistent state counted.

24. TWO MODEL VERIFICATIONS

24.1. Markov-renewal orbit geometry. Let A be a primitive zero-one matrix on b symbols, with Perron root greater than one. Its legal one-sided sequences have a stationary fully supported law μ such that every legal conditional child probability is at least $\zeta > 0$. Let h_i map a compact set of diameter at most one into pairwise g -separated subsets of itself. Assume word scales r_v , $r_\emptyset = 1$, satisfy

$$\begin{aligned} D_0^{-1} r_v |x - y| &\leq |h_v x - h_v y| \leq r_v |x - y|, \\ D_0^{-1} r_u r_v &\leq r_{uv} \leq r_u r_v, \quad 0 < r_- \leq r_i \leq r_+ < 1. \end{aligned} \quad (24.1)$$

The injective coding map identifies the legal sequences with a compact set K , on which T deletes the first symbol. Let $f : K \rightarrow \mathbb{R}^m$ obey

$$|f(x) - f(y)| \leq H|x - y|^\beta, \quad \sum_{j=0}^J |f(T^j x) - f(T^j y)|^2 \geq c_0 |x - y|^{2\beta}, \quad 0 < \beta \leq 1. \quad (24.2)$$

The renewal age weights satisfy $0 < q_- \leq w_{k+1}/w_k \leq q_+ < 1$. Set $\tau = 2\beta$ and

$$V(v) = \sum_{k < |v|} w_k r_{v_k}^\tau + W(|v|), \quad a_v = \mu[v]V(v), \quad Z(x) = (\sqrt{w_k} f(T^k x))_{k \geq 0}. \quad (24.3)$$

Corollary 24.1 (Markov renewal as a structural instance). *The data (24.1)–(24.3) satisfy Theorem 23.1 in its deletion orientation. In particular $e_M^2(Z) \asymp R_M \asymp G_M$, with a stationary finite-tree observer and an unrestricted randomized, nonstationary converse. Forbidden edges are not completed, and unary vertices are charged.*

Proof. Every suffix scale is at most one and appending a letter decreases its scale. If $|u| = m$ and $|v| = n$, monotonicity of w gives

$$\begin{aligned} V(v) - \left(\sum_{j < n} w_{m+j} r_{v_j}^\tau + W(m+n) \right) &\geq r_v^\tau \sum_{k < m} w_k, \\ V(uv) - \left(\sum_{j < n} w_{m+j} r_{v_j}^\tau + W(m+n) \right) &\leq r_+^\tau r_v^\tau \sum_{k < m} w_k. \end{aligned}$$

Therefore $V(uv) < V(v)$. Stationarity gives only $\mu[uv] \leq \mu[v]$, which is enough to give the strict energy inequality. Equality of cylinder masses at a unique predecessor causes no problem.

Appending a legal letter multiplies old spatial terms by a factor between $D_0^{-\tau} r_-^\tau$ and r_+^τ . The old tail $W(n)$ becomes $w_n r_i^\tau + W(n+1)$. Since $w_n \geq (1 - q_+)W(n)$ and $W(n+1) \geq q_-W(n)$, we have

$$\ell V(v) \leq V(vi) \leq \rho V(v), \quad \ell = \min(D_0^{-\tau} r_-^\tau, q_-), \quad \rho = 1 - (1 - r_+^\tau)(1 - q_+) < 1.$$

Use $a_- = \zeta \ell$ and sum the conditional masses to verify (23.1). These inequalities also give monotonicity of $D_v = \sqrt{V(v)}$. In a primitive graph with Perron root greater than one, a deterministic path cannot run for b repeated unary vertices: a repeated vertex would generate a closed deterministic cycle. Thus one can take $U = b$.

Introduce $d_\beta(x, y)^2 = \sum_k w_k |T^k x - T^k y|^{2\beta}$. Hölder continuity bounds the orbit distance from above by $H d_\beta$. Summing the delayed lower bound (24.2), and using $w_{k-j} \leq q_-^{-j} w_k$, bounds d_β from above by a constant times the orbit distance. Points in one cylinder have squared d_β distance at most $V(v)$. Points whose longest common prefix is u have, before its end, distances at least $D_0^{-1} g r_{u_k}$, and at its end distance at least g . The last term controls $W(|u|)$ since $W(n) \leq w_n/(1 - q_+)$. This proves both image comparisons. Finally the Hölder bound on each suffix and the bounded tail prove (23.6). The structural theorem now supplies all combinatorial, quantization, and causal conclusions. $\square$

This corollary recovers the earlier Markov-renewal profile without repeating its tree argument as a new model-specific proof. Gibbs pressure identifies the profile exponent under the additional thermodynamic assumptions of the previous Markov theorem, whose full proof is retained in the companion. Pressure is a computation of this instance, not an assumption of the foundational realization theorem. For clarity, the passage from power bounds to an exponent uses only the following elementary fact.

Proposition 24.2 (Logarithmic squeeze). *If for every $\delta > 0$ a positive sequence R_M satisfies $c_\delta M^{-\alpha-\delta} \leq R_M \leq C_\delta M^{-\alpha+\delta}$ eventually, then $-\log R_M / \log M \rightarrow \alpha$.*

Proof. Taking logarithms gives lower and upper limits between $\alpha - \delta$ and $\alpha + \delta$, because fixed logarithmic constants divided by $\log M$ vanish. Let δ decrease to zero. In a pressure-root proof one first chooses the auxiliary root parameter on either side of its zero and only then lets M tend to infinity; root continuity then supplies the arbitrary δ used here. $\square$

24.2. An infinite hidden shift with noisy continual reports. Let $(B_t)_{t \in \mathbb{Z}}$ be independent identically distributed Bernoulli variables. The hidden state at time t is the whole sequence $X_t = (B_t, B_{t-1}, \dots)$. A binary memoryless channel reports Y_t about B_t . Write

$$\gamma = \mathbb{P}(Y_t = 1) \in (0, 1), \quad a_i = \mathbb{P}(B_t = 1 \mid Y_t = i), \quad a_0 \neq a_1.$$

Fix $0 < r < 1$. The Hilbert-valued prediction target and its mean are

$$U_t = (\sqrt{1 - r^2} r^j B_{t-j})_{j \geq 0}, \quad m_t = (\sqrt{1 - r^2} r^j a_{Y_{t-j}})_{j \geq 0}. \quad (24.4)$$

This squared loss can equally be implemented by drawing a fresh terminal query j with probability $(1 - r^2)r^{2j}$ and scoring the prediction of B_{t-j} . The independent query is a task index, not a new observation of the hidden state. Its distribution and interface are fixed in advance.

Theorem 24.3 (Sharp register law for an infinite posterior). *Put $p_0 = 1 - \gamma$, $p_1 = \gamma$ and let $s \in (0, 1)$ be the unique solution of*

$$r^{2s}(p_0^s + p_1^s) = 1, \quad \alpha = (1 - s)/s. \quad (24.5)$$

For the actual noisy shift experiment,

$$R_M = B + \Theta((a_1 - a_0)^2 M^{-\alpha}), \quad B = \sum_{i=0}^1 p_i a_i (1 - a_i). \quad (24.6)$$

The constants in the excess comparison depend on r, p_0, p_1 , not on a_0, a_1 . No posterior truncation is assumed in deriving this law. For a fair hidden bit and a binary symmetric channel of error $\varepsilon < 1/2$, it becomes

$$R_M = \varepsilon(1 - \varepsilon) + \Theta((1 - 2\varepsilon)^2 M^{2 \log r / \log 2}).$$

At $\varepsilon = 1/2$ the predictive mean is constant and the excess is zero with one state.

Proof. The full posterior is the product of Bernoulli laws with parameters $a_{Y_{t-j}}$. Thus (24.4) is the actual conditional mean, and orthogonality gives the displayed B . Normalize its nonconstant part by $a_1 - a_0$. On the full binary past language take $p_v = \prod p_{v_j}$, $D_v = r^{|v|}$ and $a_v = p_v r^{2|v|}$. Children multiply energy by $p_i r^2$, their sum multiplies it by r^2 , and every nonempty prefix attachment strictly decreases it. If two pasts first differ at position n , their normalized squared image distance lies between $(1 - r^2)r^{2n}$ and r^{2n} . All hypotheses of Theorem 23.1 hold in the insertion orientation.

For the exact order of G_M , set $q_i = (p_i r^2)^s$. Equation (24.5) says $q_0 + q_1 = 1$. For every finite complete cut, consistency of this Bernoulli measure gives $\sum_v a_v^s = 1$. By greedy balance, for a cut with L leaves and maximum θ , $L(a_- \theta)^s \leq 1 \leq L\theta^s$. Hence $G_M \asymp L\theta \asymp L^{-(1-s)/s} \asymp M^{-\alpha}$. The root exists uniquely because the left side of (24.5) decreases continuously from two to $r^2 < 1$. The fair case solves it explicitly, and the constant-mean case is immediate. If only reports from time zero are supplied, the unobserved past contributes at most $r^{2(t+1)}$ to the squared target norm at time t . Its average vanishes; the finite-tree initialization in the structural theorem still gives (24.6). $\square$

This is not an exactly closed finite list of conditional means. Every past report changes its own posterior coordinate, and arbitrarily many coordinates can be queried. It also has no fixed-step common minorizer: after k hidden updates the old tail remains intact, and two different initial tails give disjoint supports. A measure dominated by all those kernels must be zero. As always, a bare Borel encoding of an infinite sequence into one real number is possible; it is not a finite-cardinality register and does not change the theorem.

25. FINITE PRESENTATIONS OF MEASURE-VALUED FILTERS

The preceding sharp example has a separated tree geometry. We next give a constructive approximation theorem requiring no such separation and no finite-dimensional exact posterior coordinate.

Lemma 25.1 (A finite Wasserstein presentation). *For $n \geq 1$ let $\mathcal{C}_n$ consist of empirical probability measures with n equally weighted atoms on $\{0, 1/n, \dots, 1\}$. Then*

$$|\mathcal{C}_n| = \binom{2n}{n} \leq 4^n, \quad \sup_{\pi \in \mathcal{P}([0,1])} W_1(\pi, Q_n \pi) \leq \frac{3}{2n} \quad (25.1)$$

for a specified Borel quantizer Q_n .

Proof. Take the generalized quantiles of π at $(j - 1/2)/n$, $1 \leq j \leq n$, and round them to nearest grid points, breaking ties toward zero. Quantiles are Borel functions of a probability law. If q is its monotone quantile function, the integral of the error from replacing q by its midpoint value on each of the n equal subintervals is at most $1/n$: bound each integral by its interval length times the oscillation and sum the monotone variations, at most one. Rounding adds at most $1/(2n)$. In one dimension W_1 is the L^1 distance of quantile functions, giving the bound. The possible sorted

lists are multisets of size n from $n + 1$ points, counted by $\binom{2n}{n}$. The binomial theorem bounds this by 4^n . $\square$

Theorem 25.2 (A charged measure-valued recursive approximation). *Let the actual filter obey $\pi_t = \Phi_{y_t, a_t}(\pi_{t-1})$ on $\mathcal{P}([0, 1])$, and suppose every allowed update is ρ -contractive in W_1 , with $\rho < 1$. Suppose an implementable update has W_1 defect at most η . The initialized register*

$$\hat{\pi}_t = Q_n \tilde{\Phi}_{y_t, a_t}(\hat{\pi}_{t-1}), \quad \hat{\pi}_0 \in \mathcal{C}_n$$

has at most $\binom{2n}{n}$ persistent values and satisfies, for every actual observation path,

$$W_1(\pi_t, \hat{\pi}_t) \leq \rho^t + \frac{3/(2n) + \eta}{1 - \rho}. \quad (25.2)$$

This bounds the error of every 1-Lipschitz test expectation, uniformly over the choice of test. Thus for $M \geq 4$, $n = \lfloor \log M / \log 4 \rfloor$ gives error $O((\log M)^{-1} + \eta)$ and squared error $O((\log M)^{-2} + \eta^2)$ after the initialization term. If these test expectations are conditional means, the latter is a Bayes excess bound. Report-law comparison additionally requires the report Lipschitz hypothesis of Theorem 22.1; it is not automatic from W_1 contraction.

Proof. Apply Theorem 22.1 with the true metric state $S = \mathcal{P}([0, 1])$, radius $3/(2n)$, and diameter one. The dual formula for W_1 gives the assertion for every test; independent selection of a terminal test does not enlarge its bound. The codebook count and the logarithmic substitution are exact. Only an index of a codebook law persists between updates, not its exact unquantized posterior or a numerical residual. $\square$

25.1. A nonlinear physical realization. Let B_t be independent fair bits and define

$$h_0(x) = x/5 + x^2/50, \quad h_1(x) = 3/5 + x/5 + x^2/50, \quad X_t = h_{B_t}(X_{t-1}). \quad (25.3)$$

The observation Y_t reports B_t correctly with probability $3/4$, independently at each time. A permitted terminal probe $g : [0, 1] \rightarrow [0, 1]$ returns a Bernoulli variable of probability $g(X_t)$; the tested family is 1-Lipschitz probes and their bounded finite executable continuations. The actual update-report instrument on measures is

$$\mathsf{T}_y \pi = \frac{1}{2} \left\{ \frac{3}{4} (h_y)_\# \pi + \frac{1}{4} (h_{1-y})_\# \pi \right\}.$$

Its mass is $1/2$, independent of the old state. The normalized filter is therefore

$$\Phi_y \pi = \frac{3}{4} (h_y)_\# \pi + \frac{1}{4} (h_{1-y})_\# \pi. \quad (25.4)$$

Corollary 25.3 (Nonlinear posterior, explicit register). *The actual measure-valued filter (25.4) is $6/25$ -contractive in W_1 . Theorem 25.2 applies with $\eta = 0$ to an exact rational implementation on $\mathcal{C}_n$. The model has no fixed-step nonzero common minorizer. For every integer $D \geq 1$, the first D polynomial moments do not determine their own exact next-step update on the space of all prior laws.*

Proof. The two images lie respectively in $[0, 11/50]$ and $[3/5, 41/50]$. Each derivative is between $1/5$ and $6/25$. Couple two old laws optimally and use the same branch draw in their mixtures; this gives $W_1(\Phi_y \pi, \Phi_y \pi') \leq (6/25) W_1(\pi, \pi')$. For a codebook input, (25.4) has at most $2n$ atoms, at rational points with denominator dividing $50n^2$, with weights multiples of $1/(4n)$. Sorted cumulative weights give the exact quantiles, which are rounded back to the fixed grid. Thus no numerical limit is being hidden in $\eta = 0$. The transient list of at most $2n$ atoms is discarded; its entries and weights have $O(\log n)$ -bit rational descriptions. The designated persistent object is the single codebook index. Approximate arithmetic instead obeys the separately charged η bound of the theorem.

After k hidden transitions the law from a fixed x has finite support $\{h_v(x) : |v| = k\}$. For a second initial point x' , each equality $h_v(x') = h_u(x)$ excludes at most one x' , since h_v is strictly

increasing. Choose x' outside this finite set. The two supports are disjoint, so a measure dominated by all k -step laws must be zero.

For the moment assertion, the D th updated moment integrates $\frac{3}{4}h_y(x)^D + \frac{1}{4}h_{1-y}(x)^D$, a polynomial of degree $2D$ with nonzero leading coefficient 50^{-D} . It is not in the span of $1, x, \dots, x^D$. Linear algebra on finitely many distinct points therefore supplies a signed measure annihilating this span but not the new polynomial. Its positive and negative parts have equal mass; normalizing them yields two prior probabilities with identical first D moments and different updated D th moment. The argument is about ordinary finite moment closure, not the impossibility of a Borel encoding of measures by real numbers. $\square$

The quotient here is genuinely the posterior law for the specified probe family: bounded Lipschitz functions separate probability measures on a compact interval, and constants do not affect that separation. Updates respect this quotient by (25.4). The advancing reports have the exact state-independent fair law, while a terminal Bernoulli probe has total-variation defect at most $W_1(\pi_t, \hat{\pi}_t)$. Thus the same finite presentation defines a causal experiment simulation with a controlled terminal error and an exactly charged register. Multiple probes would update the posterior again; no claim for an unverified enlarged probe protocol is implicit. This example connects positive instruments, predictive quotient, recursive approximation and resource transformation within one model.

26. SHARP FINITE-EXPERIMENT BUDGETS

For experiments $\mathcal{E} = \{P_\theta\}$ and $\mathcal{F} = \{Q_\theta\}$ with the same parameter set, write

$$\delta(\mathcal{E}, \mathcal{F}) = \inf_K \sup_\theta \|KP_\theta - Q_\theta\|_{\text{TV}}, \quad \Delta(\mathcal{E}, \mathcal{F}) = \max\{\delta(\mathcal{E}, \mathcal{F}), \delta(\mathcal{F}, \mathcal{E})\}.$$

All parameterized laws in this section are Borel families. The infimum is over parameter-independent Markov kernels. These are terminal experiment comparisons in the sense of [11, 12]; only the encoding statistic, not the comparison randomization, is part of the physical data reduction.

Let $\mathcal{G} = \{N(z, \Gamma) : z \in K\}$, where $K \subset \mathbb{R}^r$ is compact and Γ is positive definite. In the lower bound assume that K contains a nonempty Euclidean ball.

Theorem 26.1 (Gaussian finite-experiment cardinality). *There are constants $c, C > 0$, depending on K, Γ, r , such that*

$$cM^{-2/r} \leq \inf_{\mathcal{F}: |\mathcal{F}| \leq M} \Delta(\mathcal{F}, \mathcal{G}) \leq CM^{-2/r}, \quad M \geq 1. \quad (26.1)$$

The lower bound ranges over all experiments on at most M outcomes, not just deterministic Gaussian quantizers. The upper bound is attained in order by a deterministic statistic with J^r labels, where $J = \lfloor M^{1/r} \rfloor$, followed by a specified positive reconstruction kernel. No overflow label is needed.

26.1. The cardinality obstruction.

Lemma 26.2 (A decision-theoretic lower bound). *Under the preceding assumptions, $\Delta(\mathcal{F}, \mathcal{G}) \geq cM^{-2/r}$ whenever $\mathcal{F}$ has at most M outcomes.*

Proof. Restrict the mean parameter to a closed cube K_0 of positive side length contained in K and put the uniform prior on K_0 . Restriction can only decrease an experiment distance. Use squared-error estimation of z , with actions restricted to K_0 ; its loss range is bounded by $L = \text{diam}(K_0)^2$. Let $m(y) = \mathbb{E}[z | y]$ in this Gaussian experiment. Differentiation under the compact prior integral gives

$$Dm(y) = \text{Cov}(z | y)\Gamma^{-1}. \quad (26.2)$$

The posterior density is positive on the cube. Hence its covariance is positive definite, and $Dm(y)$ is nonsingular for every y . The inverse function theorem on a small neighborhood of any fixed

y_0 , together with the positive continuous marginal density of y , shows that the law of $m(y)$ has a density bounded below by a positive constant on some smaller cube in $\mathbb{R}^r$.

This implies $e_M^2(m) \geq c_0 M^{-2/r}$. To see it directly, put balls of radius $t = (|Q|/(2Mv_r))^{1/r}$ about the M proposed centres, where Q is the smaller cube and v_r the volume of the unit ball. At least half of Q remains outside them, and contributes at least a fixed multiple of t^2 . This argument permits arbitrary centres.

Every M -outcome garbling S of the Gaussian observation therefore has Bayes risk at least $B_{\mathcal{G}} + c_0 M^{-2/r}$, by conditional orthogonality. If $\Delta(\mathcal{F}, \mathcal{G}) < \epsilon$, choose a kernel K from $\mathcal{G}$ to $\mathcal{F}$ with error below ϵ . The experiment $K\mathcal{G}$ has at most M outputs, and bounded-loss comparison gives

$$B_{K\mathcal{G}} \leq B_{\mathcal{F}} + L\epsilon \leq B_{\mathcal{G}} + 2L\epsilon.$$

The second inequality uses a kernel in the reverse direction. Comparing these bounds and taking the infimum over ϵ proves the result. Randomized actions do not improve a Bayes squared-error minimum. Notice that $\mathcal{F}$ itself was not assumed to be a Gaussian garbling. $\square$

26.2. Positive second-order reconstruction. Put

$$\tau(x) = \frac{1}{2} + \pi^{-1} \arctan x, \quad u = \tau^{\otimes r}(y) \in (0, 1)^r. \quad (26.3)$$

Let p_z be the transformed Gaussian density. If $x_j = \tan \pi(u_j - 1/2)$, then

$$p_z(u) = \phi_{\Gamma}(x - z) \prod_{j=1}^r \pi(1 + x_j^2). \quad (26.4)$$

It extends by zero across every face to a smooth periodic density. Indeed each fixed-order derivative is a polynomial in x times the Gaussian density, which tends to zero faster than any polynomial as a face is approached. Uniformity holds for z in the fixed compact set. For use below define the finite constants

$$H_2 = \sup_{z, u} \|D^2 p_z(u)\|_{\text{op}}, \quad H_0 = \max_j \sup_{z, u_j} p_{z, j}(u_j), \quad (26.5)$$

where $p_{z, j}$ is a transformed marginal density. These suprema, or any upper bounds obtained from (26.4), fix all the reconstruction and boundary constants; they do not depend on M or sample size. For instance, differentiating twice introduces only polynomials and the eigenvalue bounds of Γ , so bounded mean norm and fixed covariance conditioning suffice.

Partition the flat unit torus into J^r cubes of side $h = 1/J$, with centres c_i , and write $Q_J(u)$ for the cube label. For $J \geq 2$ let Φ_i be the periodic tensor product of the piecewise-linear hat functions with value one at c_i , zero at its neighboring mesh centres, and support within one mesh step in each coordinate. Then

$$\Phi_i \geq 0, \quad \int \Phi_i = h^r, \quad \sum_i \Phi_i(u) = 1.$$

On each mesh interval the neighbouring centres reproduce the affine coordinate, using local periodic lifts across the seam. Define the reconstruction from label i to have density $h^{-r} \Phi_i(u)$, then apply τ^{-1} coordinatewise. Values on the faces have probability zero.

Lemma 26.3 (A quadratic reconstruction estimate). *Uniformly in $z \in K$, this kernel H_J satisfies*

$$\|H_J Q_J \tau^{\otimes r} N(z, \Gamma) - N(z, \Gamma)\|_{\text{TV}} \leq \frac{r H_2}{12 J^2}. \quad (26.6)$$

Proof. Let p_i be the probability of cube i . Symmetric Taylor expansion about its centre gives $|p_i/h^r - p_z(c_i)| \leq r H_2 h^2/24$. The linear interpolant $I_h p_z = \sum_i p_z(c_i) \Phi_i$ has error at most $r H_2 h^2/8$: its weights reproduce constants and affine functions and the sum of their coordinate variances is at most $r h^2/4$. The second Taylor remainder is half this variance times H_2 . Both calculations hold across the seam because p_z has a smooth periodic extension. The reconstructed density is

$\sum_i (p_i/h^r) \Phi_i$. Nonnegativity and the partition of unity give supremum error at most $rH_2h^2/6$. Integrating over volume one and using the factor $1/2$ for total variation proves the claim. Inverting (26.3) preserves this distance. $\square$

Proof of Theorem 26.1. Lemma 26.2 gives the lower bound. The deterministic statistic $Q_J \tau^{\otimes r}$ has J^r outputs, so the forward deficiency is zero; Lemma 26.3 gives the reverse deficiency. For $M \geq 2^r$, $J = \lfloor M^{1/r} \rfloor \geq M^{1/r}/2$. For the remaining finite range use any one-outcome experiment and the trivial bound $\Delta \leq 1$, increasing C . The reconstruction uses positive probabilities; it is not a signed interpolation operator. $\square$

The use of a compactified grid avoids a growing Gaussian overflow radius. The use of hats rather than constant cells cancels the first Taylor term. Both choices are necessary for the stated proof of a logarithm-free quadratic budget rate.

27. A SHARP BUDGET FOR THE PHYSICAL COUNT-TO-CONTACT EXPERIMENT

We now specify the actual local protocol, following the known-mark experiment of A2 [8]. Fix $\lambda_0 \in (0, \infty)^m$, a matrix $B \in \mathbb{R}^{m \times d}$, a compact set $H \subset \mathbb{R}^d$, and a full-row-rank matrix $C \in \mathbb{R}^{r \times m}$. Observe independent counts

$$K_{N,j} \sim \text{Pois}(N\lambda_{0,j} + \sqrt{N}(Bh)_j), \quad h \in H. \quad (27.1)$$

Write $\Lambda = \text{diag } \lambda_0$, $A = CB$, and $\Gamma = C\Lambda C^T$. The contact score and limiting experiment are

$$S_N = C(K_N - N\lambda_0)/\sqrt{N}, \quad \mathcal{G} = \{N(Ah, \Gamma) : h \in H\}.$$

The physical finite statistic is

$$T_{N,J} = Q_J(\tau^{\otimes r}(S_N)), \quad |T_{N,J}| \leq J^r. \quad (27.2)$$

There is no physical jitter, no retained real score, and no omitted overflow event. Every real score belongs to one of the J^r cells of the open compactified cube, with deterministic boundary tie-breaking.

Theorem 27.1 (Joint sample and register law). *For the fixed protocol (27.1), there are $N_0, C_* < \infty$ such that uniformly for $N \geq N_0$ and $J \geq 2$,*

$$\Delta(\mathcal{E}_{N,J}, \mathcal{G}) \leq C_* \left(N^{-1/2} + \frac{J}{\sqrt{N}} + J^{-2} \right), \quad (27.3)$$

where $\mathcal{E}_{N,J} = \{\text{Law}_h(T_{N,J}) : h \in H\}$. If AH contains an r -dimensional ball, every experiment with at most M outputs has distance from $\mathcal{G}$ at least $c_* M^{-2/r}$. Consequently the statistic with $J = \lfloor M^{1/r} \rfloor$ satisfies

$$c_* M^{-2/r} \leq \Delta(\mathcal{E}_{N,J}, \mathcal{G}) \leq C_{*,K} M^{-2/r}, \quad 2^r \leq M \leq KN^{r/6}, \quad (27.4)$$

for every fixed $K < \infty$. In particular $M \asymp N^{r/6}$ gives distance of order $N^{-1/3}$; achieving distance of order $N^{-1/5}$ requires and is achieved by a budget of order $N^{r/10}$. All constants may depend on the fixed matrices, compact parameter set and covariance conditioning. No uniformity at rank degeneration or unknown marks is asserted.

We prove the uniform estimates in five steps, keeping comparison randomization separate from (27.2).

Lemma 27.2 (Jittered Poisson local normal comparison). *There is an absolute $C_P < \infty$ such that for $\mu \geq 1$, independent $U \sim \text{Unif}(-1/2, 1/2)$, and $K \sim \text{Pois}(\mu)$,*

$$\|\text{Law}(K + U) - N(\mu, \mu)\|_{\text{TV}} \leq C_P \mu^{-1/2}. \quad (27.5)$$

Proof. It suffices to consider $\mu \geq 64$, increasing the constant for the remaining range. The jittered density is $p_\mu(k) = e^{-\mu} \mu^k / k!$ on $[k - 1/2, k + 1/2)$, $k \geq 0$. On the central cells $|k - \mu| \leq \mu^{2/3}/4$, put $t = k - \mu$, $x = k + u$, $|u| \leq 1/2$, and $z = (x - \mu)/\sqrt{\mu}$. Stirling's formula with remainder $0 < R_k < 1/(12k)$ gives

$$\log \frac{p_\mu(k)}{\phi_\mu(x)} = -\frac{1}{2} \log(k/\mu) - R_k - \{k \log(k/\mu) - k + \mu\} + \frac{(x - \mu)^2}{2\mu},$$

where ϕ_μ is the $N(\mu, \mu)$ density. For $|t| \leq \mu/2$, Taylor's theorem applied to $(1 + v) \log(1 + v) - (1 + v) + 1$ gives remainder at most $|t|^3/\mu^2$ after its quadratic term. Also $|\log(1 + t/\mu)| \leq 2|t|/\mu$ and $|(x - \mu)^2 - t^2| \leq |t| + 1/4$. Substitution therefore gives the convenient bound

$$\left| \log \frac{p_\mu(k)}{\phi_\mu(x)} \right| \leq 32\mu^{-1/2}(1 + |z|^3), \quad \left| \log \frac{p_\mu(k)}{\phi_\mu(x)} \right| \leq 64. \quad (27.6)$$

The constants are deliberately loose. For example $|z| \leq \mu^{1/6}/4 + 1/(2\sqrt{\mu})$ on these cells verifies the second bound directly. The inequality $|e^v - 1| \leq e^{|v|}|v|$ now bounds the integrated central density error by $32e^{64}\mu^{-1/2}(1 + \mathbb{E}|G|^3)$, $G \sim N(0, 1)$.

For completeness the omitted tails are exponentially smaller. For $|s| \leq 1$, the centered Poisson log moment generating function is $\mu(e^s - 1 - s) \leq \mu s^2$. Exponential Markov with $s = \mu^{-1/3}/16$, and its negative, bounds the probability outside the central cells, allowing the half-unit jitter margin, by $2 \exp(-\mu^{1/3}/128)$ after adjusting an absolute factor for $64 \leq \mu$ in the bounded initial interval. The Gaussian bound follows by its moment generating function. Thus an absolute choice such as $C_P = 10^8 e^{64}$ dominates the central error, the tails and the finite initial range. This proves a uniform $\mu^{-1/2}$ bound rather than an unquantified local-limit assertion. $\square$

Lemma 27.3 (Variance replacement and projected comparison). *Let U_j be independent uniform half-cell jitters, used only in this proof, and set $S_N^o = S_N + CU/\sqrt{N}$. Uniformly for $h \in H$,*

$$\| \text{Law}_h(S_N^o) - N(Ah, \Gamma) \|_{\text{TV}} \leq C_0 N^{-1/2}. \quad (27.7)$$

One may choose C_0 as a function only of m , the minimum of λ_0 , $\max_{h \in H} \|Bh\|$, and fixed matrix norms.

Proof. Put $b_* = \max_{h \in H} \|Bh\|_\infty$ and $\lambda_* = \min_j \lambda_{0,j}$. Choose $N_0 \geq \max(128/\lambda_*, 4b_*^2/\lambda_*^2)$. Then each count mean is at least $N\lambda_*/2$. Product comparison using Lemma 27.2, followed by centering and scaling, gives distance at most $mC_P\sqrt{2/\lambda_*}N^{-1/2}$ from the normal vector with mean Bh and covariance $\Lambda + N^{-1/2} \text{diag}(Bh)$.

For the replacement of this covariance by Λ , the relative variance eigenvalues are $1 + t_j$, with $|t_j| \leq b_*/(\lambda_*\sqrt{N}) \leq 1/2$. The exact Gaussian relative entropy is $\frac{1}{2} \sum_j \{t_j - \log(1 + t_j)\} \leq \sum_j t_j^2$. The elementary entropy-variation inequality $\|P - Q\|_{\text{TV}} \leq \sqrt{D_{\text{KL}}(P\|Q)/2}$ gives a further bound $\sqrt{m/2} b_*/(\lambda_*\sqrt{N})$. Taking $C_0 = mC_P\sqrt{2/\lambda_*} + \sqrt{m/2} b_*/\lambda_*$ is sufficient. Projection by C contracts total variation and sends the fixed Gaussian covariance to Γ , proving the assertion. $\square$

Lemma 27.4 (Deterministic cell crossing). *Let H_0 be the transformed marginal bound in (26.5), now for the means AH , and put $d_N = \|C\|_{\infty,1}/(2\pi\sqrt{N})$. Then*

$$\sup_{h \in H} \| \text{Law}_h(T_{N,J}) - Q_J \tau^{\otimes r} N(Ah, \Gamma) \|_{\text{TV}} \leq 2C_0 N^{-1/2} + 2rH_0 J d_N. \quad (27.8)$$

Here $\|C\|_{\infty,1}$ is the maximum absolute row sum.

Proof. The coupled scores satisfy $\|S_N - S_N^o\|_\infty \leq \|C\|_{\infty,1}/(2\sqrt{N})$. Since $|\tau'| \leq 1/\pi$, their compactified coordinates differ by at most d_N . Distinct grid labels require a coordinate of the jittered score to be within d_N of a grid boundary. On the torus there are J boundaries per coordinate;

under the transformed Gaussian their total probability is at most $2rH_0Jd_N$, or its truncation by one. Replacing the jittered law by that Gaussian adds at most $C_0N^{-1/2}$, by (27.7). The coupling inequality then compares theunjittered and jittered labels. One more data-processing comparison of the jittered labels to the Gaussian labels adds the second $C_0N^{-1/2}$. Boundary tie events are already in the strips. This argument never asserts that the unjittered real score has a Gaussian law in total variation. $\square$

Proof of Theorem 27.1. Use the deterministic Gaussian-to-cell kernel for the direction $\mathcal{G} \rightarrow \mathcal{E}_{N,J}$. Lemma 27.4 bounds its error by $b_{N,J} = 2C_0N^{-1/2} + 2rH_0Jd_N$. For the reverse direction, use the positive kernel H_J from Lemma 26.3. Kernel contraction and the triangle inequality give

$$\sup_h \|H_J \text{Law}_h(T_{N,J}) - N(Ah, \Gamma)\|_{\text{TV}} \leq b_{N,J} + rH_2/(12J^2).$$

This proves (27.3), with, for example, $C_* = 2C_0 + rH_0\|C\|_{\infty,1}/\pi + rH_2/12$ increased to one. Every kernel is parameter independent. Jitter was introduced only to bound a difference of laws, not to modify the physical statistic.

If AH contains a ball, restrict to an affine parameter section over a smaller mean cube and apply Lemma 26.2. More generally one may simply index the Gaussian family by its mean and choose a Borel representative preimage for each mean in the cube (the compact fibres of the continuous map on H admit a measurable section); the lower comparison uses only that subfamily. For $J = \lfloor M^{1/r} \rfloor$ and $M \geq 2^r$, $J^{-2} \leq 4M^{-2/r}$. If $M \leq KN^{r/6}$, then $M^{3/r}/\sqrt{N} \leq K^{3/r}$; hence $J/\sqrt{N}$ and $N^{-1/2}$ are also at most fixed multiples of $M^{-2/r}$. This proves the window. Taking M respectively of order $N^{r/6}$ and $N^{r/10}$ gives the two rates. Conversely, an error at most $CN^{-1/5}$ forces $M \geq cN^{r/10}$ by the universal cardinality lower bound. $\square$

27.1. The A2 matrices and the scope of the consequence. In A2's known-mark local protocol, the moment derivative D has $\Sigma = D\Lambda D^T > 0$, and its calibrated exposure direction is $B = \Lambda D^T \Sigma^{-1}$, so $DB = I$. If L maps to the contact coordinates and F is an orthonormal frame of its range, the matrices above are

$$C = F^T L D, \quad A = C B = F^T L, \quad \Gamma = C \Lambda C^T > 0.$$

Here A has full row rank r . A compact local parameter neighborhood therefore has an image containing a mean ball. Both sides of Theorem 27.1 apply to that actual protocol. This invokes a structural experiment-cardinality theorem verbatim, not just an analogy between a root singularity and a contact observation.

The result is a matched budget theorem in the displayed joint window. It is not a lower bound saying that $N^{-1/3}$ is the best sample-size comparison achievable with arbitrarily many labels. It also does not assign a stochastic noise law to a conductor scheme or establish uniformity at a rank-changing physical stratum. The geometric source, the exposure protocol, the finite statistic and the comparison kernels are distinct typed objects throughout.

REFERENCES

- [1] D. Blackwell, Equivalent comparisons of experiments, *Ann. Math. Statist.* **24** (1953), 265–272. doi:10.1214/aoms/1177729032.
- [2] O. Kallenberg, *Foundations of Modern Probability*, third ed., Probability Theory and Stochastic Modelling, vol. 99, Springer, 2021. doi:10.1007/978-3-030-61871-1.
- [3] T. Linder and S. Yüksel, On optimal zero-delay coding of vector Markov sources, *IEEE Trans. Inform. Theory* **60** (2014), 5975–5991. doi:10.1109/TIT.2014.2346780.
- [4] R. G. Wood, T. Linder and S. Yüksel, Optimal zero delay coding of Markov sources: stationary and finite memory codes, *IEEE Trans. Inform. Theory* **63** (2017), 5968–5980. doi:10.1109/TIT.2017.2692215.
- [5] H. Luschgy and G. Pagès, Sharp asymptotics of the functional quantization problem for Gaussian processes, *Ann. Probab.* **32** (2004), 1574–1599. doi:10.1214/009117904000000324.

- [6] M. K. Roychowdhury, Quantization dimension for Gibbs-like measures on cookie-cutter sets, *Kyoto J. Math.* **54** (2014), 239–257. doi:10.1215/21562261-2642377.
- [7] B. P. Tunstall, *Synthesis of noiseless compression codes*, Ph.D. dissertation, Georgia Institute of Technology, 1967.
- [8] Q. Qi, Conductor strata and nonreduced multiplication failure schemes, A2 revision 118, manuscript, September 22, 2026. Fixed repository edition, commit 44bfc648ead0.
- [9] M. Kesseböhmer and S. Zhu, The upper and lower quantization coefficient for Markov-type measures, *Math. Nachr.* **290** (2017), 827–839. doi:10.1002/mana.201500328. Theorem 1.1 checked in the authors’ preprint, entitled *The quantization for Markov-type measures on a class of ratio-specified graph directed fractals*, arXiv:1406.3257v1.
- [10] Y. E. Demirci, A. D. Kara and S. Yüksel, Average cost optimality of partially observed MDPs: contraction of nonlinear filters and existence of optimal solutions and approximations, *SIAM J. Control Optim.* **62** (2024), 2859–2883. doi:10.1137/24M1643736. Theorem numbering in the text refers to arXiv:2312.14111v3.
- [11] L. Le Cam and G. L. Yang, *Asymptotics in Statistics: Some Basic Concepts*, second edition, Springer, New York, 2000. doi:10.1007/978-1-4612-1166-2.
- [12] E. Torgersen, *Comparison of Statistical Experiments*, Encyclopedia of Mathematics and its Applications 36, Cambridge University Press, Cambridge, 1991.
- [13] C. R. Shalizi and J. P. Crutchfield, Computational mechanics: pattern and prediction, structure and simplicity, *J. Statist. Phys.* **104** (2001), 817–879. arXiv:cond-mat/9907176.
- [14] L. Le Cam, Sufficiency and approximate sufficiency, *Ann. Math. Statist.* **35** (1964), 1419–1455. doi:10.1214/aoms/1177700372.
- [15] E. Norberg, Comparison of statistical experiments with filtered probability spaces, *Statistics & Risk Modeling* **20** (2002), 1–28. doi:10.1524/strm.2002.20.14.1.
- [16] A. Nerode, Linear automaton transformations, *Proc. Amer. Math. Soc.* **9** (1958), 541–544. doi:10.1090/S0002-9939-1958-0135681-9.
- [17] B. R. Kloeckner, A generalization of Hausdorff dimension applied to Hilbert cubes and Wasserstein spaces, *J. Topol. Anal.* **4** (2012), 203–235. doi:10.1142/S1793525312500094. Author preprint: arXiv:1105.0360v2.
- [18] J. Matoušek, *Using the Borsuk–Ulam Theorem: Lectures on Topological Methods in Combinatorics and Geometry*, Universitext, Springer, Berlin, 2003. doi:10.1007/978-3-540-76649-0.
- [19] Q. Qi, Attainable information and causal compression at exponent collisions, A1 revision 37, manuscript, September 9, 2026. Fixed repository edition, commit 90465076589f.
- [20] A. Nayyar, A. Mahajan and D. Teneketzis, Decentralized stochastic control with partial history sharing: a common information approach, *IEEE Trans. Automat. Control* **58** (2013), 1644–1658. doi:10.1109/TAC.2013.2239000. The theorem numbers cited here refer to the author preprint, arXiv:1209.1695.
- [21] H. S. Witsenhausen, On the structure of real-time source coders, *Bell System Tech. J.* **58** (1979), 1437–1451. doi:10.1002/j.1538-7305.1979.tb02263.x.
- [22] N. Saldi, A topology for team policies and existence of optimal team policies in stochastic team theory, preprint (2017), arXiv:1711.00639.
- [23] S. Yüksel and N. Saldi, Convex analysis in decentralized stochastic control, strategic measures and optimal solutions, author preprint arXiv:1609.07685v2.
- [24] S. Yüksel, On optimal causal coding of partially observed Markov sources in single and multi-terminal settings, author preprint arXiv:1010.4824v4.
- [25] S. Kiefer, Notes on equivalence and minimization of weighted automata, preprint (2020), arXiv:2009.01217v1.
- [26] M. Nisio, On a non-linear semi-group attached to stochastic optimal control, *Publ. Res. Inst. Math. Sci.* **12** (1976), 513–537. doi:10.2977/prims/1195190727.
- [27] N. Bäuerle and U. Rieder, Partially observable risk-sensitive Markov decision processes, *Math. Oper. Res.* **42** (2017), 1180–1196. doi:10.1287/moor.2016.0844. The comparison uses Theorem 3.3 and Section 4.4 of arXiv:1504.03530v3.
- [28] W.-G. Tzeng, A polynomial-time algorithm for the equivalence of probabilistic automata, *SIAM J. Comput.* **21** (1992), 216–227. doi:10.1137/0221017.